

Plots album

Prediction of Dynamic Stall on a Pitching NACA 0012 Section

A case study at a tunnel condition and at a rotor-blade-station condition

UNISTALL, a Leishman-Beddoes dynamic-stall load model with tabulated separation (Python package unistall)

Leishman-Beddoes indicial method with a tabulated separation point

Akosa Samuel Onyejekwe (independent)

7 October 2026

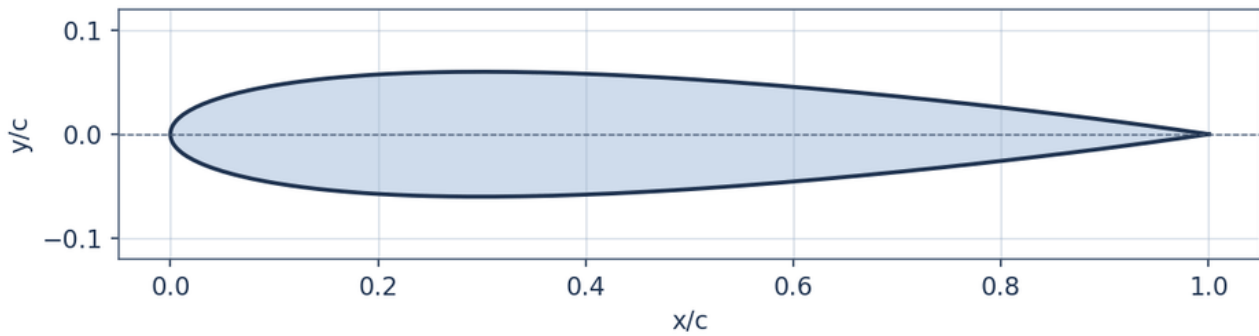
112 images: every image of the case study and of results/figures.

The flow-field images show a reconstruction, not a flow solution: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Case B is an aerofoil at a rotor-blade-station condition in a steady stream with prescribed pitch, not a rotor calculation.

Geometry (01_geometry)

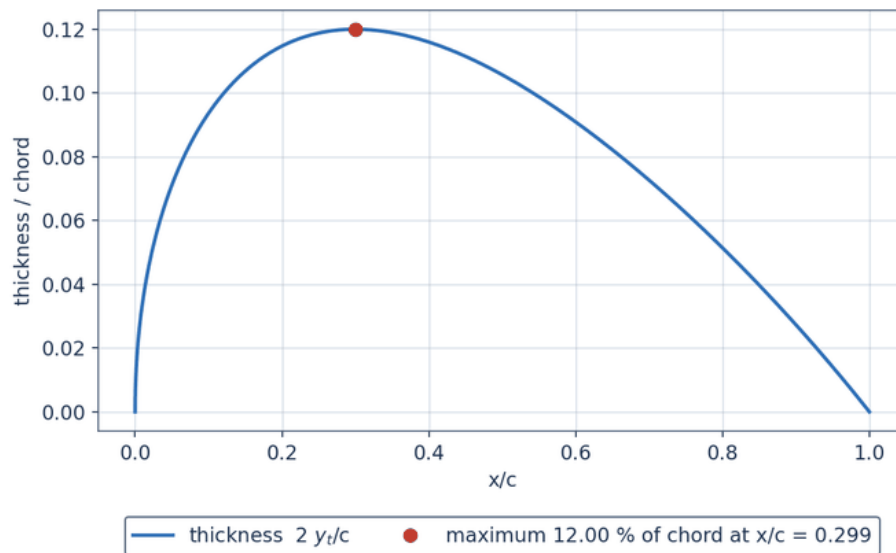
NACA 0012 section (closed trailing edge)



01_geometry/fig_geometry_profile.png

The NACA 0012 section, from the four-digit thickness form with a closed trailing edge.

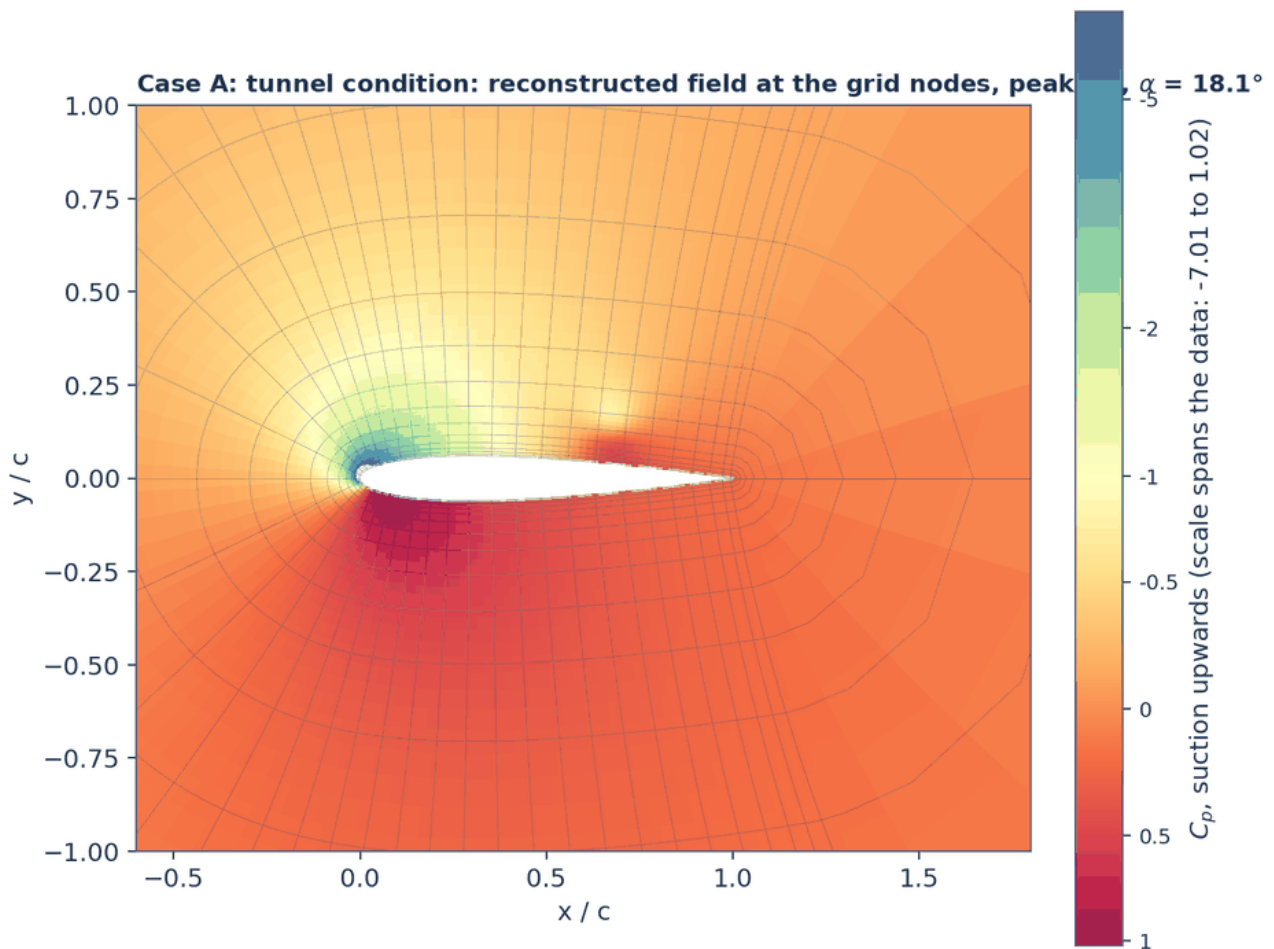
NACA 0012 thickness distribution



01_geometry/fig_geometry_thickness.png

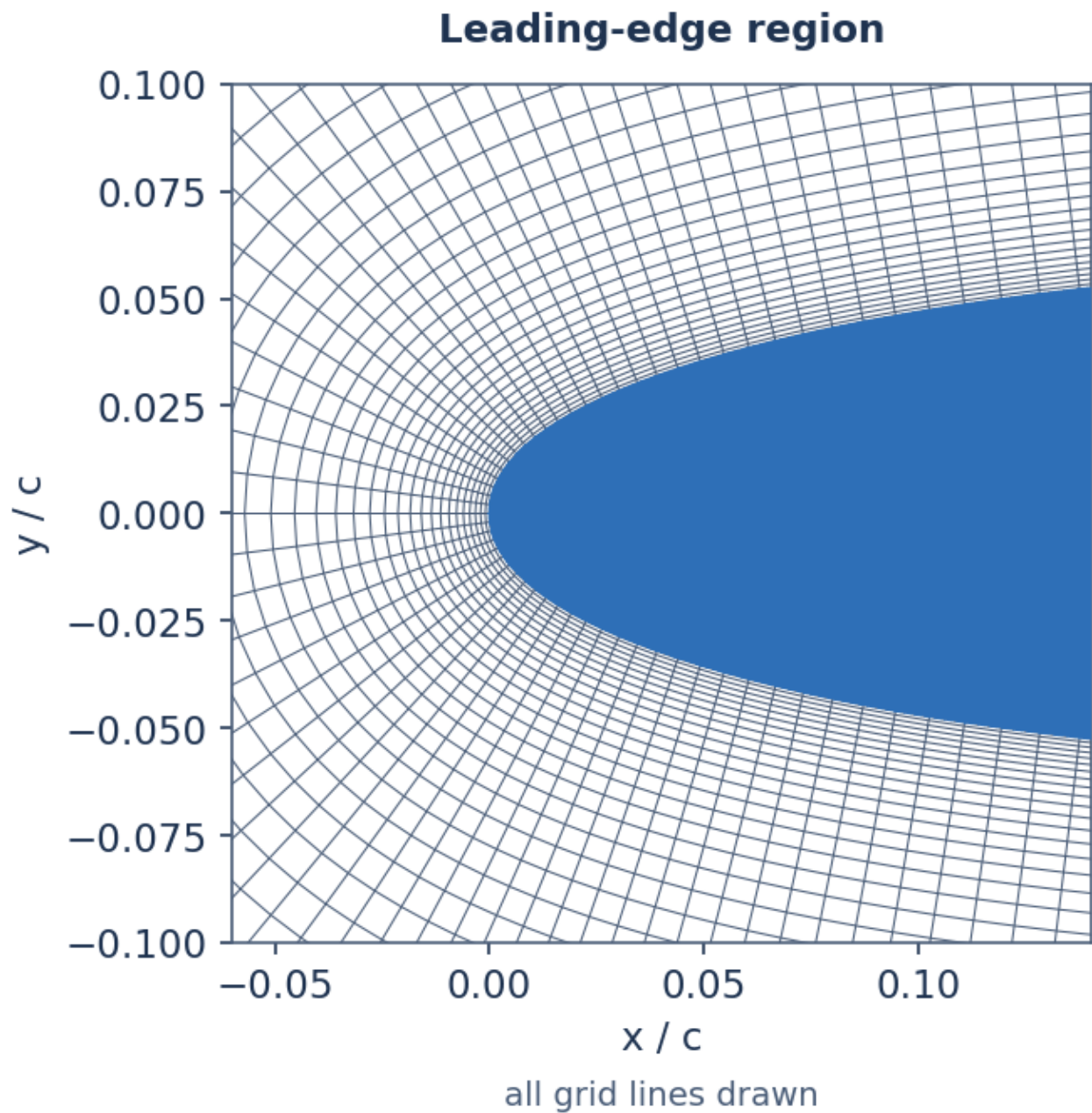
Thickness distribution of the NACA 0012 section.

The grid (02_mesh)



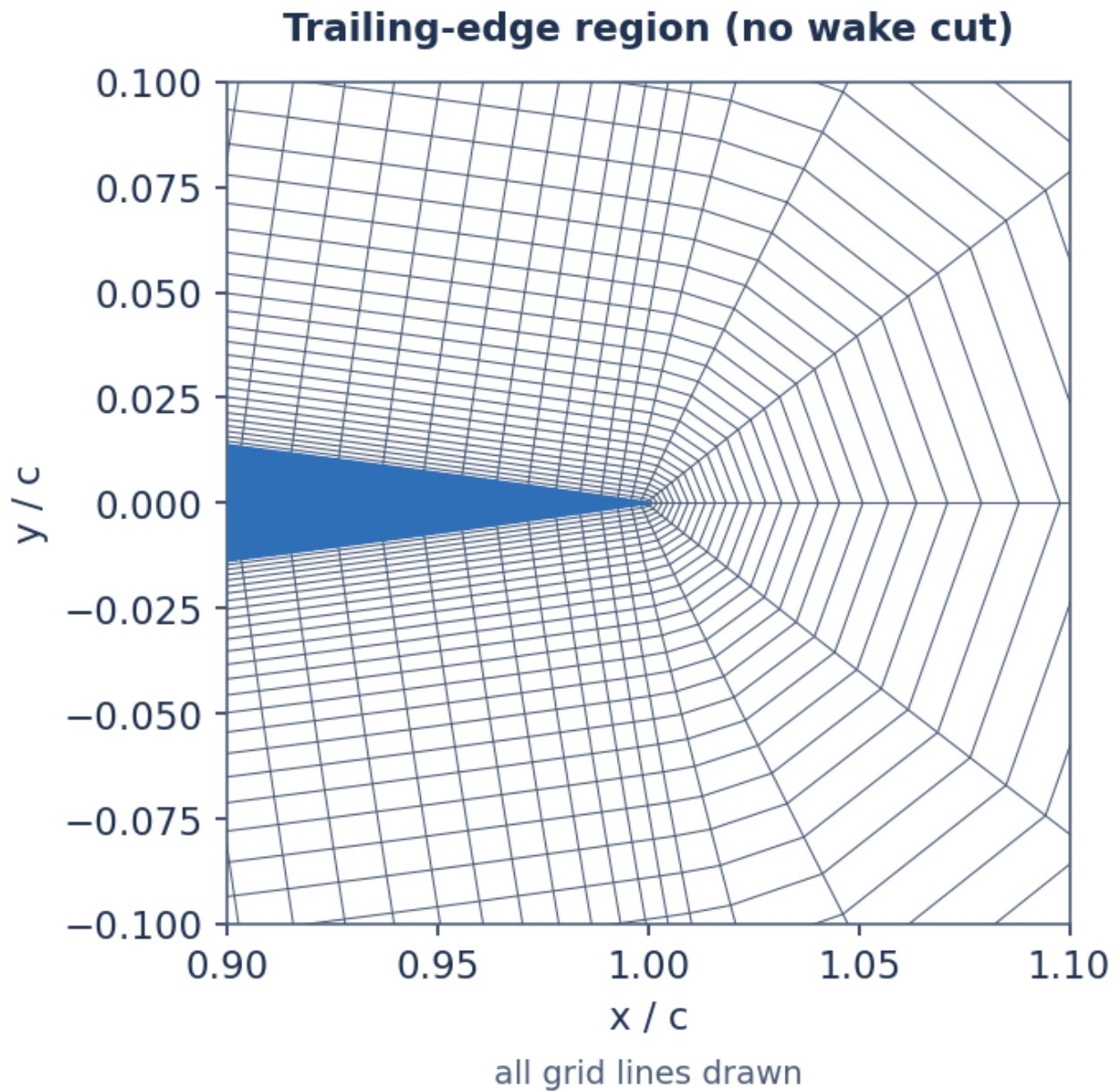
02_mesh/fig_mesh_field_A_validation.png

Case A: tunnel condition: pressure coefficient of the reconstructed field evaluated at the nodes of the O-grid, at the instant of peak lift. The grid is only the set of points; no equation of the flow is solved on it. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.



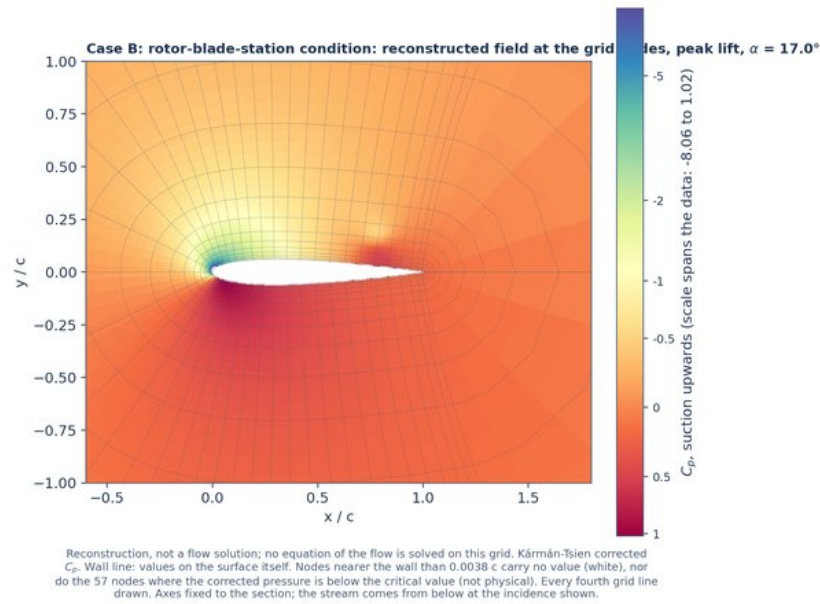
02_mesh/fig_mesh_le_zoom.png

The O-grid at the leading edge. Not read by the load model.



02_mesh/fig_mesh_te_zoom.png

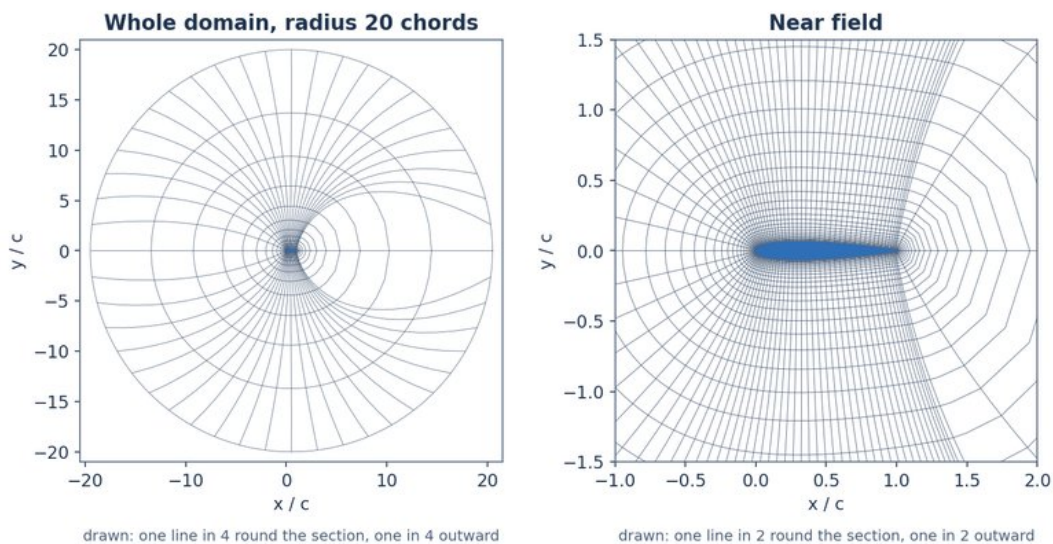
The O-grid at the trailing edge. Not read by the load model.



02_mesh/fig_mesh_field_B_application.png

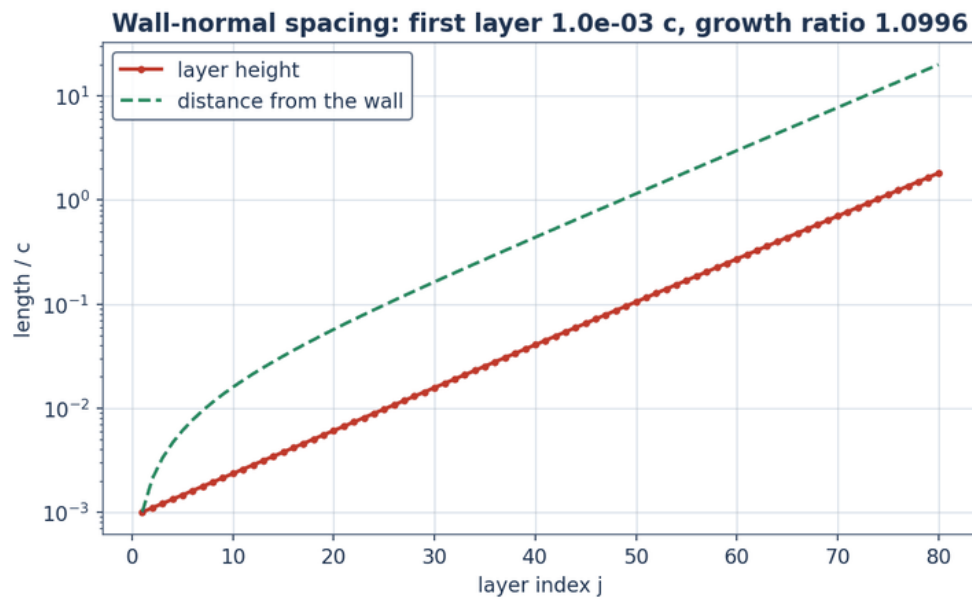
Case B: rotor-blade-station condition: pressure coefficient of the reconstructed field evaluated at the nodes of the O-grid, at the instant of peak lift. The grid is only the set of points; no equation of the flow is solved on it. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

O-grid round the NACA 0012 section: 193 x 81 nodes (illustration; no flow solved on it)



02_mesh/fig_mesh_full.png

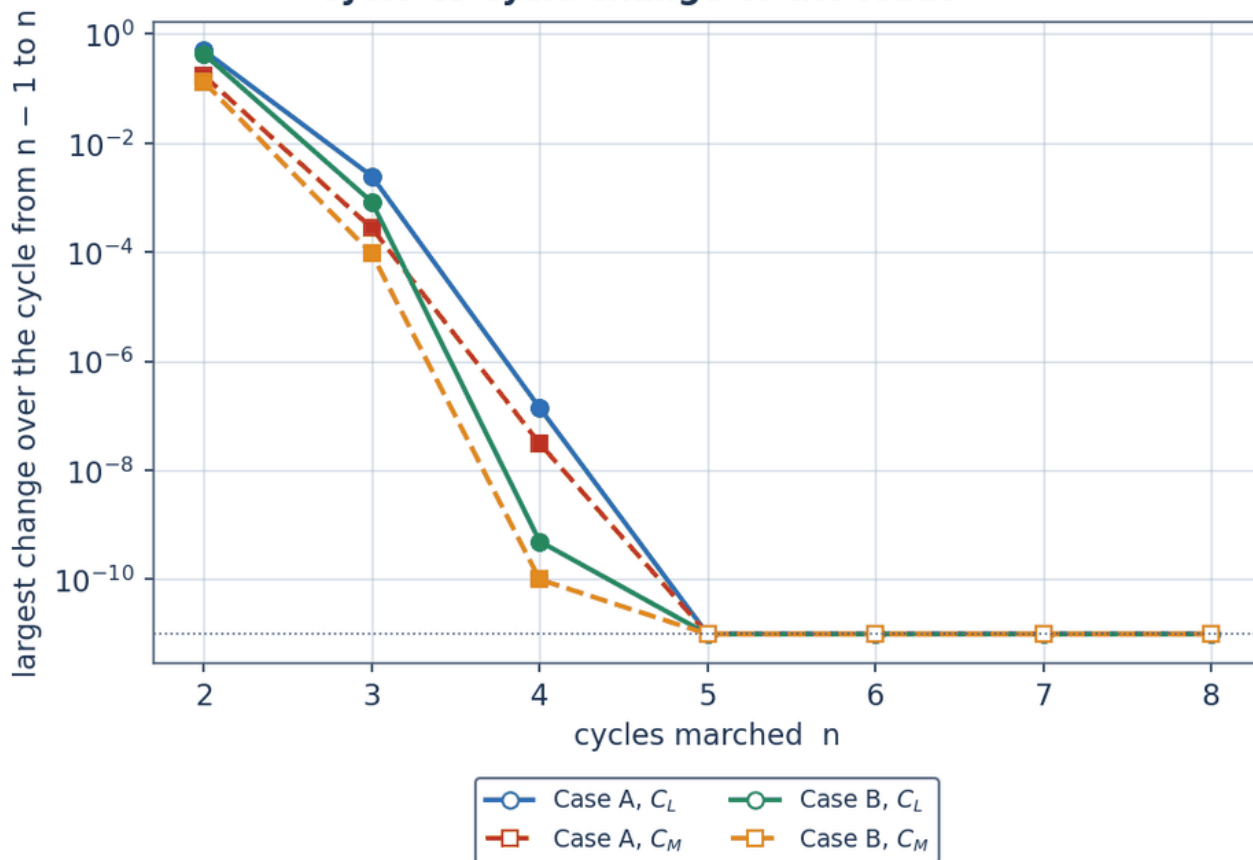
The O-grid round the section. An illustration: no equation of the flow is solved on it and the load model does not read it.

**02_mesh/fig_mesh_wall_spacing.png**

Spacing of the O-grid along the wall and away from it. Not read by the load model.

Loads, states and convergence (06_postprocessing)

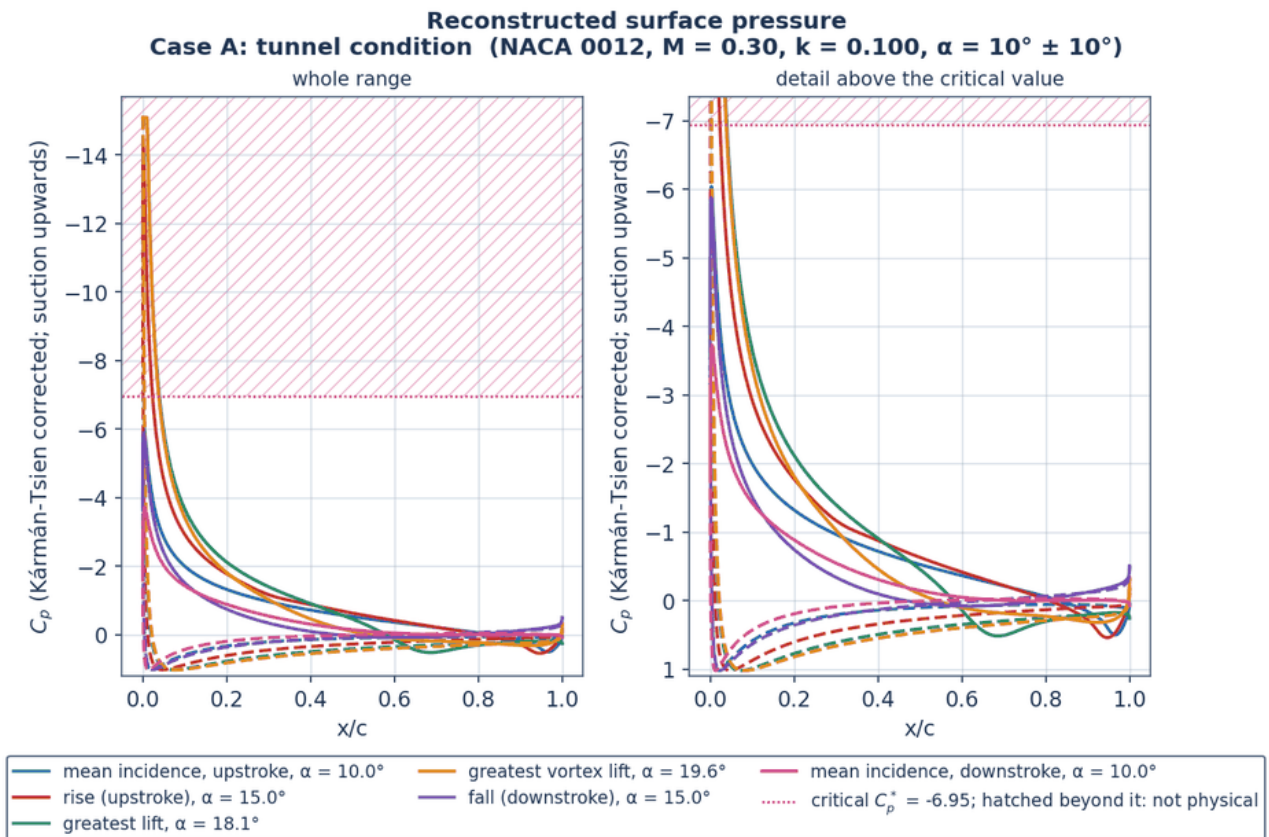
Cycle-to-cycle change of the loads



Open symbols on the dotted line: no change at all. Cycles marched for the reported results: Case A: 5, Case B: 4. A change that reappears after the loop has settled comes from the sign of the pitch rate at a turning point, which rounding can place on either side of zero.

06_postprocessing/plots/convergence_residuals.png

Largest change of the loads over the cycle when one more cycle is marched, against the number of cycles marched.



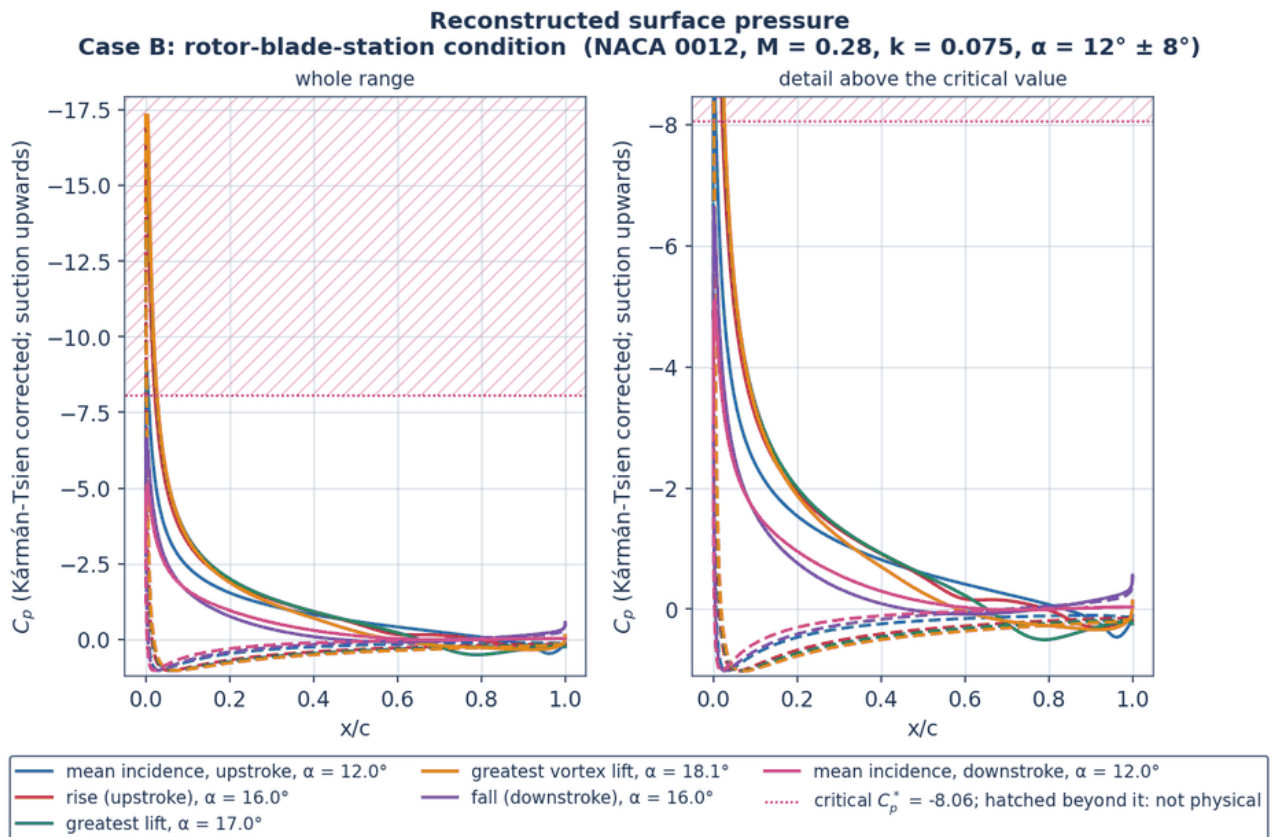
Solid: upper surface; dashed: lower surface. Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Beyond the dotted line the corrected C_p is below the critical value -6.95 (local Mach number above one): the correction does not hold there and the values are not physical; the corrected value is not allowed below -15.1 (95 % of the vacuum value). Part of the surface beyond the critical pressure: mean incidence, upstroke 0.0 %; rise (upstroke) 2.0 %; greatest lift 3.2 %; greatest vortex lift 3.1 %; fall (downstroke) 0.0 %; mean incidence, downstroke 0.0 %. Lift returned by this pressure at greatest lift: $+0.000$ % from the lift given, with the sheet carrying 0.893 of the circulation of the lift (Prandtl-Glauert factor 0.954); the incompressible pressure with the whole circulation returns it within -1.00 %.

Change of surface C_p beneath the vortex marker when it is put in: rise (upstroke) $+0.07$ ($x/c = 0.31$); greatest lift $+0.56$ ($x/c = 0.68$); greatest vortex lift -0.14 ($x/c = 1.00$). It RAISES the pressure beneath it at 2 of these 3 instants: it does not reproduce the suction footprint of a dynamic-stall vortex.

06_postprocessing/plots/cp_distribution_A_validation.png

Case A: tunnel condition: reconstructed surface pressure at instants of the cycle, suction upwards, the band beyond the critical value hatched. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.



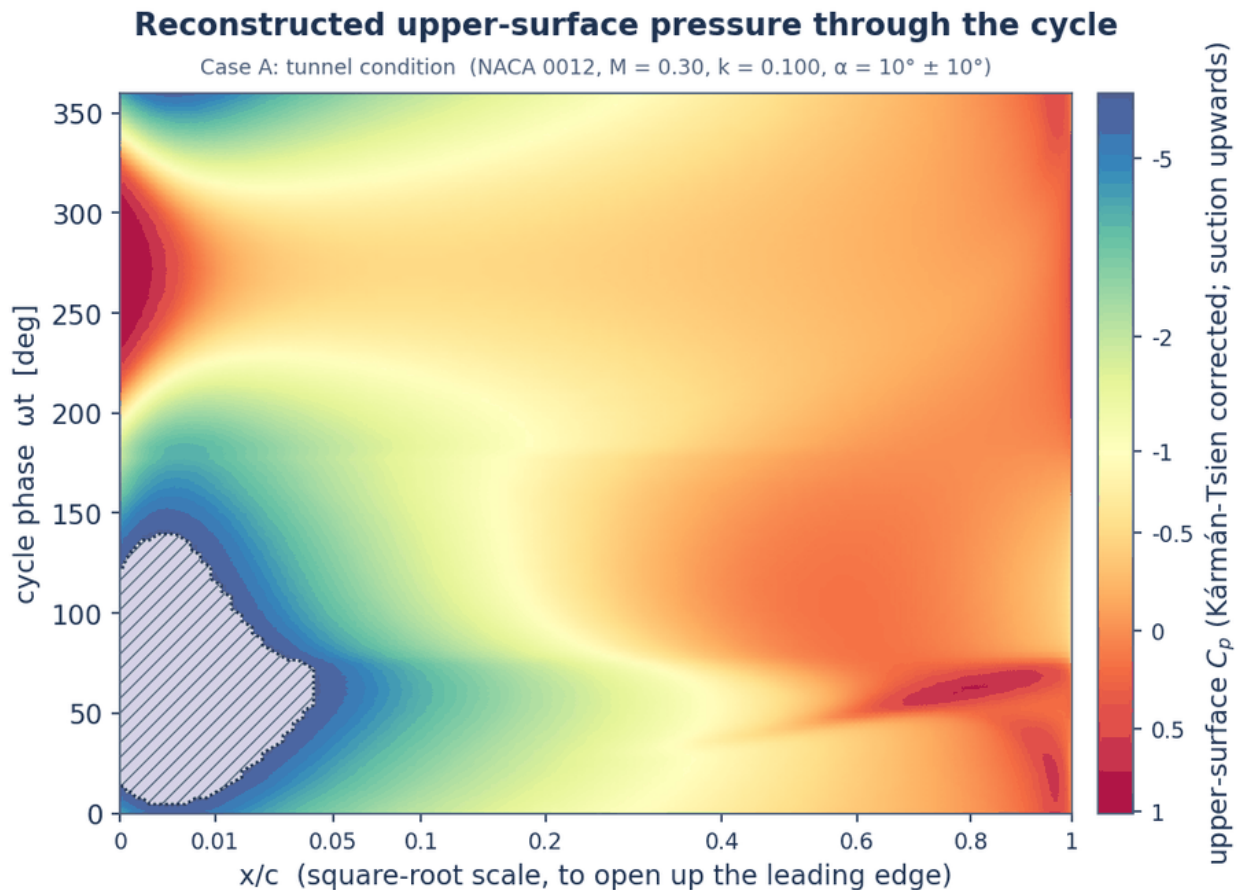
Solid: upper surface; dashed: lower surface. Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Beyond the dotted line the corrected C_p is below the critical value -8.06 (local Mach number above one): the correction does not hold there and the values are not physical; the corrected value is not allowed below -17.3 (95 % of the vacuum value). Part of the surface beyond the critical pressure: mean incidence, upstroke 0.5 %; rise (upstroke) 2.0 %; greatest lift 2.3 %; greatest vortex lift 2.3 %; fall (downstroke) 0.0 %; mean incidence, downstroke 0.0 %. Lift returned by this pressure at greatest lift: +0.000 % from the lift given, with the sheet carrying 0.889 of the circulation of the lift (Prandtl-Glauert factor 0.960); the incompressible pressure with the whole circulation returns it within -0.67 %.

Change of surface C_p beneath the vortex marker when it is put in: rise (upstroke) +0.25 ($x/c = 0.60$); greatest lift +0.36 ($x/c = 0.78$); greatest vortex lift -0.11 ($x/c = 1.00$). It RAISES the pressure beneath it at 2 of these 3 instants: it does not reproduce the suction footprint of a dynamic-stall vortex.

06_postprocessing/plots/cp_distribution_B_application.png

Case B: rotor-blade-station condition: reconstructed surface pressure at instants of the cycle, suction upwards, the band beyond the critical value hatched. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.



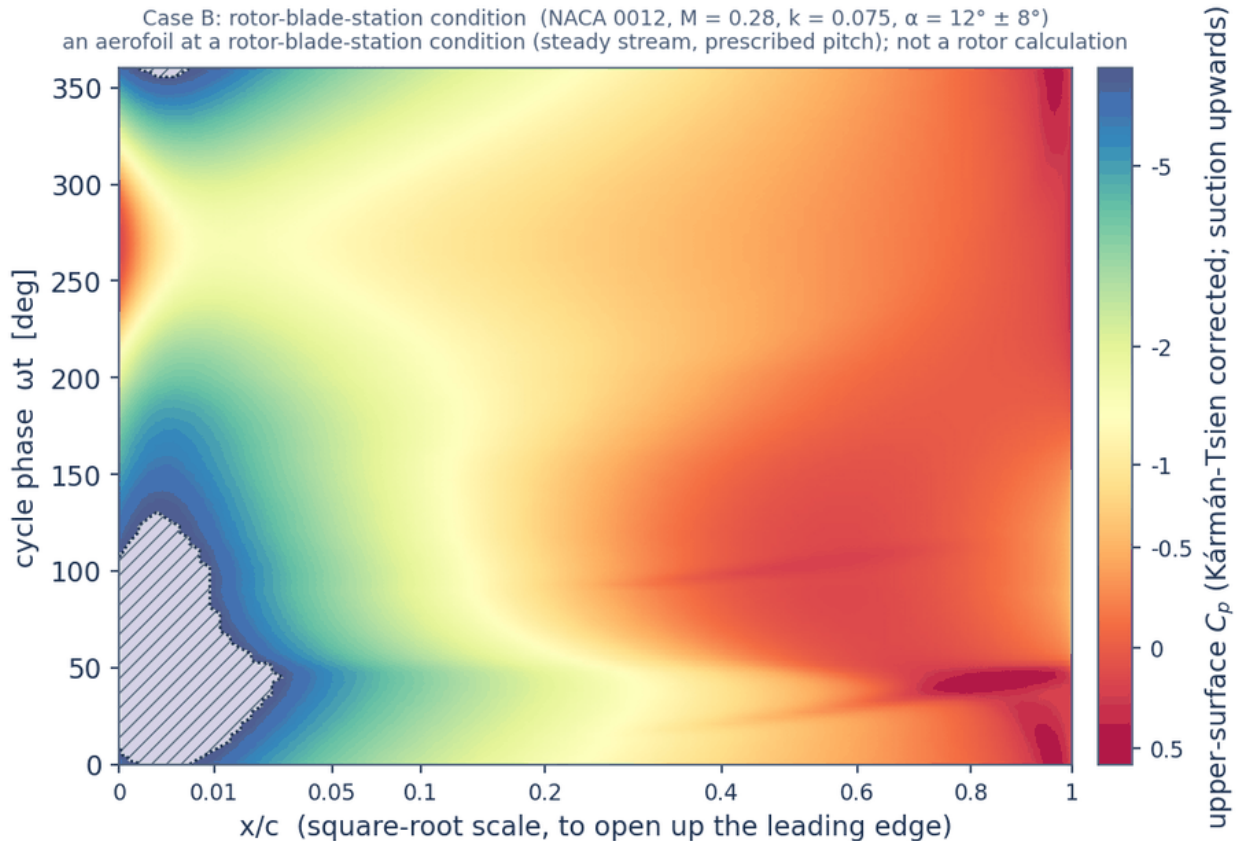
160 chordwise stations at 121 instants; the colour scale spans the values outside the hatched region, -6.9 to 1.02. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95; the region is 4.0 % of the points shown. Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/cp_phase_map_A_validation.png

Case A: tunnel condition: reconstructed upper-surface pressure over the chord and the cycle; Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Reconstructed upper-surface pressure through the cycle

Case B: rotor-blade-station condition (NACA 0012, $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$)
an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation



160 chordwise stations at 121 instants; the colour scale spans the values outside the hatched region, -8.1 to 0.59. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06; the region is 3.3 % of the points shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

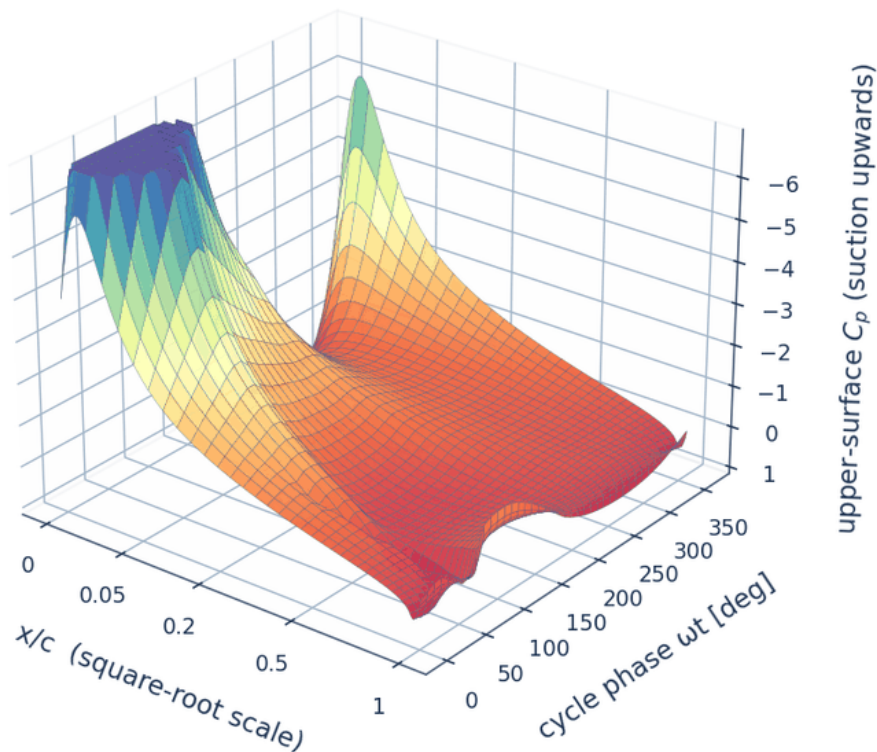
06_postprocessing/plots/cp_phase_map_B_application.png

Case B: rotor-blade-station condition: reconstructed upper-surface pressure over the chord and the cycle;

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Reconstructed upper-surface pressure through the cycle

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



160 chordwise stations over the whole chord at 41 instants; Kármán-Tsien corrected C_p . The surface is cut off at the critical value -6.95, beyond which the correction does not hold and the values are not physical (4.0 % of the points shown).

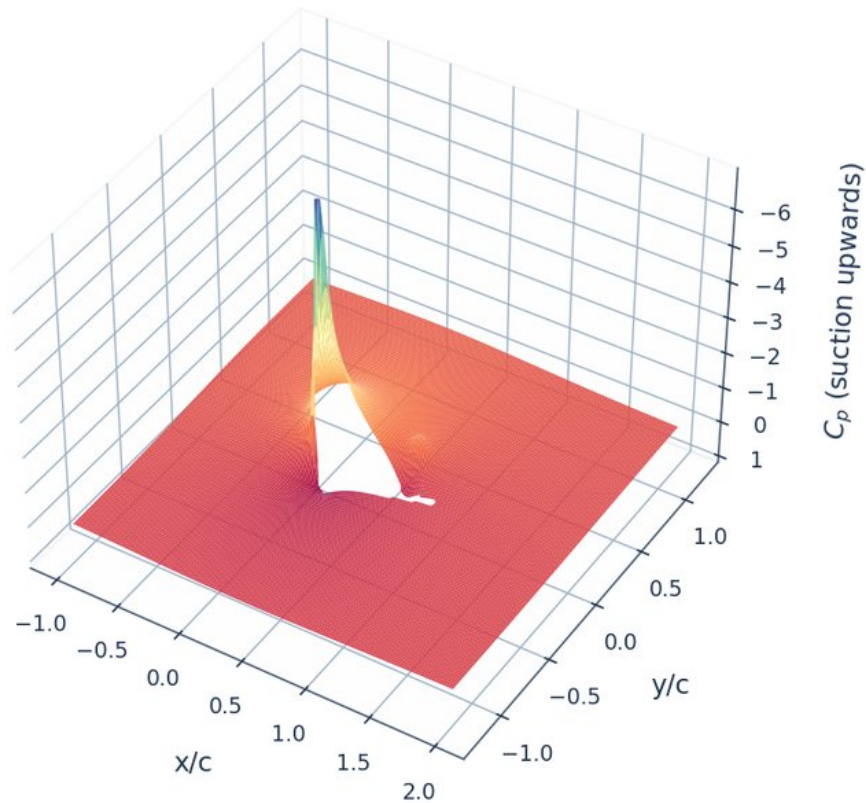
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/fig3d_cp_phase_surface.png

Reconstructed upper-surface pressure of Case A against chordwise position and cycle phase as a surface, suction upwards, cut off at the critical value. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Reconstructed field as a surface: C_p at greatest lift, $\alpha = 18.1^\circ$

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



Kármán-Tsien corrected C_p , cut off at the critical value -6.95, beyond which the correction does not hold and the values are not physical (3 of 24930 nodes). Range shown: -6.95 to 1.02. The gap is the section. Axes fixed to the section; the stream comes from below at the incidence shown.

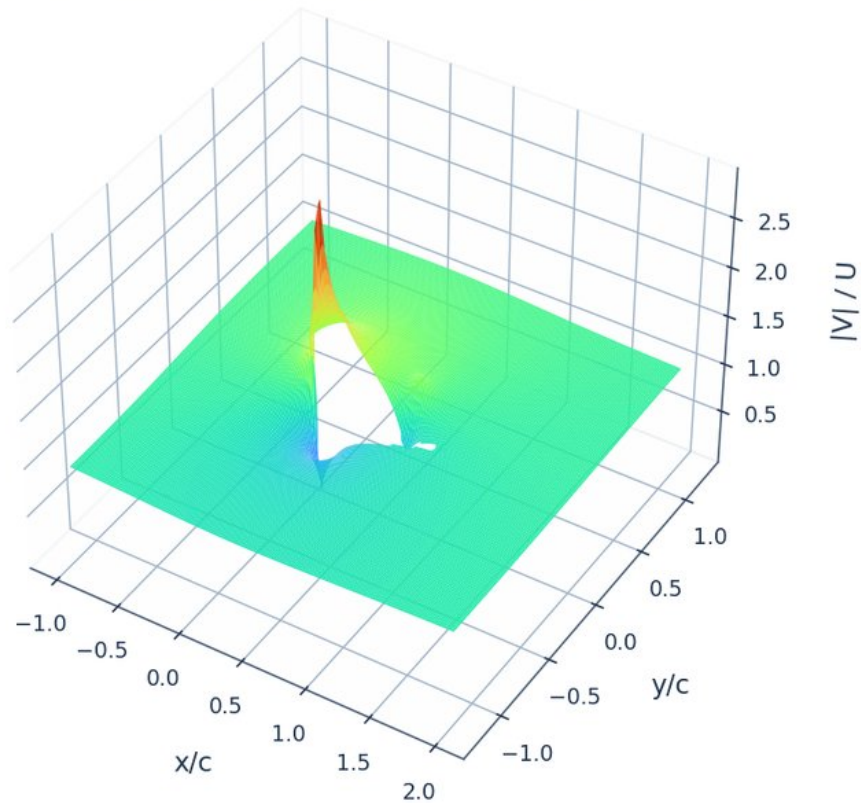
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/fig3d_field_surface_Cp.png

The corrected pressure coefficient of Case A at the instant of greatest lift, drawn as a surface over the plane, suction upwards, cut off at the critical value. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Reconstructed field as a surface: $|V| / U$ at greatest lift, $\alpha = 18.1^\circ$

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



Speed of the incompressible solution, over its whole range. Range shown: 0.0102 to 2.93. The gap is the section. Axes fixed to the section; the stream comes from below at the incidence shown.

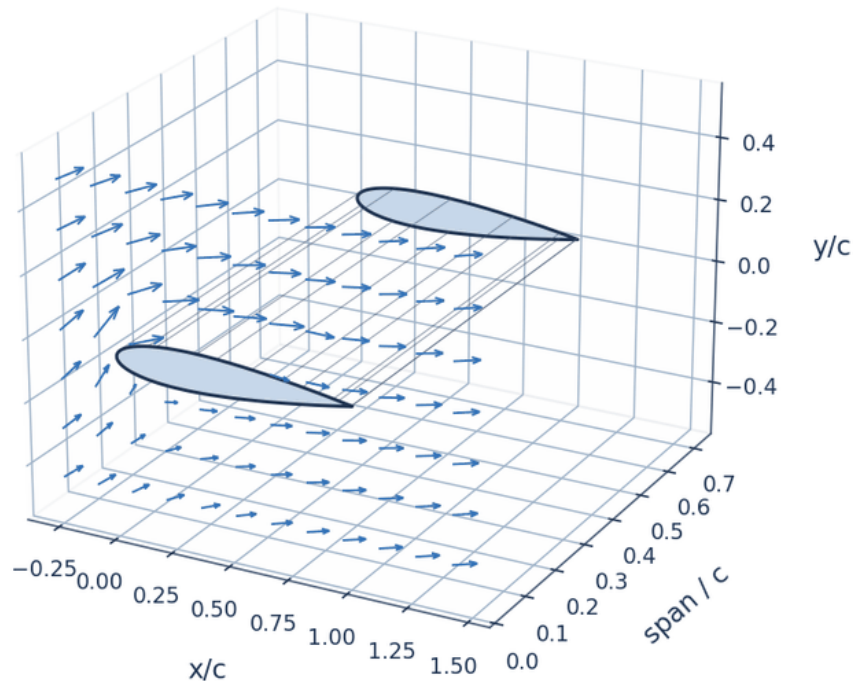
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/fig3d_field_surface_speed.png

The speed of the incompressible solution of Case A at the instant of greatest lift, drawn as a surface over the plane. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Section with the reconstructed velocity at greatest lift, $\alpha = 18.1^\circ$

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



Arrows: incompressible velocity on the end plane nearest the observer, in front of the section; the longest stands for $1.42 U$ and faster ones are drawn at that length. The span is drawn for the picture only: the calculation is two-dimensional. Axes fixed to the section; the stream comes from below at the incidence shown.

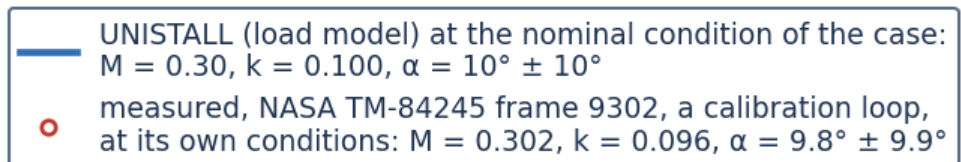
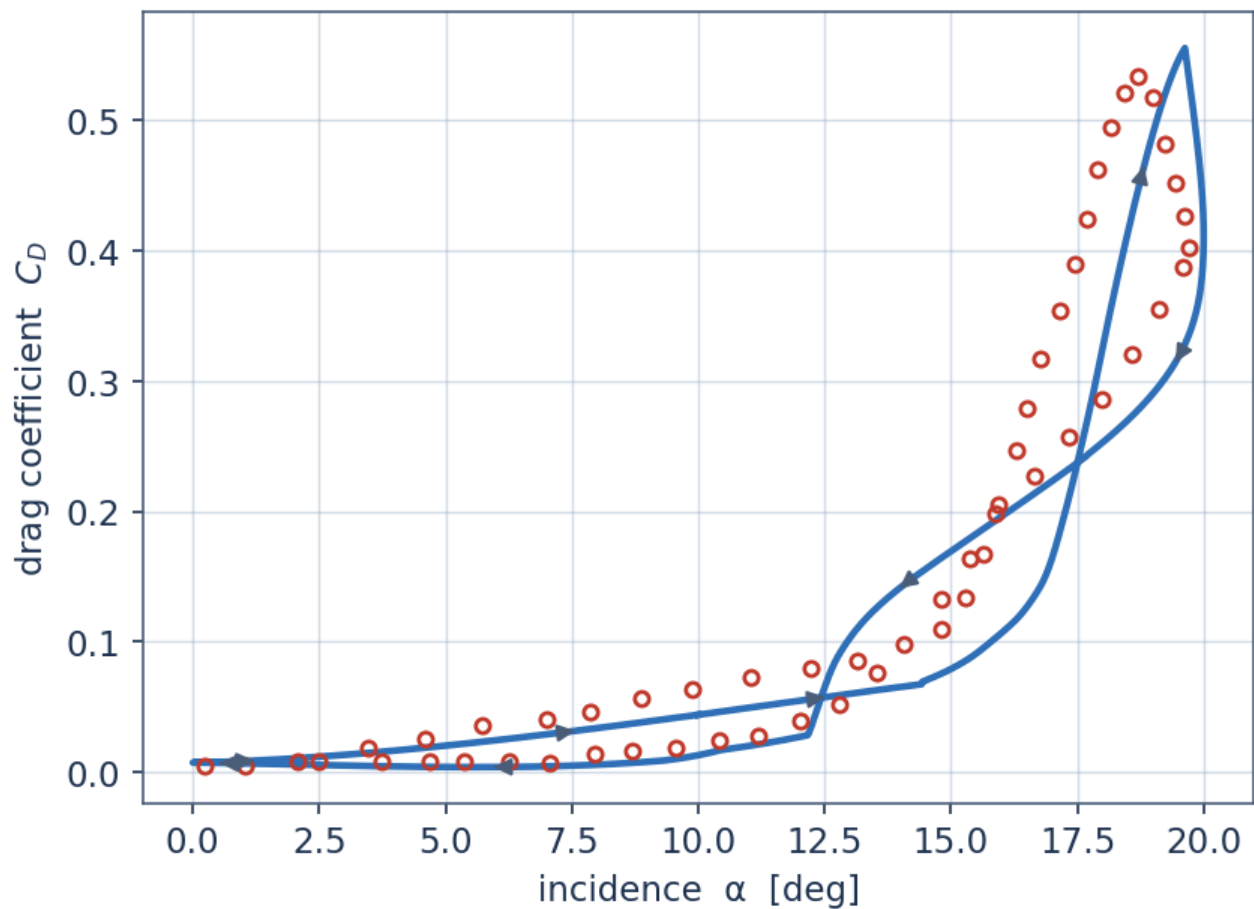
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/fig3d_section_vectors.png

The section drawn with span, with the reconstructed velocity on its near end plane. The span is for the picture only; the calculation is two-dimensional. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Load loop: drag coefficient

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)

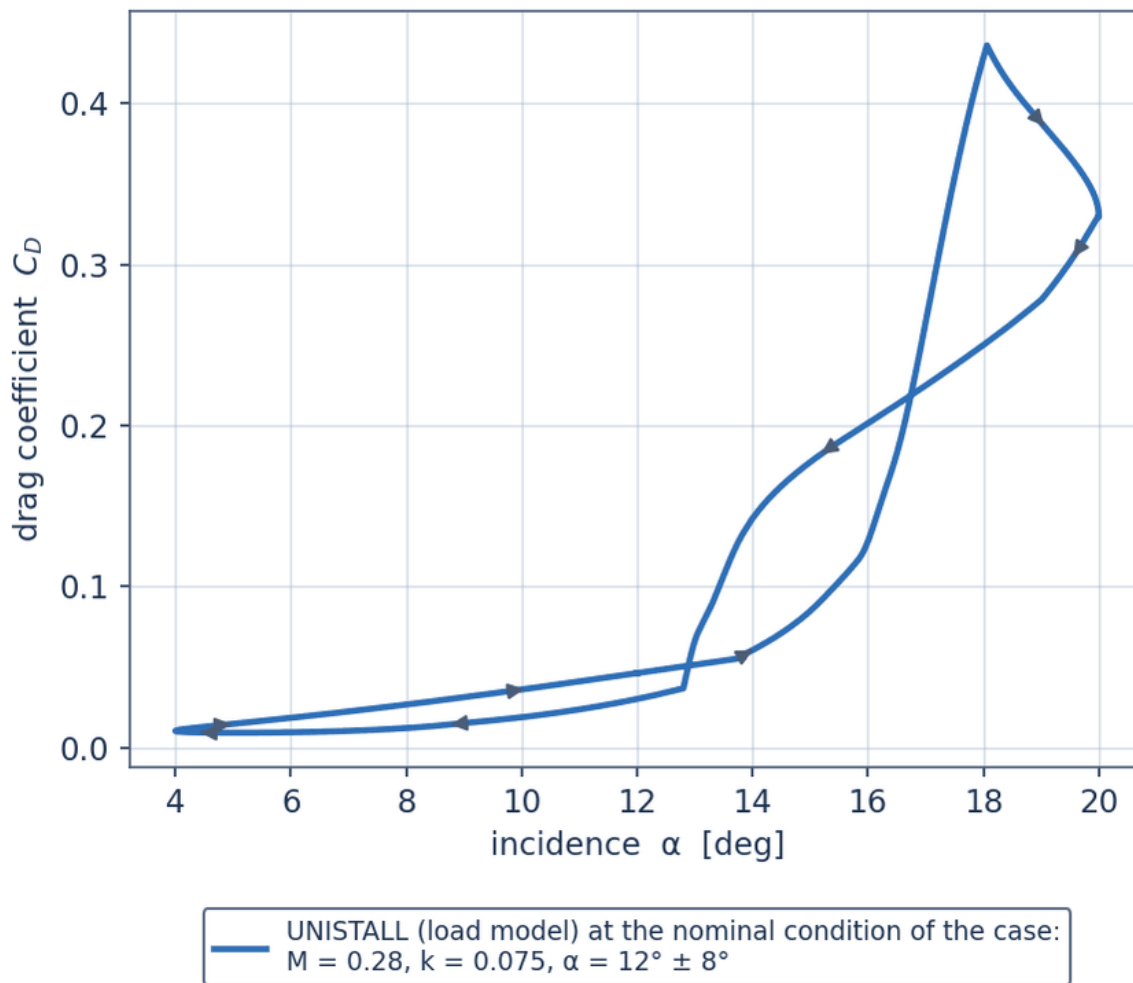


06_postprocessing/plots/hyst_cd_A_validation.png

Case A: tunnel condition: loop of drag against incidence at the nominal condition of the case, with the measured points of frame 9302 of NASA TM-84245, a calibration loop, taken at its own conditions (given in the legend).

Load loop: drag coefficient

Case B: rotor-blade-station condition (NACA 0012, $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$)
 an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation

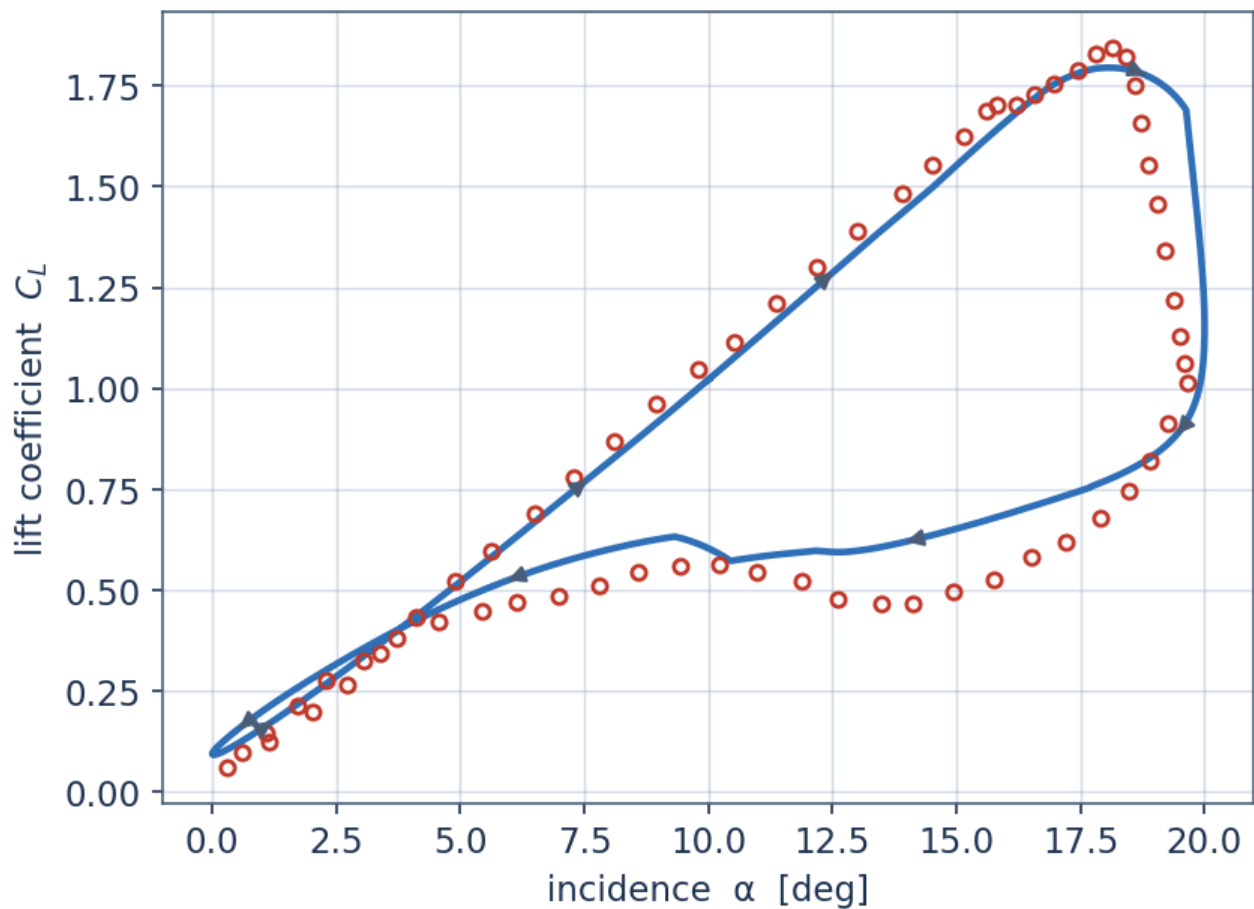


06_postprocessing/plots/hyst_cd_B_application.png

Case B: rotor-blade-station condition: loop of drag against incidence. No measurement is compared with.

Load loop: lift coefficient

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



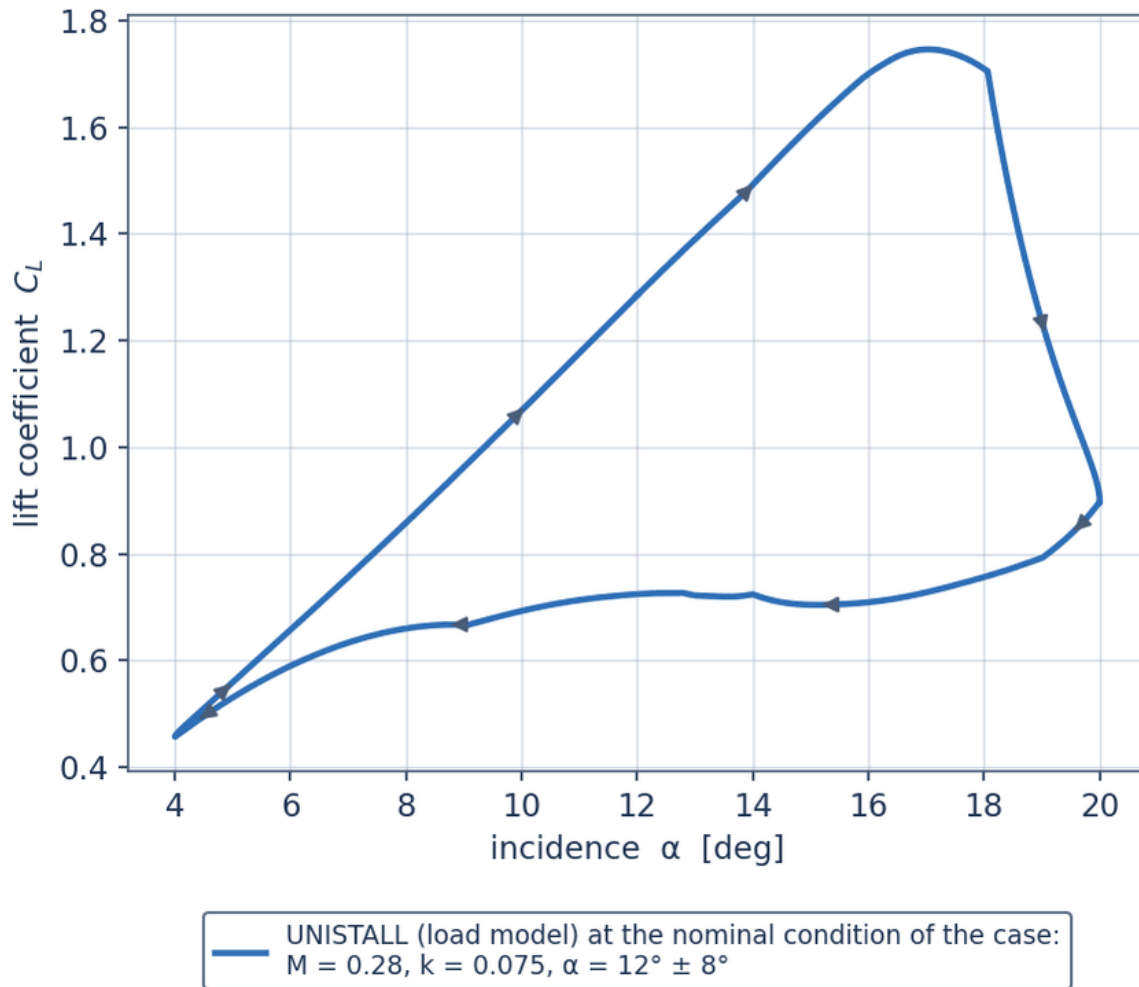
- UNISTALL (load model) at the nominal condition of the case:
 $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$
- measured, NASA TM-84245 frame 9302, a calibration loop,
at its own conditions: $M = 0.302$, $k = 0.096$, $\alpha = 9.8^\circ \pm 9.9^\circ$

06_postprocessing/plots/hyst_cl_A_validation.png

Case A: tunnel condition: loop of lift against incidence at the nominal condition of the case, with the measured points of frame 9302 of NASA TM-84245, a calibration loop, taken at its own conditions (given in the legend).

Load loop: lift coefficient

Case B: rotor-blade-station condition (NACA 0012, $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$)
 an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation

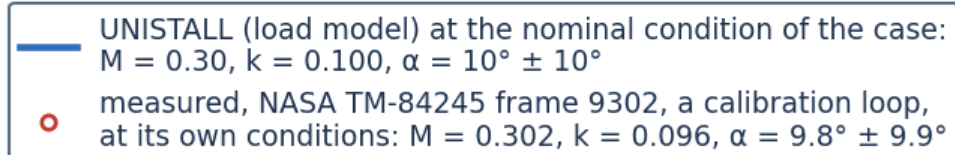
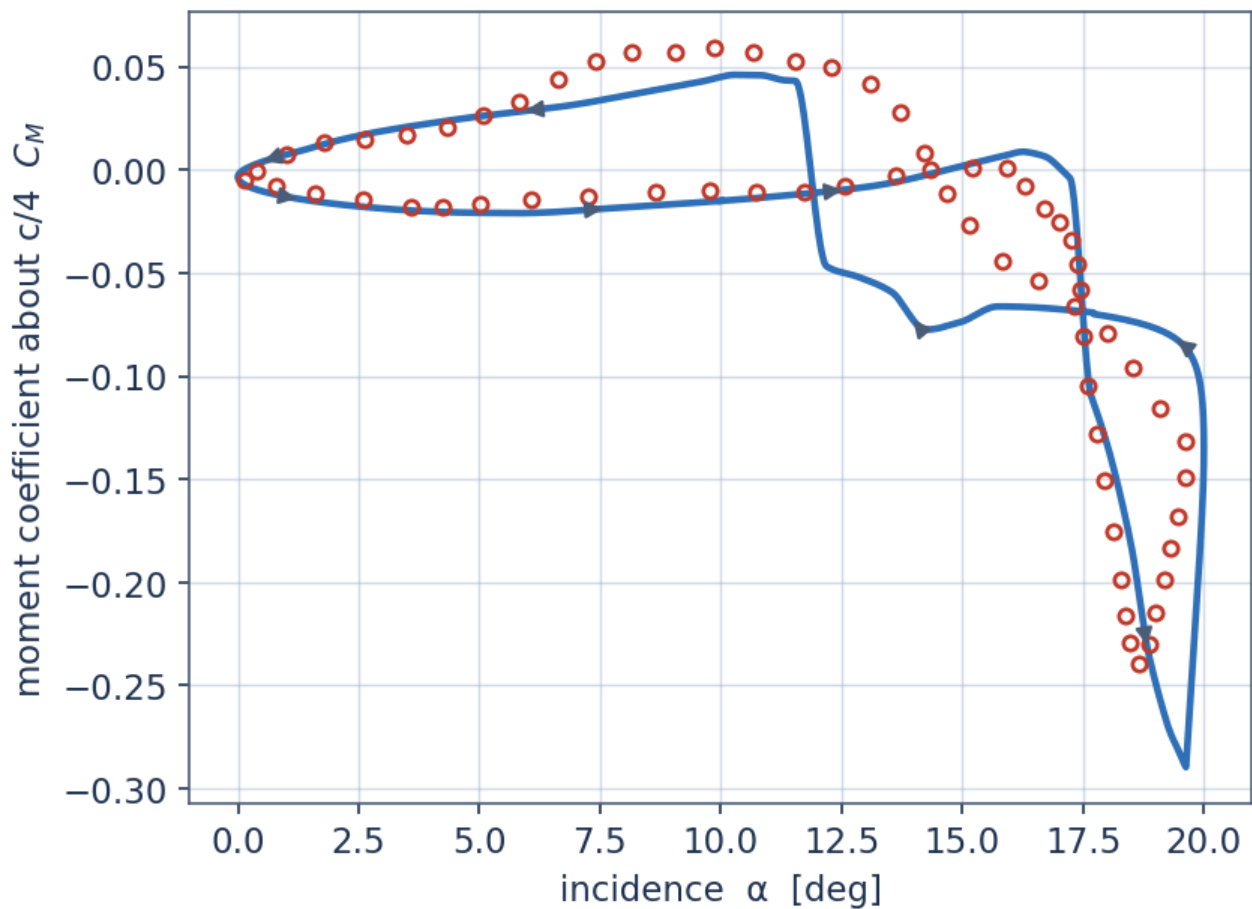


06_postprocessing/plots/hyst_cl_B_application.png

Case B: rotor-blade-station condition: loop of lift against incidence. No measurement is compared with.

Load loop: moment coefficient about c/4

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)

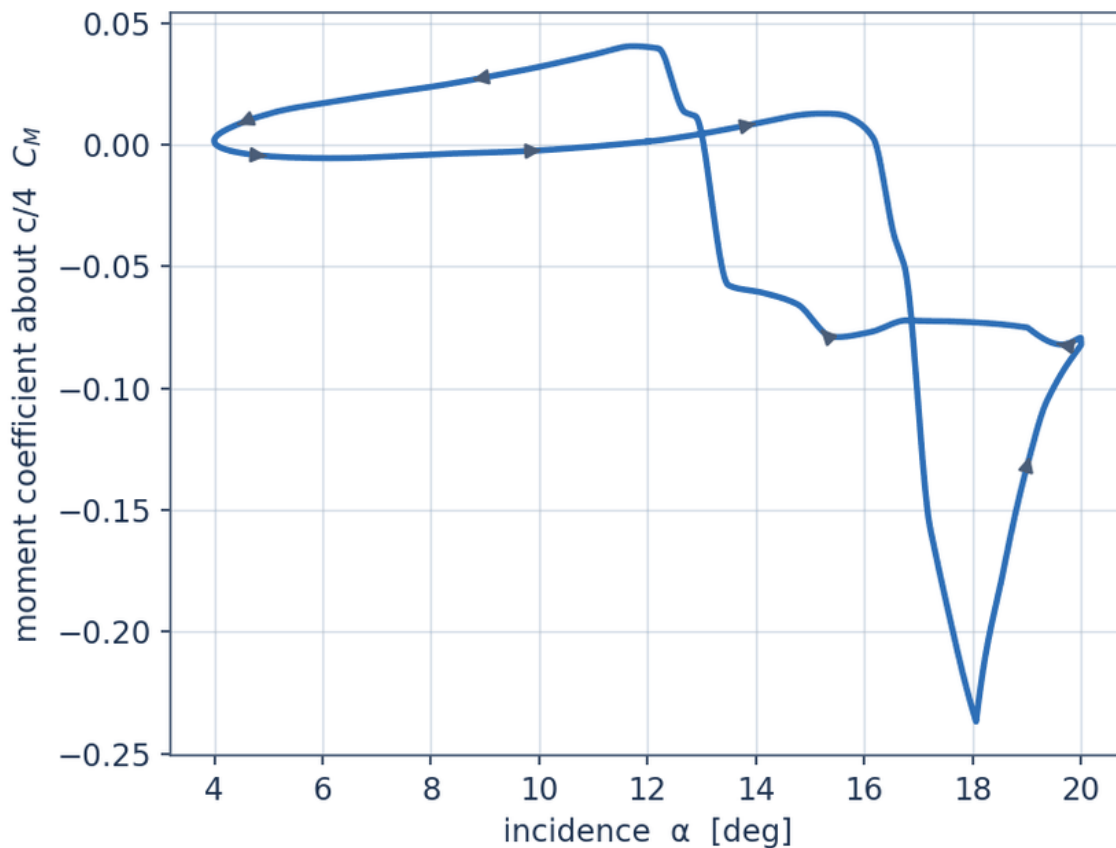


06_postprocessing/plots/hyst_cm_A_validation.png

Case A: tunnel condition: loop of moment about the quarter chord against incidence at the nominal condition of the case, with the measured points of frame 9302 of NASA TM-84245, a calibration loop, taken at its own conditions (given in the legend).

Load loop: moment coefficient about c/4

Case B: rotor-blade-station condition (NACA 0012, $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$)
 an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation

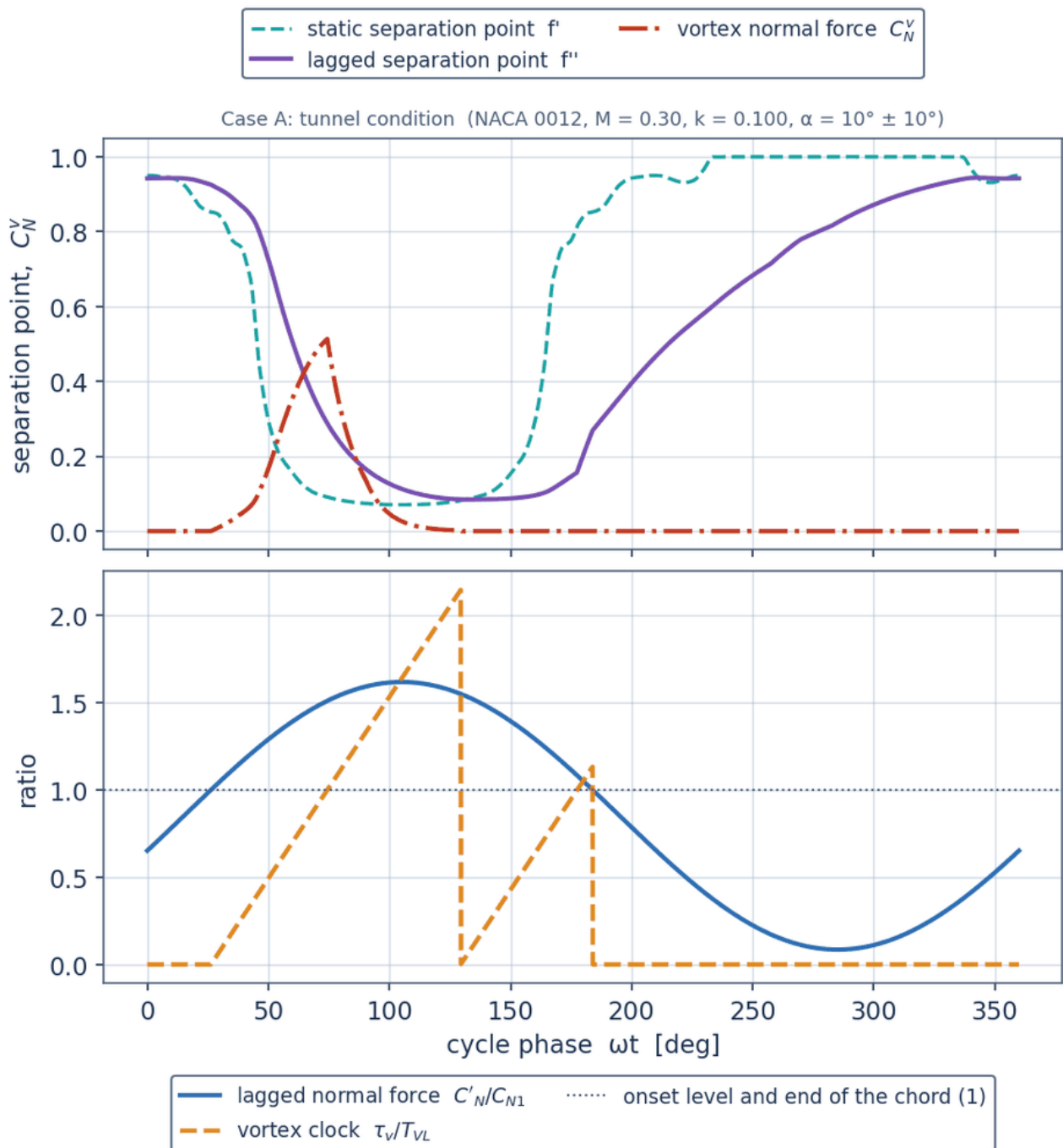


— UNISTALL (load model) at the nominal condition of the case:
 $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$

06_postprocessing/plots/hyst_cm_B_application.png

Case B: rotor-blade-station condition: loop of moment about the quarter chord against incidence. No measurement is compared with.

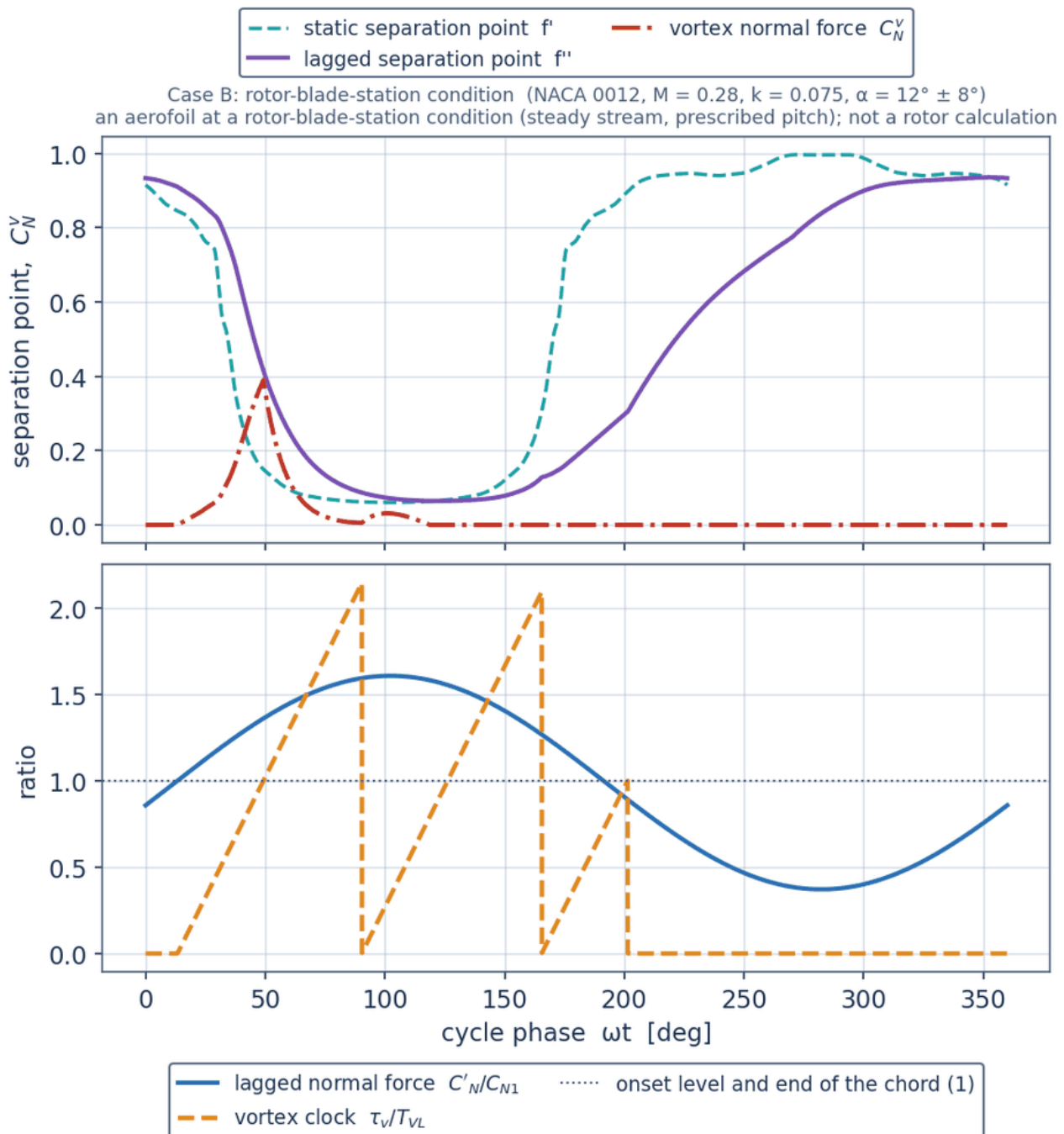
States of the load model over one cycle



06_postprocessing/plots/states_A_validation.png

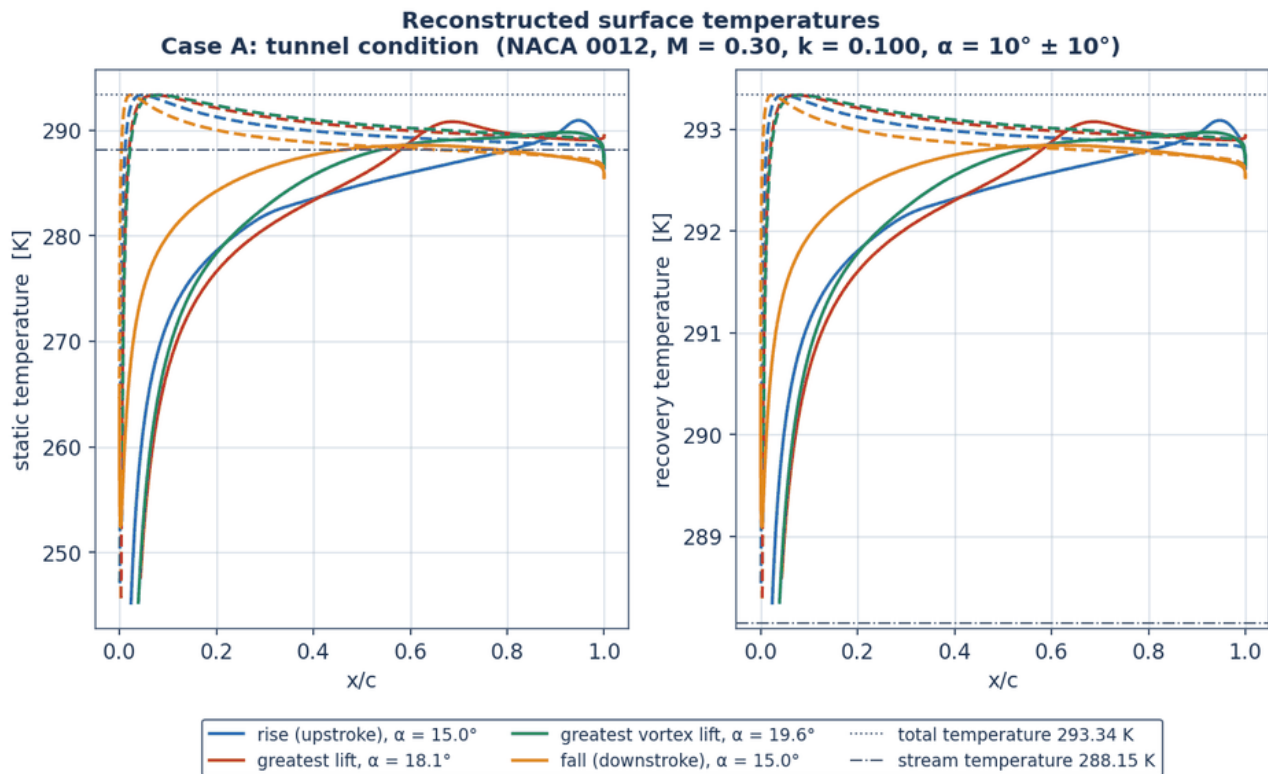
Case A: tunnel condition: states of the load model through the reported cycle: static and lagged separation point, vortex normal force, lagged normal force over its onset level, and the vortex clock.

States of the load model over one cycle



06_postprocessing/plots/states_B_application.png

Case B: rotor-blade-station condition: states of the load model through the reported cycle: static and lagged separation point, vortex normal force, lagged normal force over its onset level, and the vortex clock.

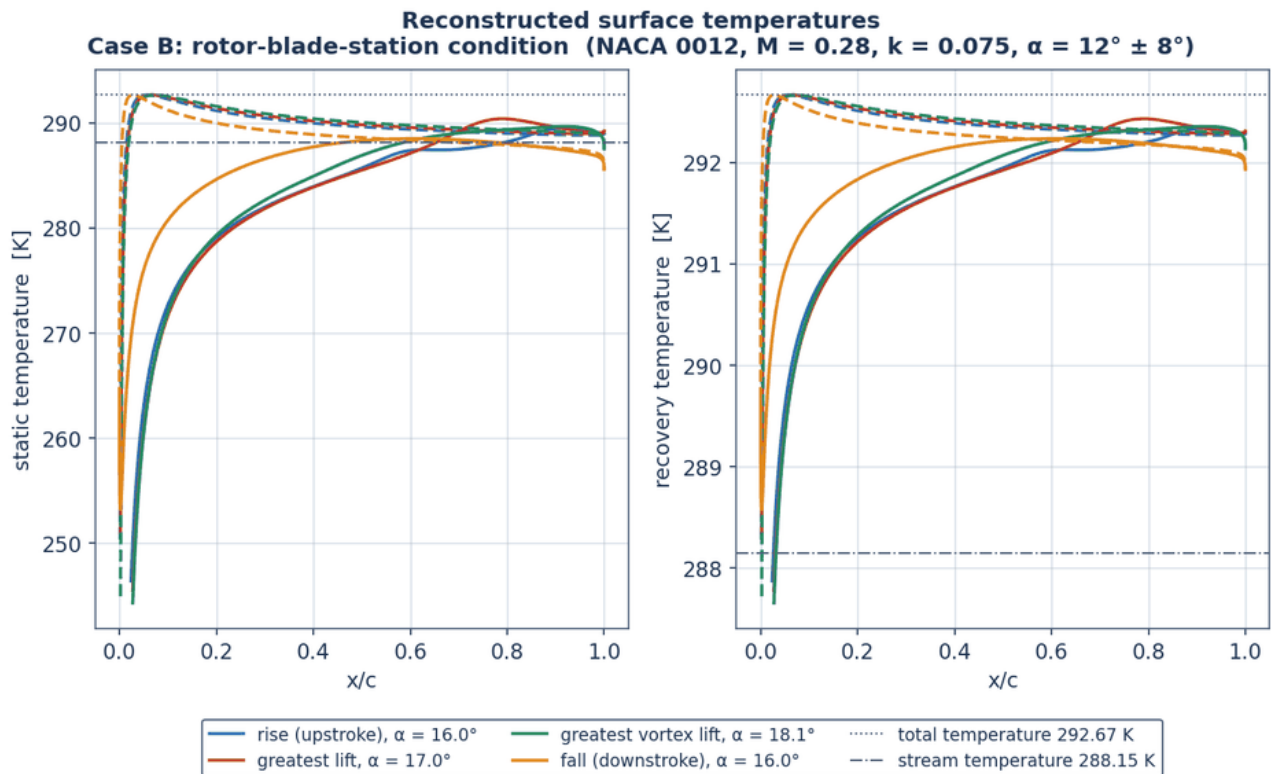


Solid: upper surface; dashed: lower surface. From the corrected pressure by the isentropic relations with the stagnation state of the stream; recovery factor $0.896 (Pr^{1/3})$, the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). The recovery temperature is that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have. The curves stop where the corrected pressure is below the critical value: the values there are not physical. Part of the surface beyond the critical pressure: rise (upstroke) 2.0 %; greatest lift 3.2 %; greatest vortex lift 3.1 %; fall (downstroke) 0.0 %.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/temperature_profile_A_validation.png

Case A: tunnel condition: static and recovery temperature along the surface at instants of the cycle, from the corrected pressure by the isentropic relations; the curves stop where that pressure is below the critical value. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.



Solid: upper surface; dashed: lower surface. From the corrected pressure by the isentropic relations with the stagnation state of the stream; recovery factor $0.896 (Pr^{1/3})$, the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). The recovery temperature is that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have. The curves stop where the corrected pressure is below the critical value: the values there are not physical. Part of the surface beyond the critical pressure: rise (upstroke) 2.0 %; greatest lift 2.3 %; greatest vortex lift 2.3 %; fall (downstroke) 0.0 %.

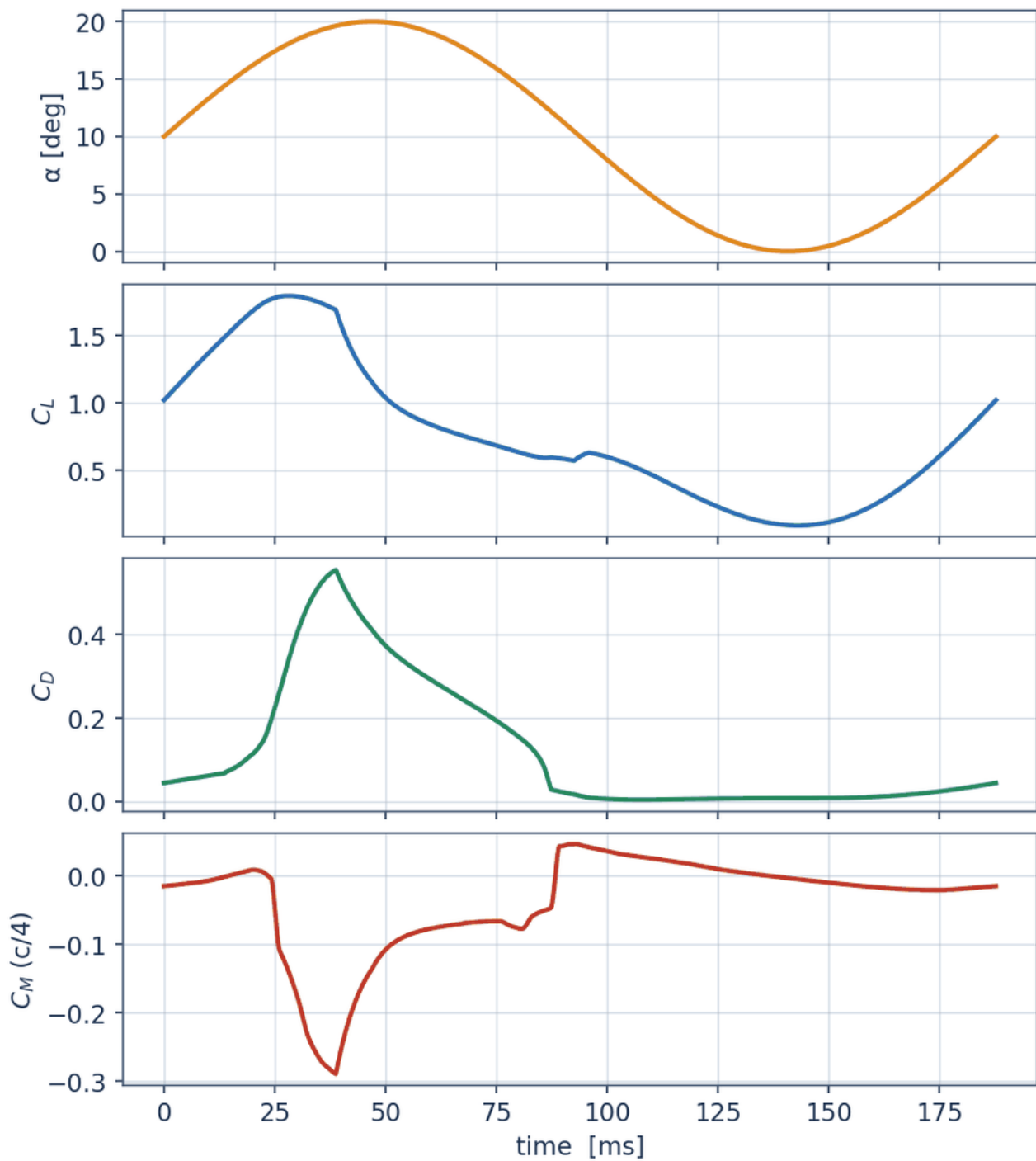
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/temperature_profile_B_application.png

Case B: rotor-blade-station condition: static and recovery temperature along the surface at instants of the cycle, from the corrected pressure by the isentropic relations; the curves stop where that pressure is below the critical value. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Loads over one cycle

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)

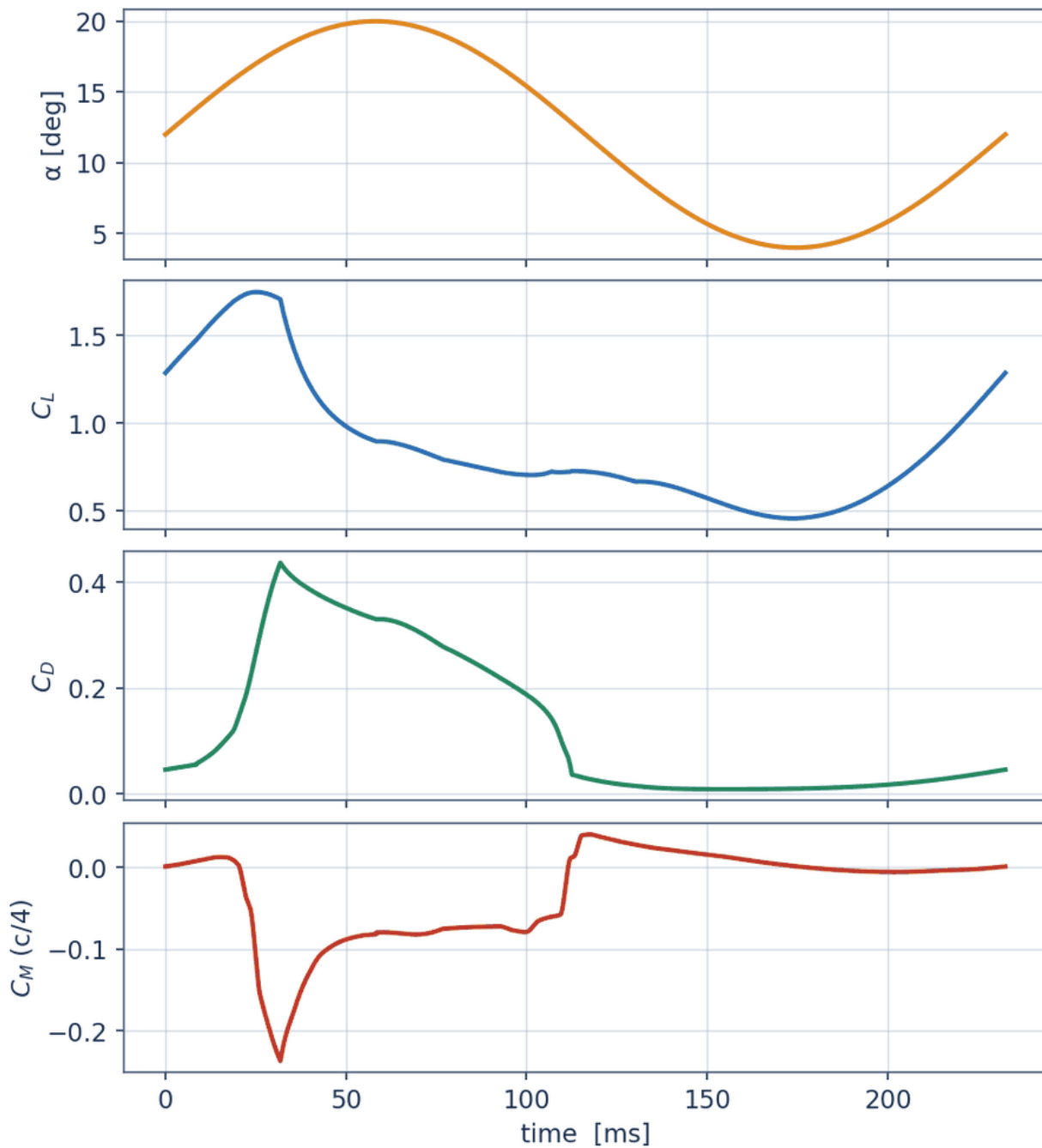


06_postprocessing/plots/timehist_loads_A_validation.png

Case A: tunnel condition: lift, drag and moment through the reported cycle.

Loads over one cycle

Case B: rotor-blade-station condition (NACA 0012, $M = 0.28$, $k = 0.075$, $\alpha = 12^\circ \pm 8^\circ$)
an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation

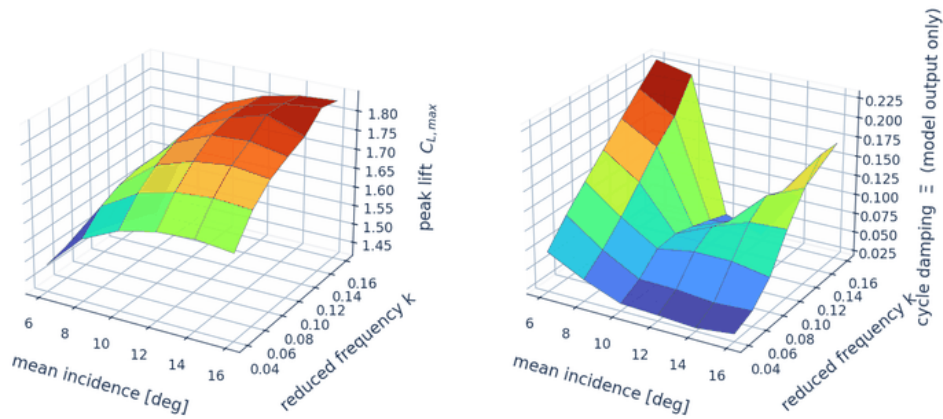


06_postprocessing/plots/timehist_loads_B_application.png

Case B: rotor-blade-station condition: lift, drag and moment through the reported cycle.

Response of the load model to mean incidence and reduced frequency

NACA 0012, $M = 0.28$, pitch amplitude 8° (the Mach number and amplitude of Case B); a steady stream and prescribed pitch, not a rotor calculation

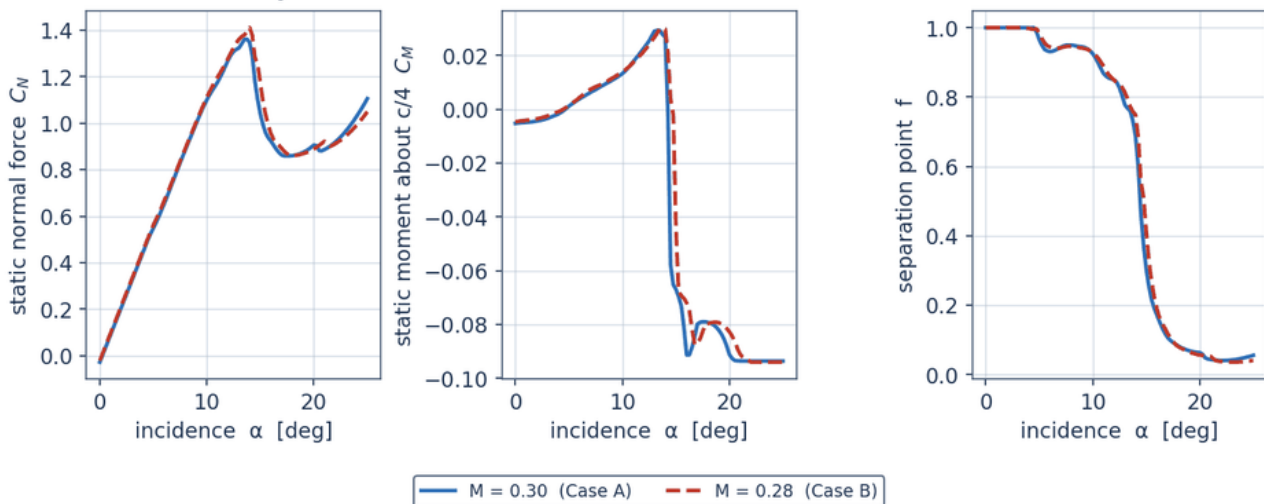


42 conditions. The highest peak incidence is 24° ; the measured loops the model was compared with reach 25° . Cycle damping is model output only. It is negative at 0 of these 42 conditions. On the 45 held-out measured loops with a closed moment loop the model gets the SIGN of the damping right on a fraction 0.60 (a prediction of positive damping everywhere would score 0.67); 15 of those loops measure negative damping. The damping surface is no evidence of what a measurement would show.

06_postprocessing/plots/fig3d_response_surface.png

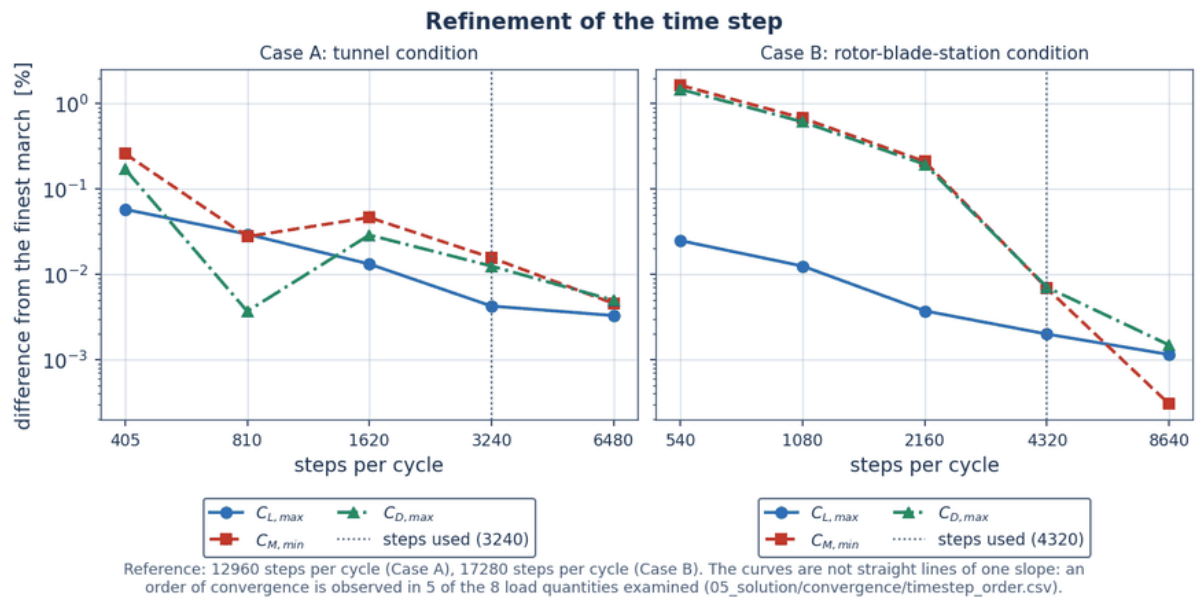
Peak lift and cycle damping given by the load model over mean incidence and reduced frequency, at the Mach number and amplitude of Case B. The cycle damping is model output only: the model does not predict cycle damping through stall. The figure prints how often the model gets the sign of the damping right on the held-out loops.

Static inputs of the load model at the Mach numbers of the two cases



06_postprocessing/plots/static_inputs.png

Static inputs of the load model at the Mach numbers of the cases: normal force, separation point and moment against incidence.

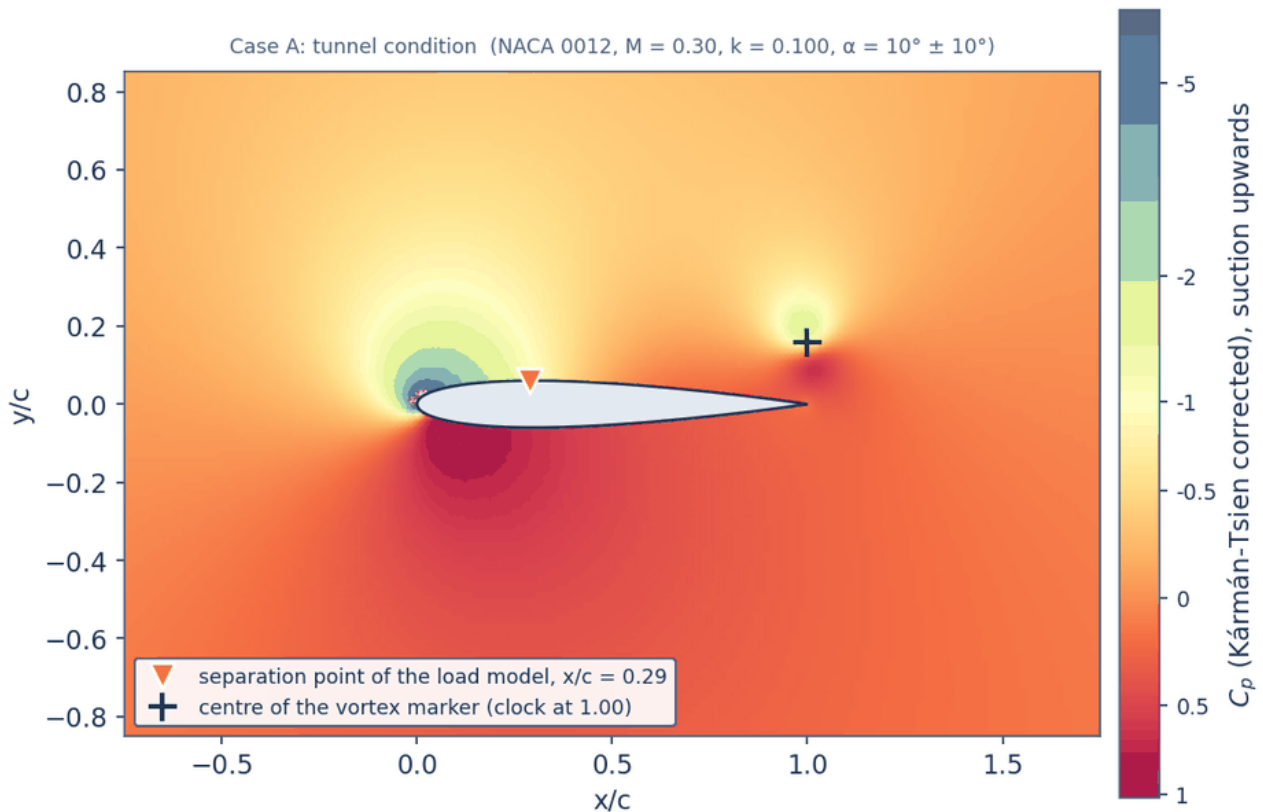


06_postprocessing/plots/timestep_refinement.png

Refinement of the time step: difference of the peak loads from the finest march, against the number of steps per cycle.

Reconstructed fields, Case A: tunnel condition (06_postprocessing)

Pressure coefficient at greatest vortex lift, $\alpha = 19.6^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: -6.94 to 1.02. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.1 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it).

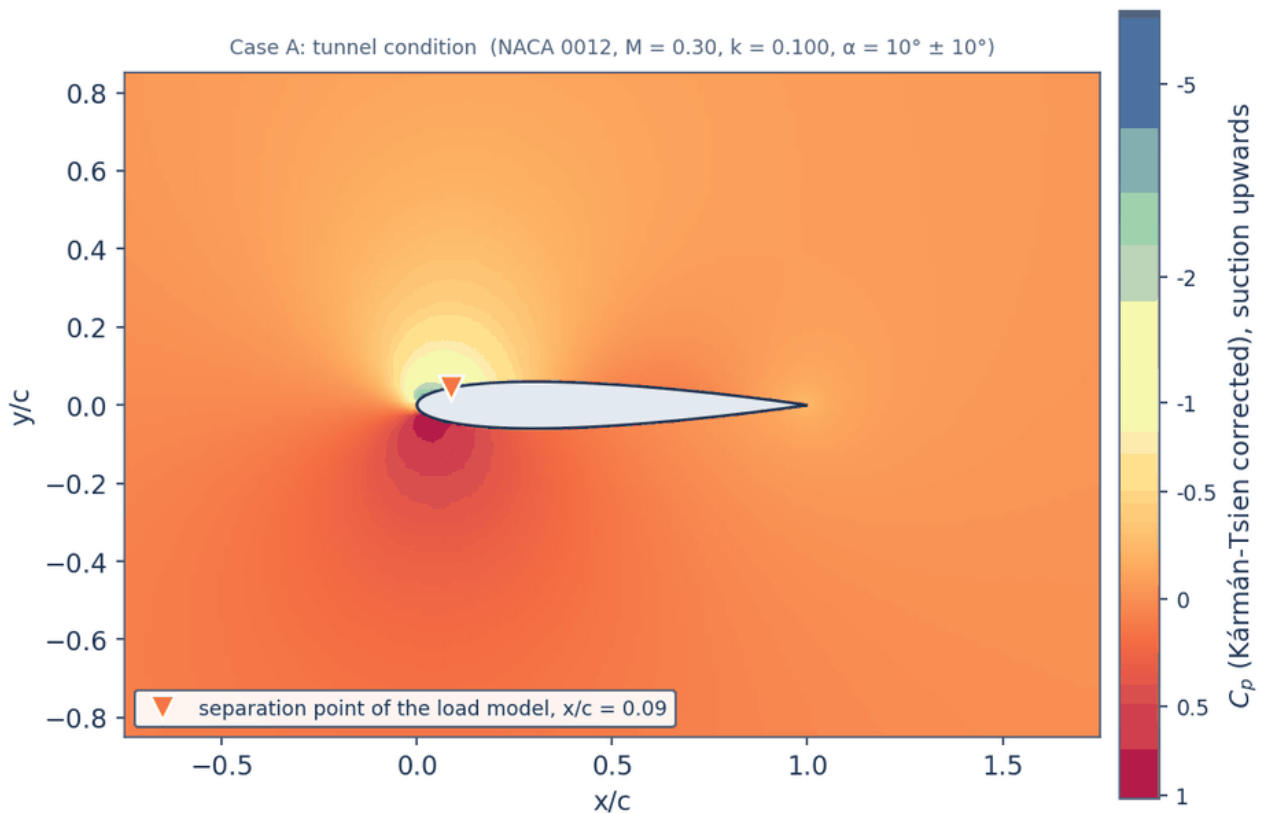
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.29$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Cp_A_validation_dsv_a20.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.1 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

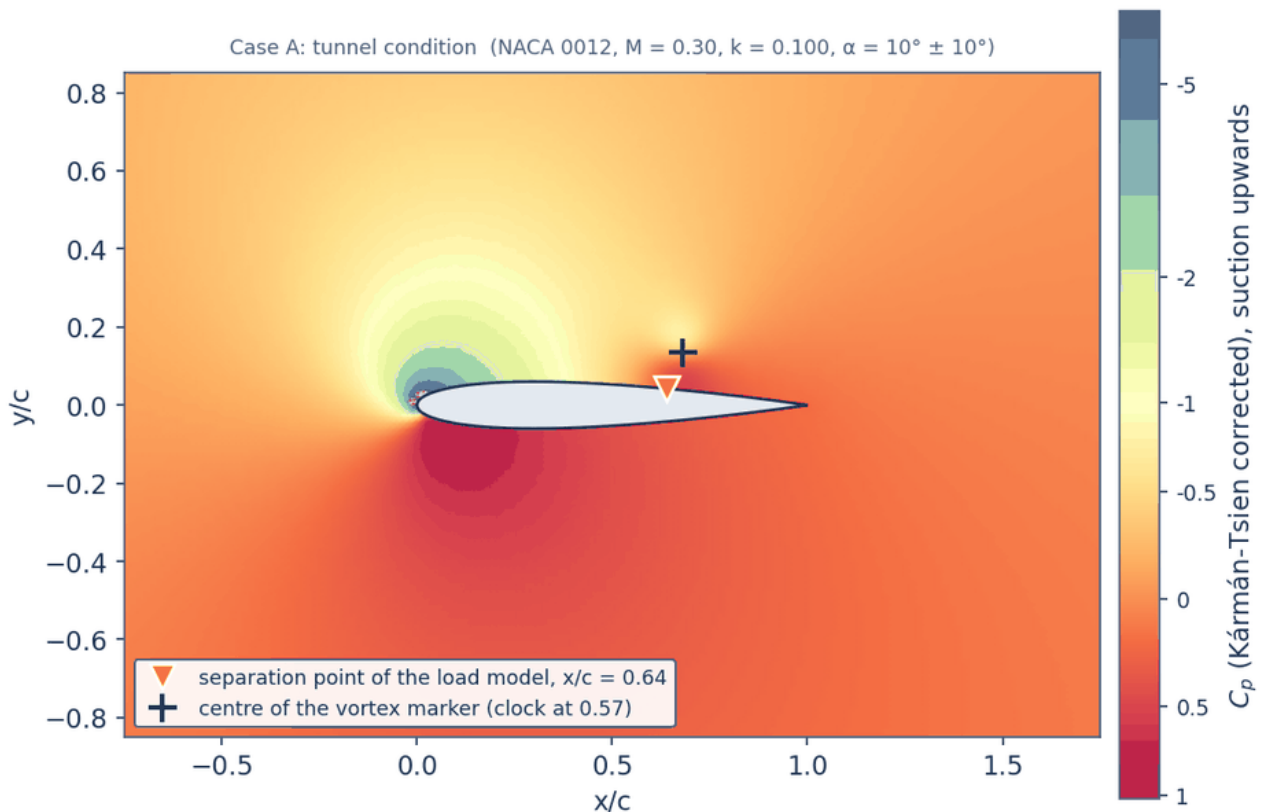
Pressure coefficient at fall (downstroke), $\alpha = 15.0^\circ$



06_postprocessing/plots/contour_Cp_A_validation_fall_a15.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

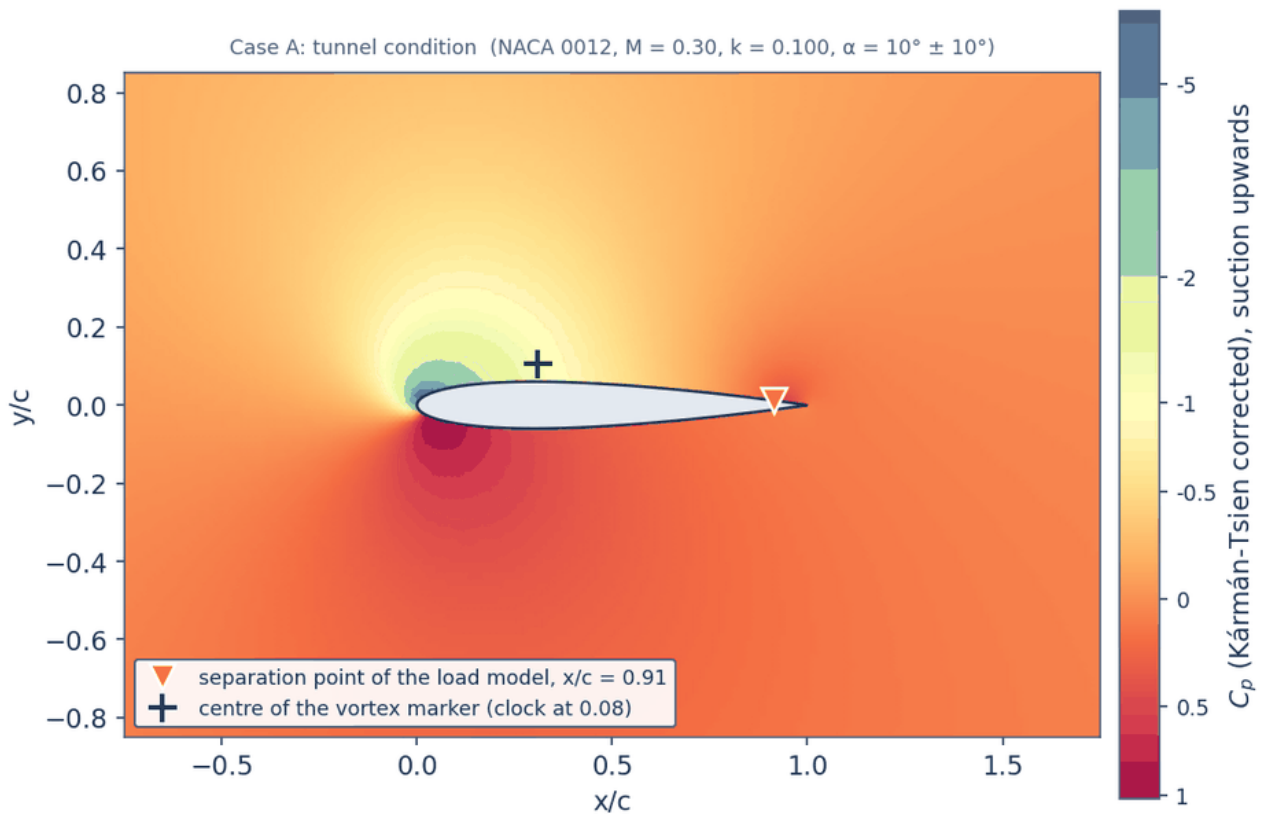
Pressure coefficient at greatest lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_Cp_A_validation_peak_a18.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.2 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

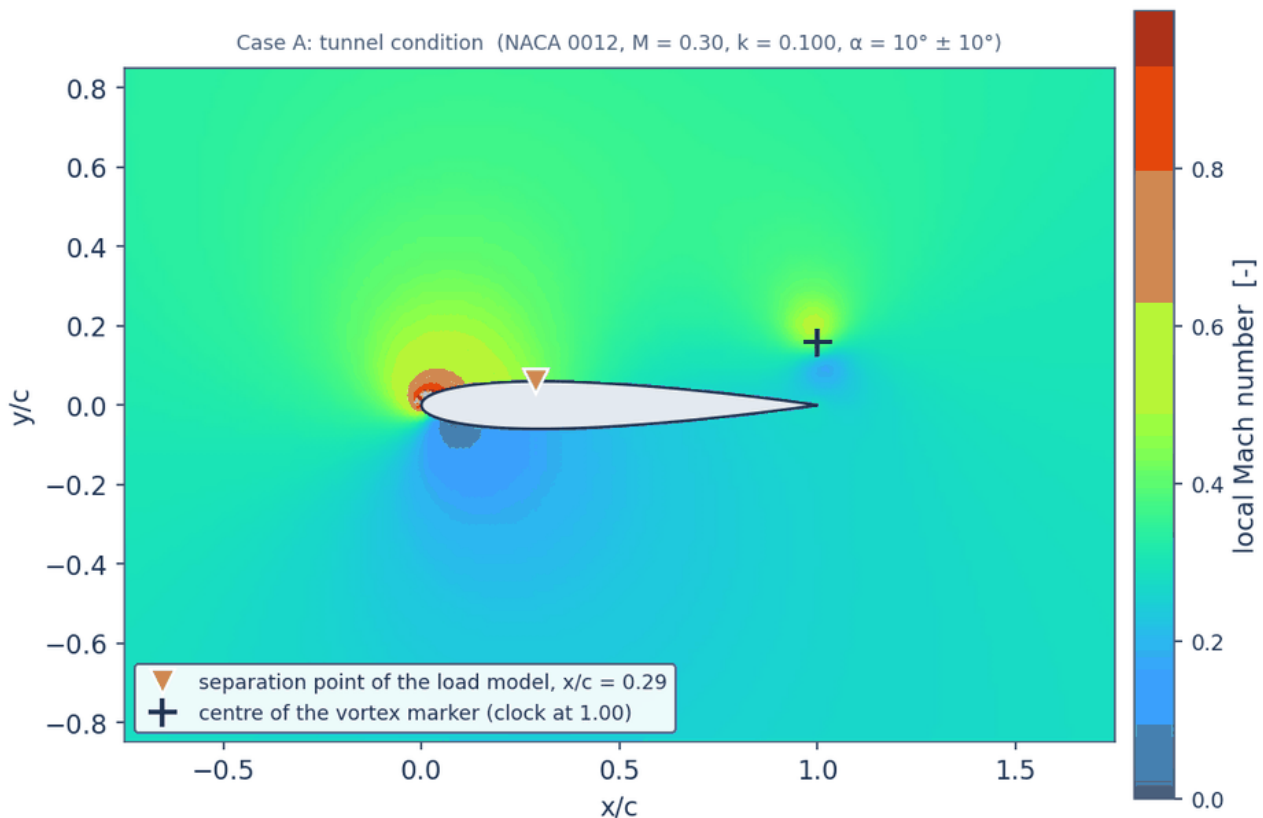
Pressure coefficient at rise (upstroke), $\alpha = 15.0^\circ$



06_postprocessing/plots/contour_Cp_A_validation_rise_a15.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at greatest vortex lift, $\alpha = 19.6^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 1.00. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.1 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it).

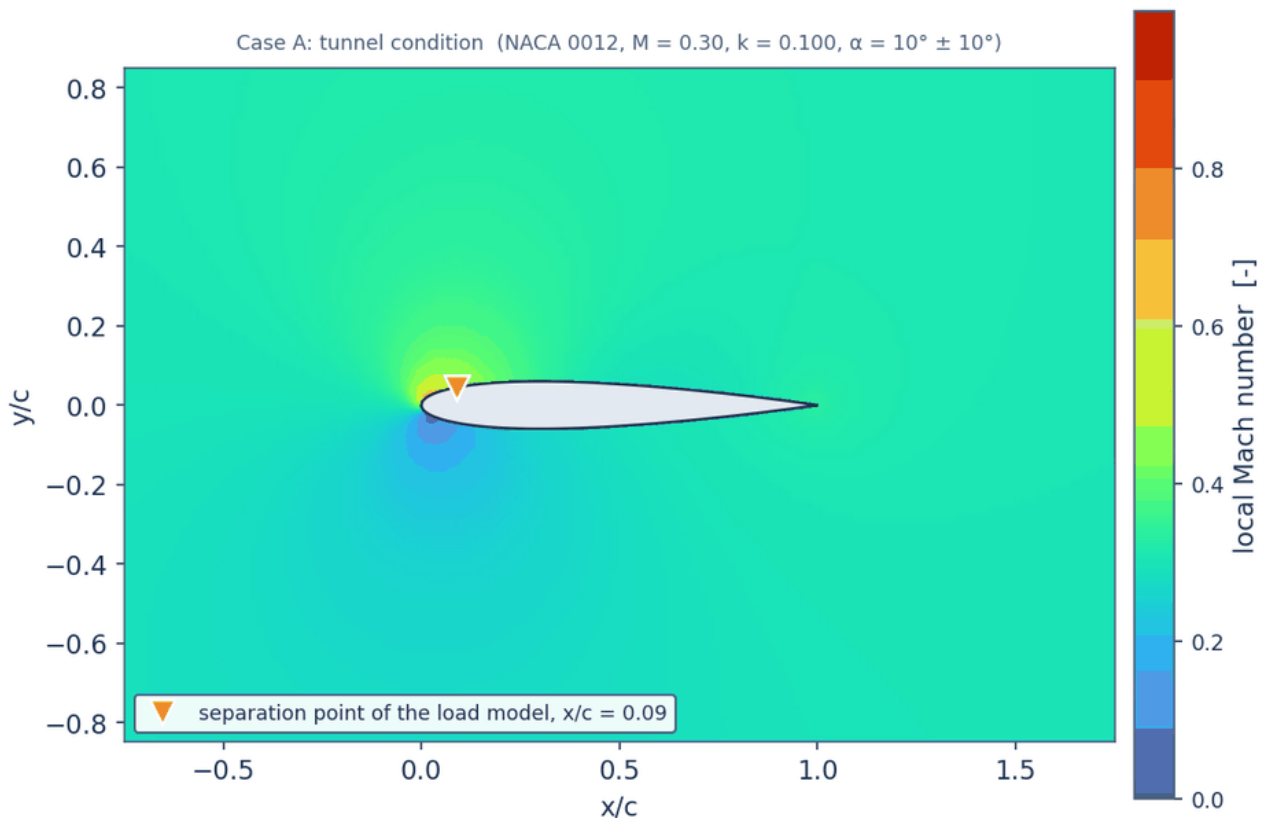
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.29$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_A_validation_dsv_a20.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.1 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at fall (downstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 1.00. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 0.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it).

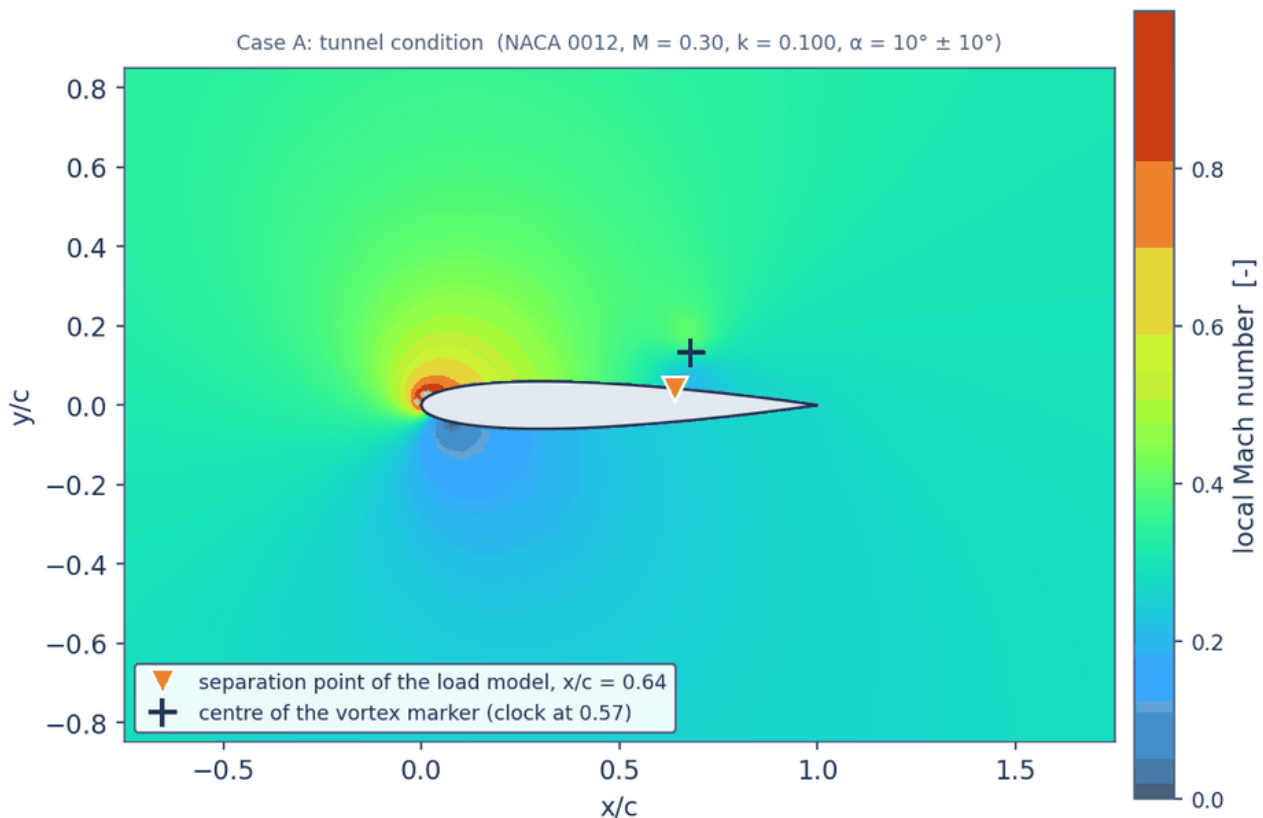
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.09$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_A_validation_fall_a15.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at greatest lift, $\alpha = 18.1^\circ$



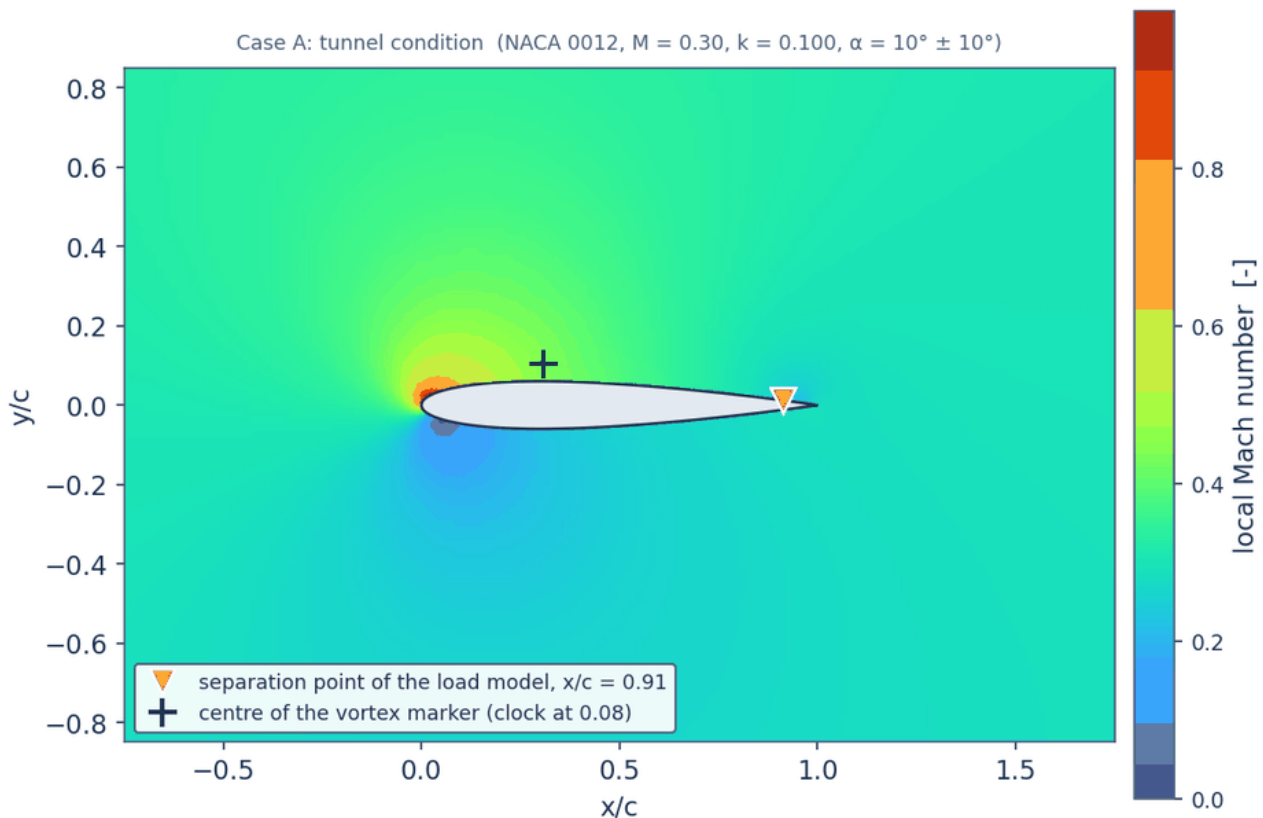
Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 1.00. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.2 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it).

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.64$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_A_validation_peak_a18.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.2 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at rise (upstroke), $\alpha = 15.0^\circ$ 

Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 1.00. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 2.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it).

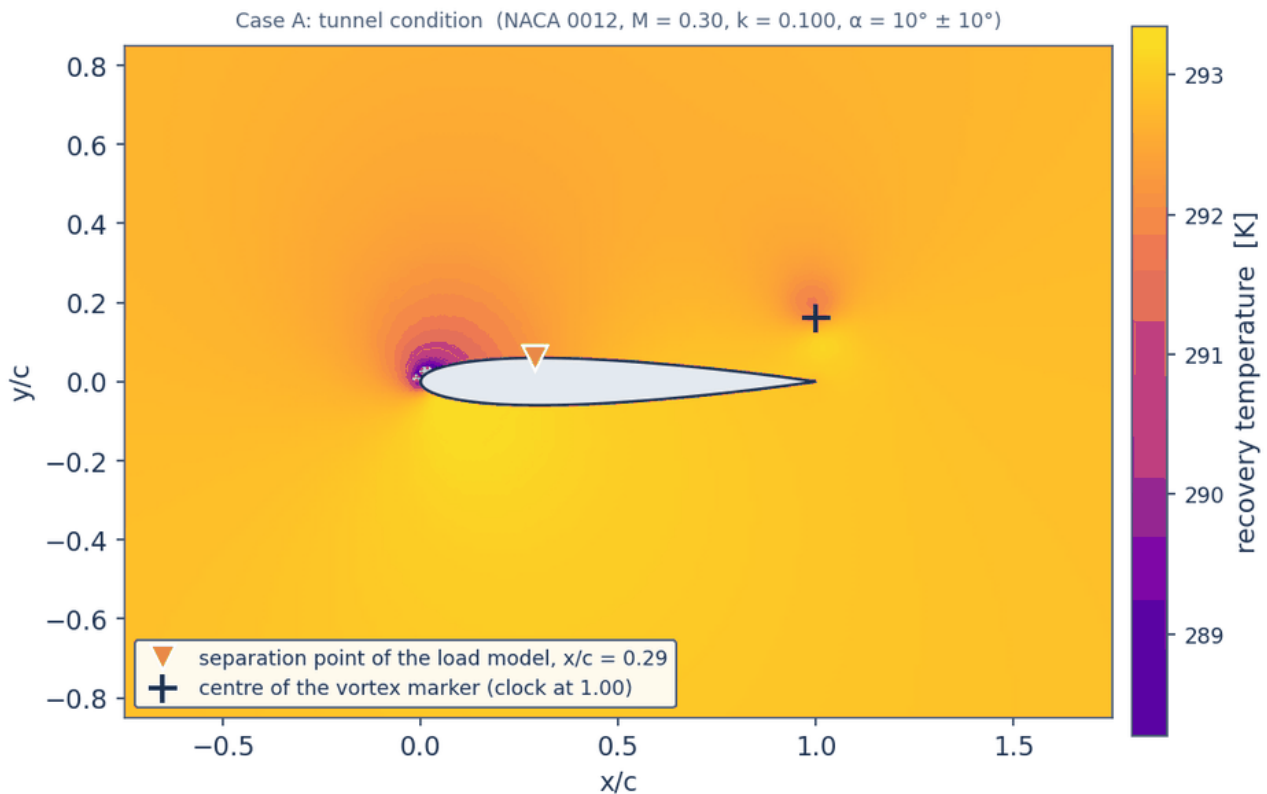
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.91$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_A_validation_rise_a15.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Recovery temperature at greatest vortex lift, $\alpha = 19.6^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 288.27 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.1 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). A recovery temperature belongs to an adiabatic wall: away from the wall the map shows what such a wall would take at the local state.

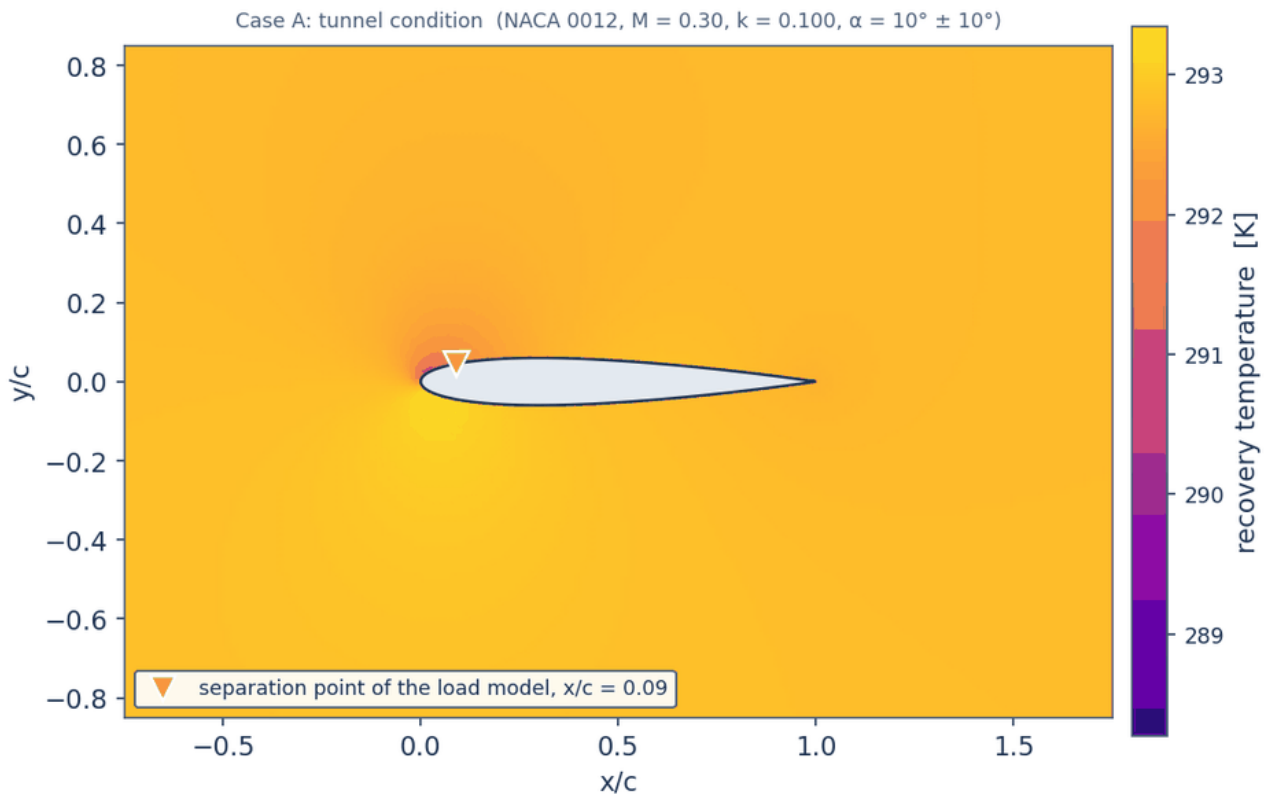
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.29$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Trecovery_A_validation_dsv_a20.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.1 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Recovery temperature at fall (downstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 288.27 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 0.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). A recovery temperature belongs to an adiabatic wall: away from the wall the map shows what such a wall would take at the local state.

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.09$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

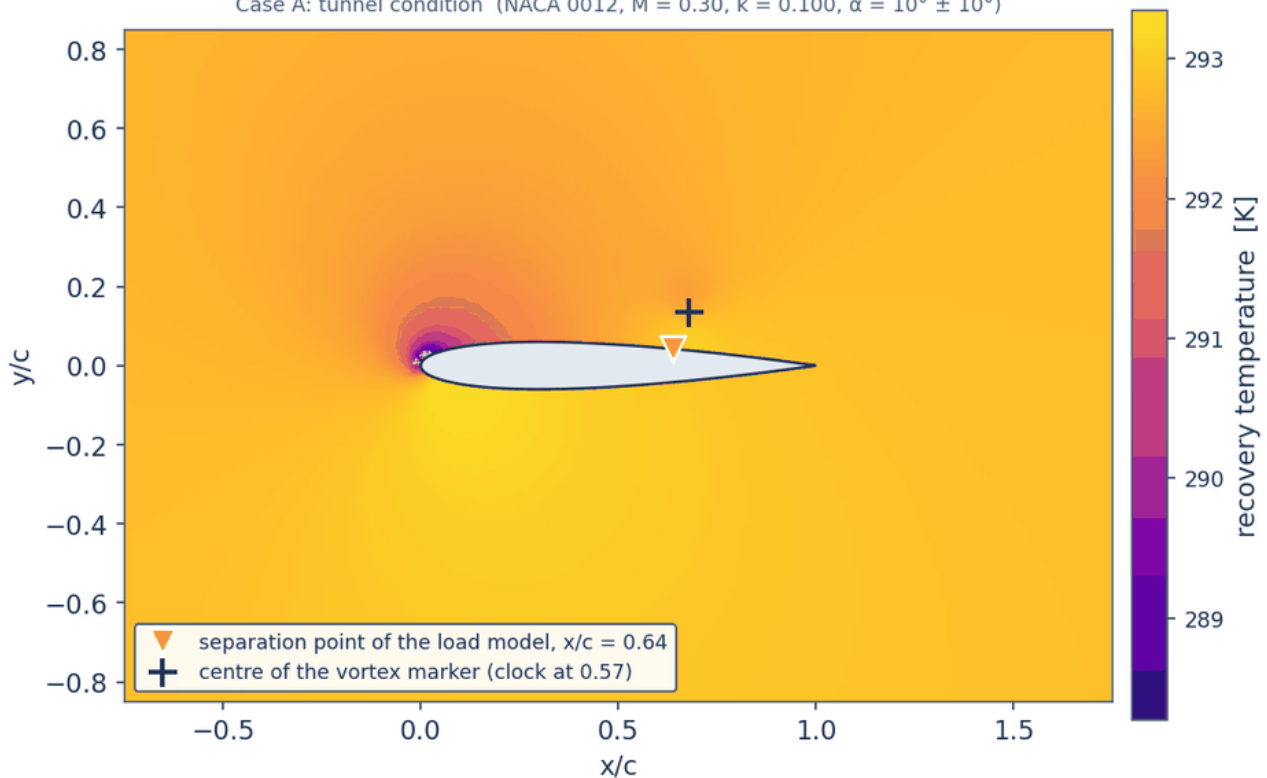
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Trecovery_A_validation_fall_a15.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Recovery temperature at greatest lift, $\alpha = 18.1^\circ$

Case A: tunnel condition (NACA 0012, $M = 0.30$, $k = 0.100$, $\alpha = 10^\circ \pm 10^\circ$)



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 288.27 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.2 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). A recovery temperature belongs to an adiabatic wall: away from the wall the map shows what such a wall would take at the local state.

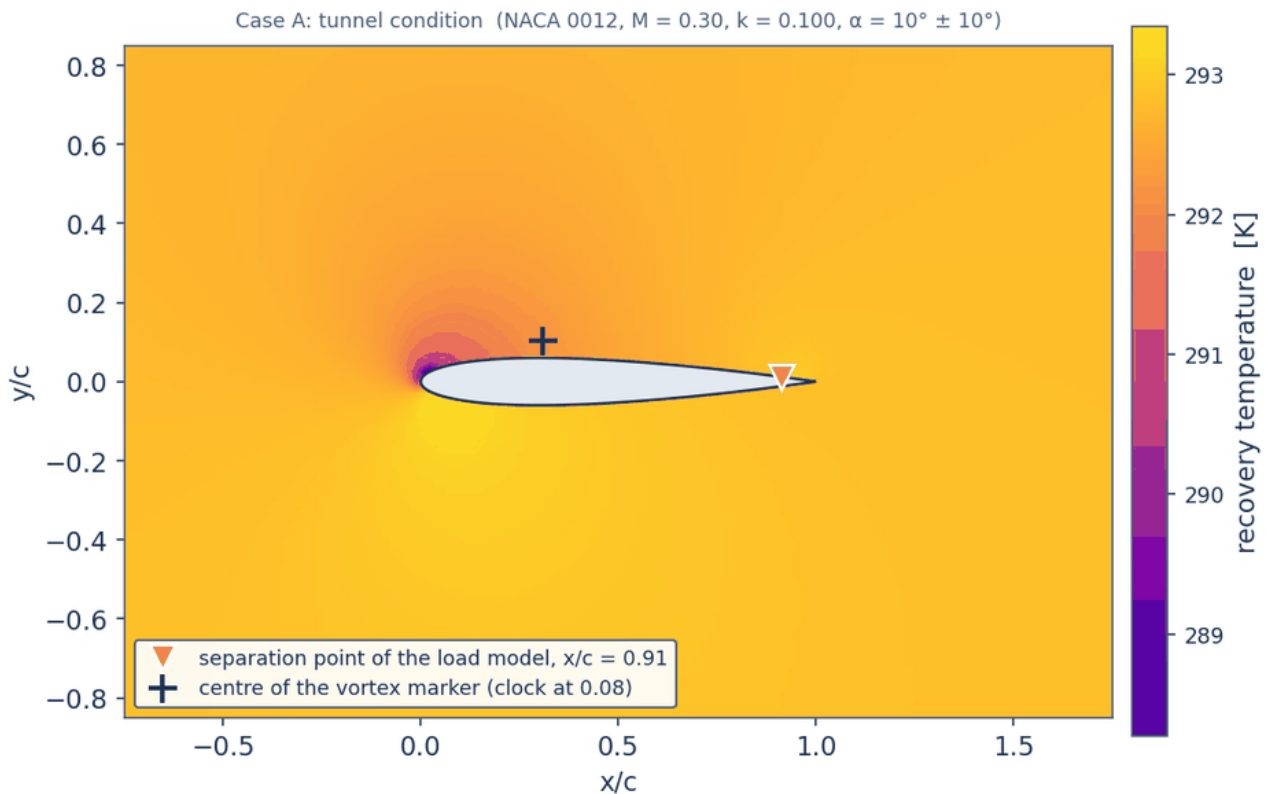
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.64$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Trecovery_A_validation_peak_a18.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.2 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Recovery temperature at rise (upstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 288.27 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 2.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air). A recovery temperature belongs to an adiabatic wall: away from the wall the map shows what such a wall would take at the local state.

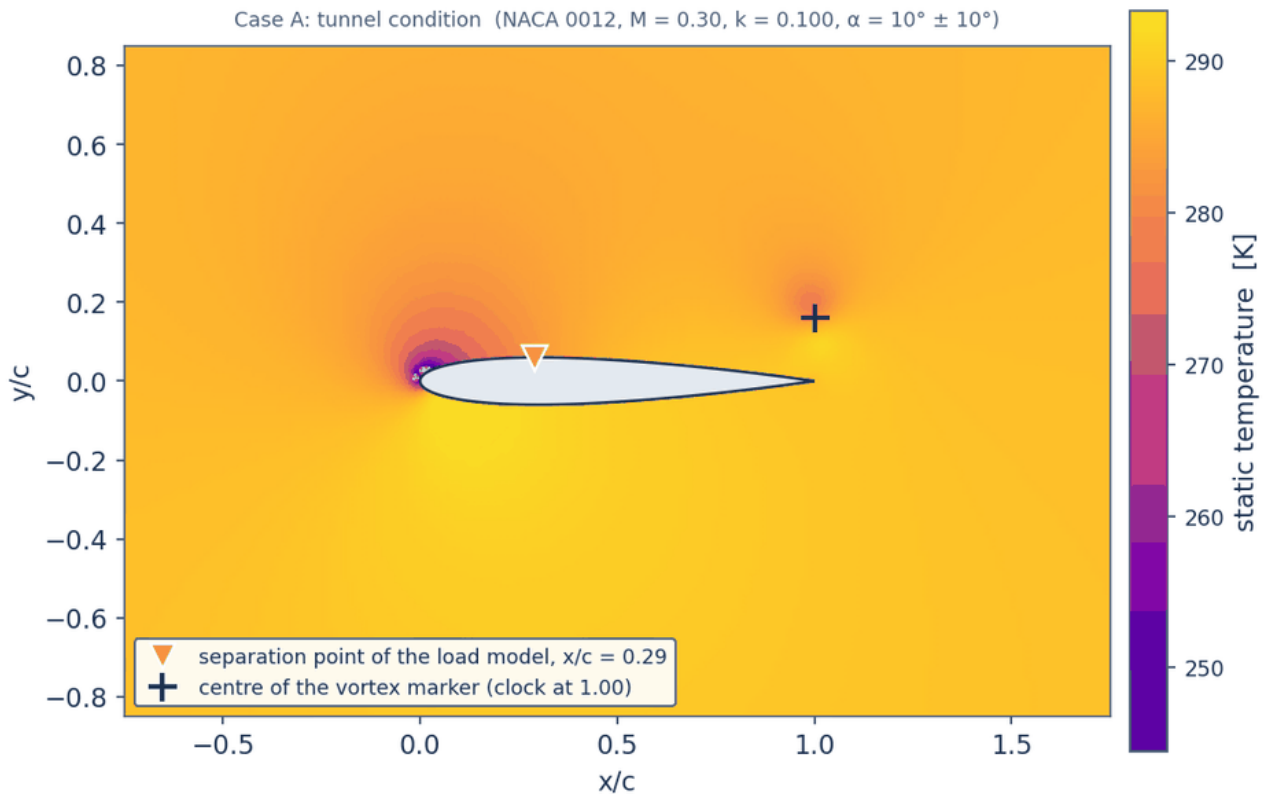
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.91$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Trecovery_A_validation_rise_a15.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at greatest vortex lift, $\alpha = 19.6^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 244.47 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.1 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.29$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

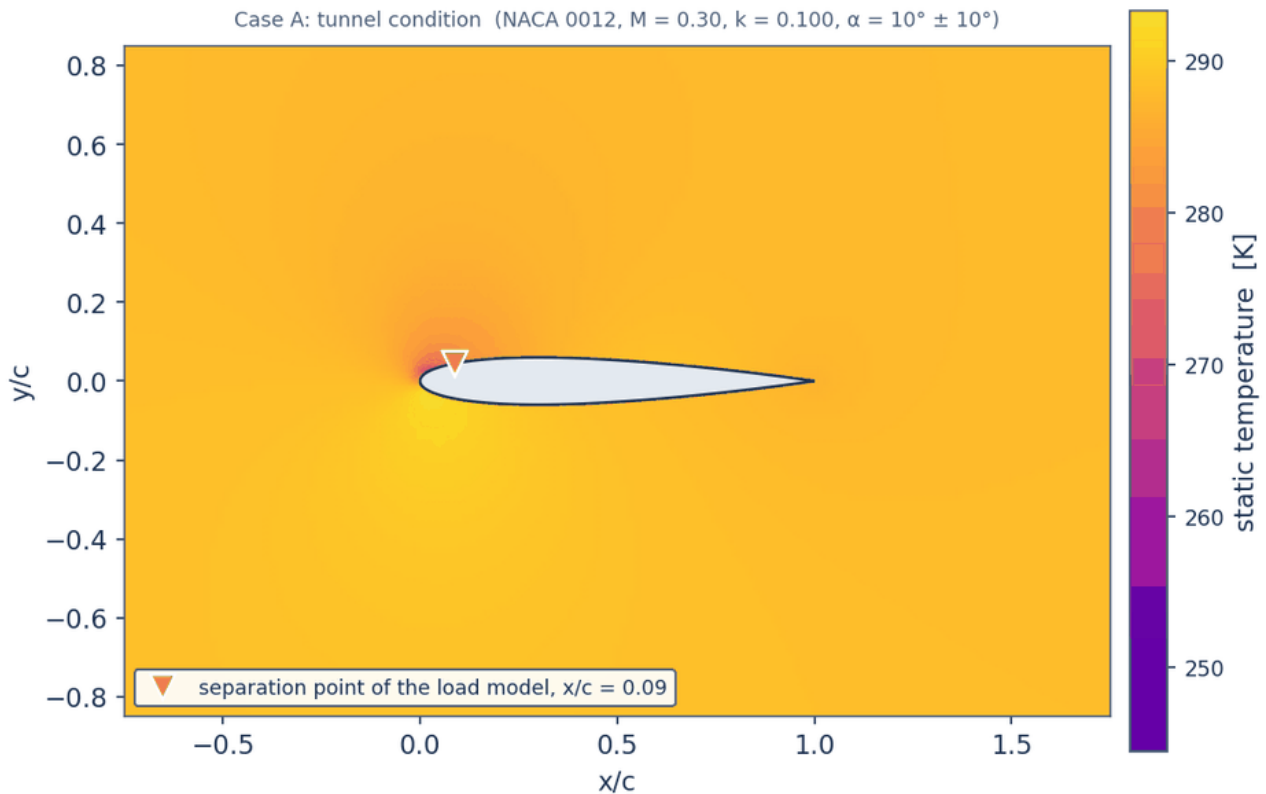
06_postprocessing/plots/contour_Tstatic_A_validation_dsv_a20.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name).

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.1 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised

(Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at fall (downstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 244.47 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 0.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

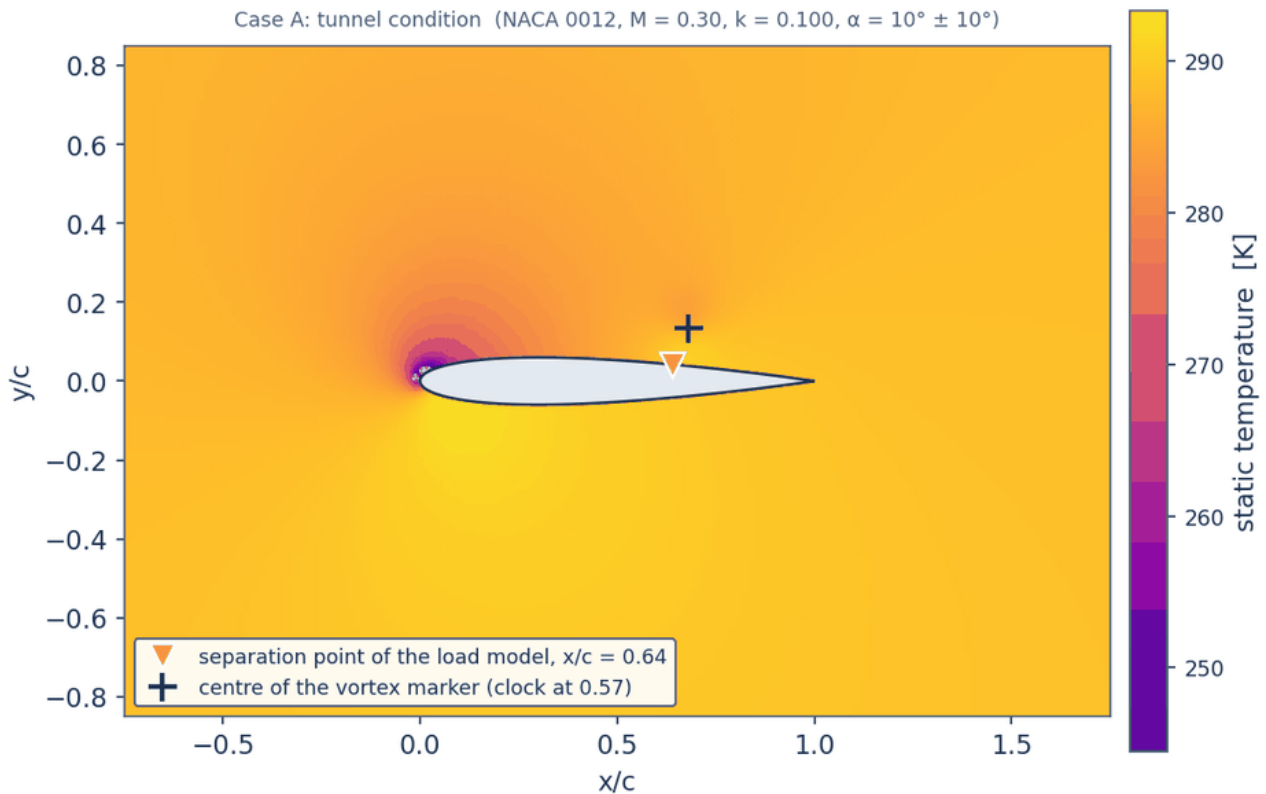
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.09$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Tstatic_A_validation_fall_a15.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at greatest lift, $\alpha = 18.1^\circ$



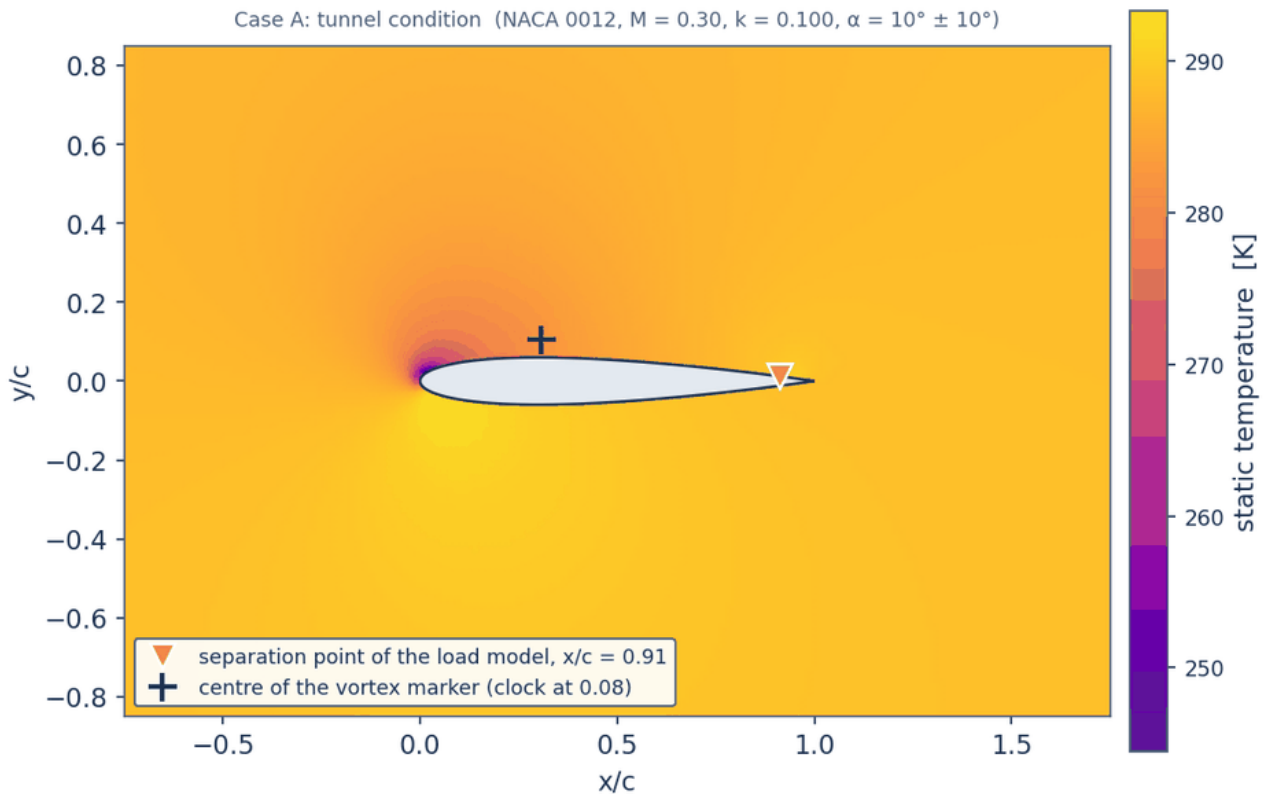
Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 244.47 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 3.2 % of the surface and 0.0009 c^2 of the stored field (0.012 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.64$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Tstatic_A_validation_peak_a18.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 3.2 % of the surface and 0.012 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at rise (upstroke), $\alpha = 15.0^\circ$ 

Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 244.47 to 293.34. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -6.95. At this instant the region is 2.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

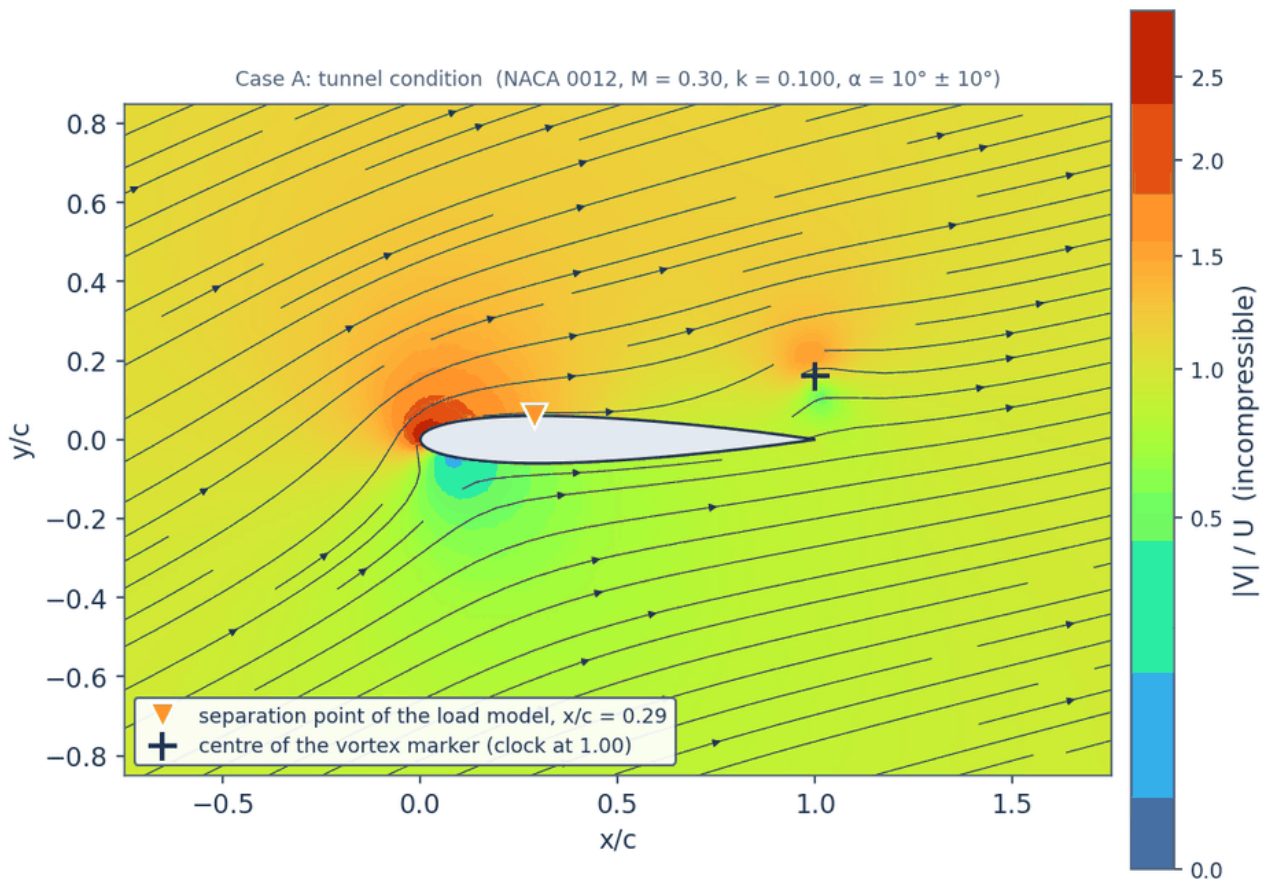
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.91$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Tstatic_A_validation_rise_a15.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

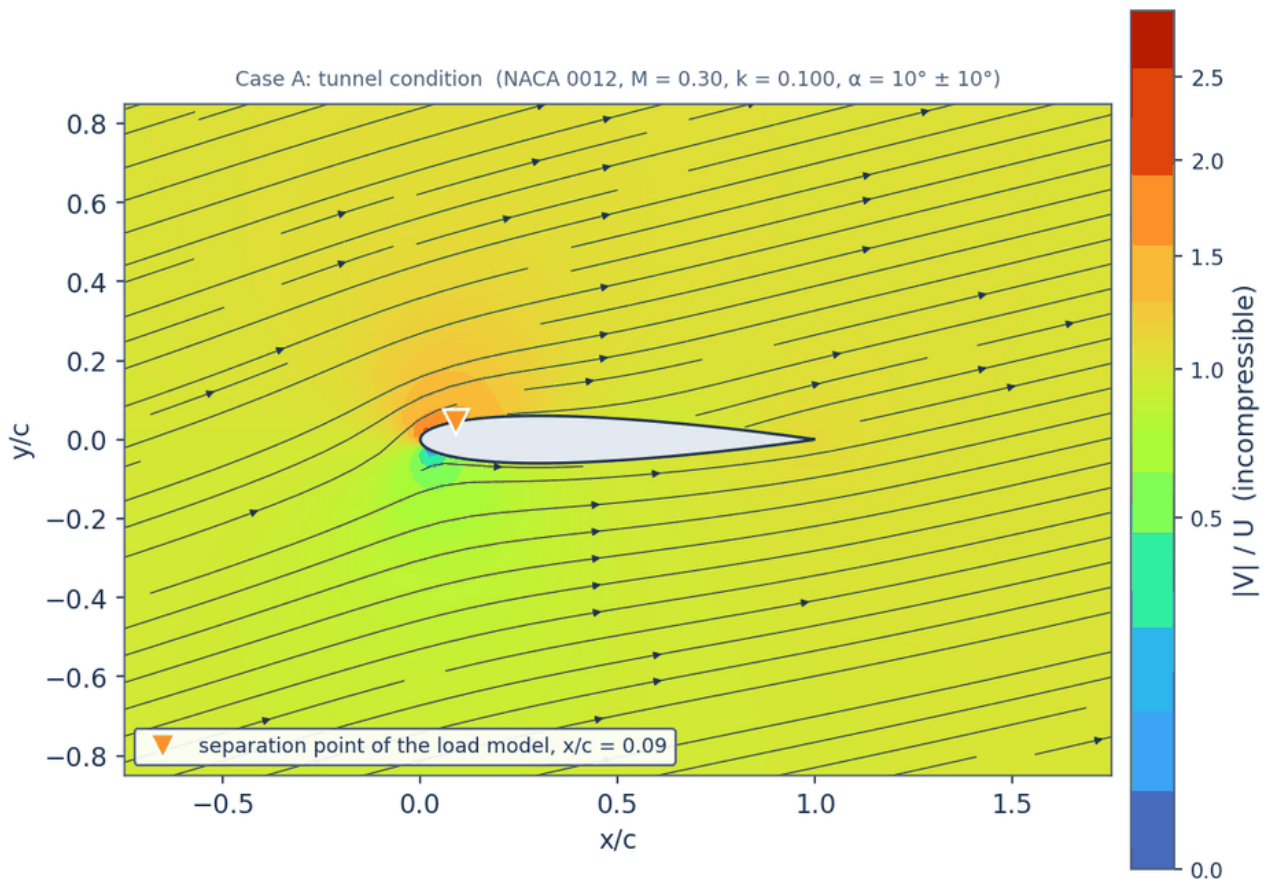
Speed and streamlines at greatest vortex lift, $\alpha = 19.6^\circ$



06_postprocessing/plots/contour_speed_stream_A_validation_dsv_a20.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at fall (downstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.93.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

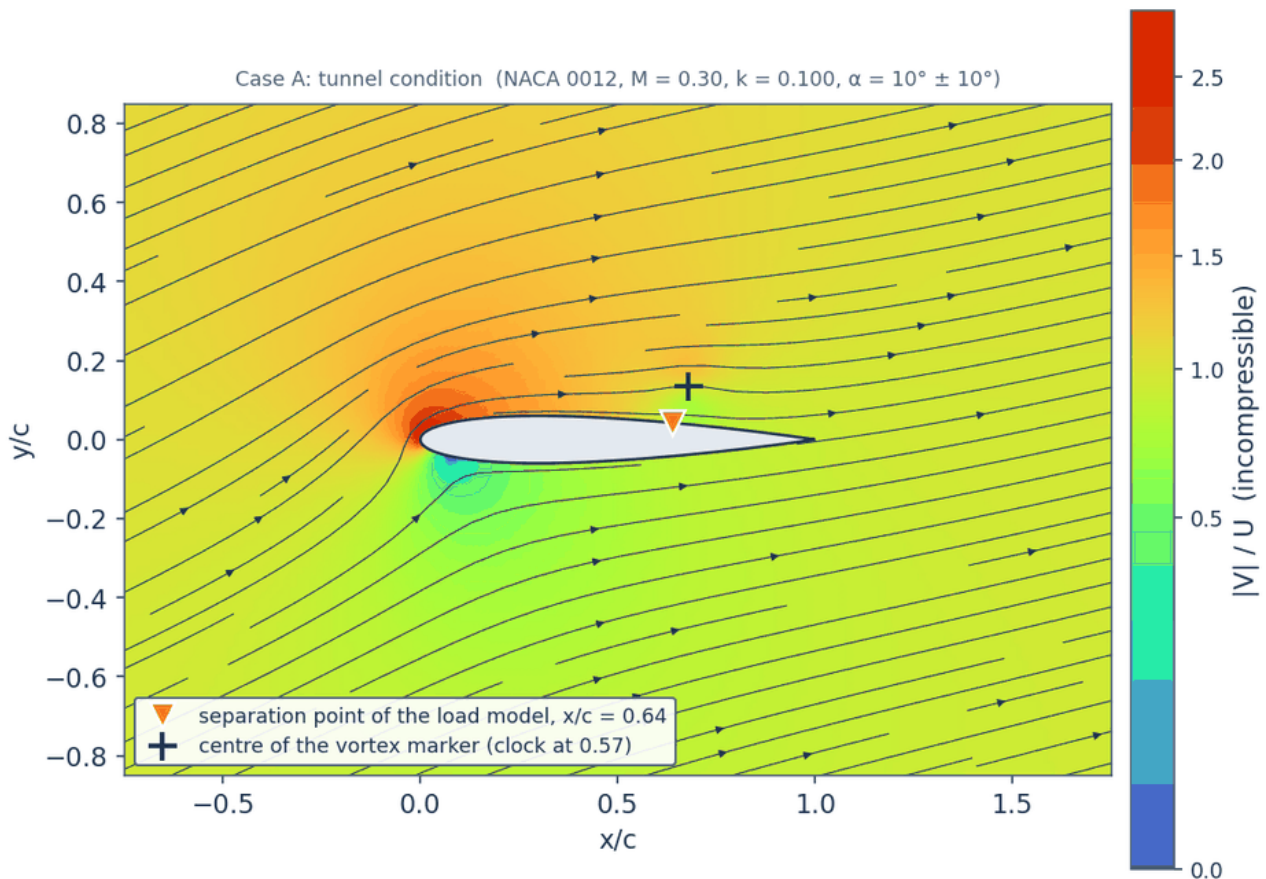
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.09$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_A_validation_fall_a15.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at greatest lift, $\alpha = 18.1^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.93.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

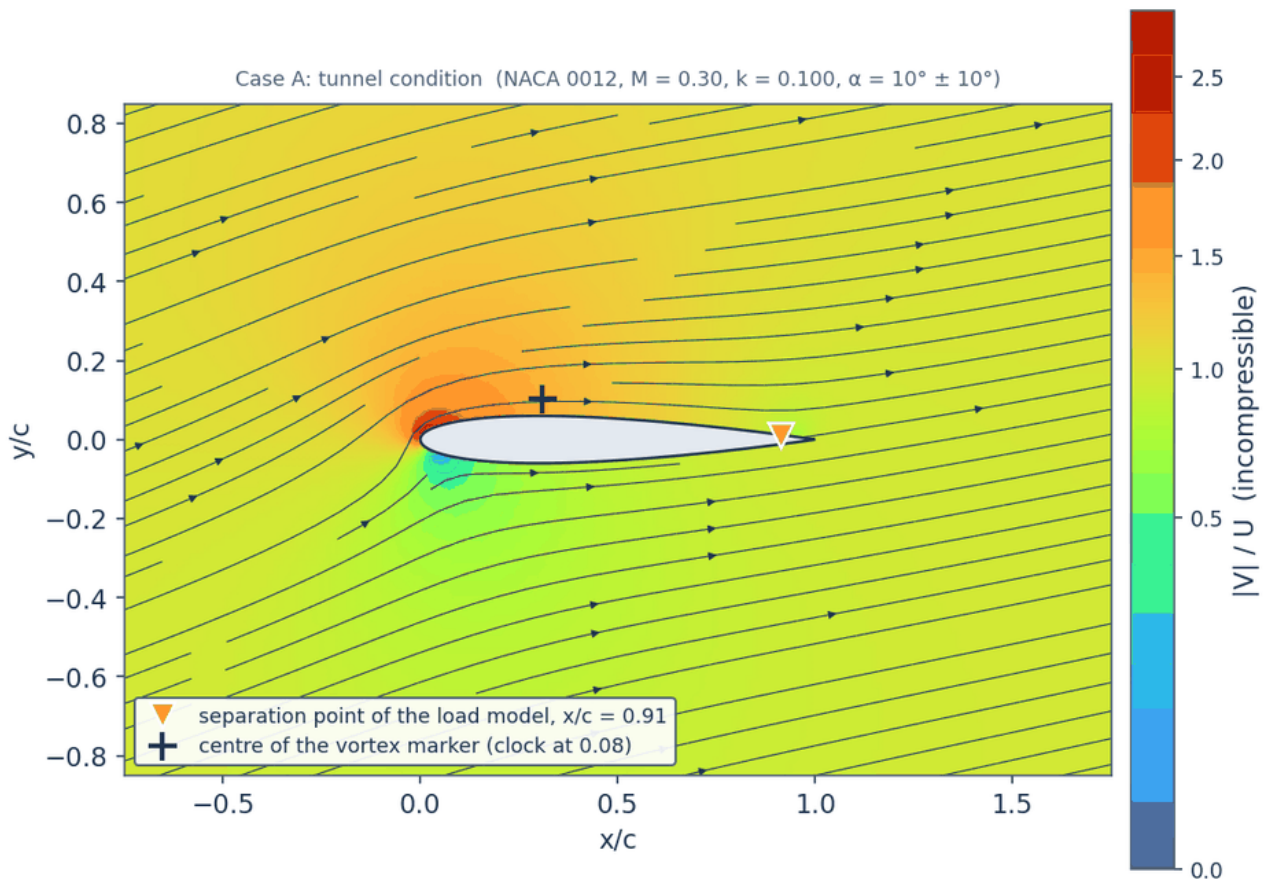
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.64$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_A_validation_peak_a18.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at rise (upstroke), $\alpha = 15.0^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.93.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

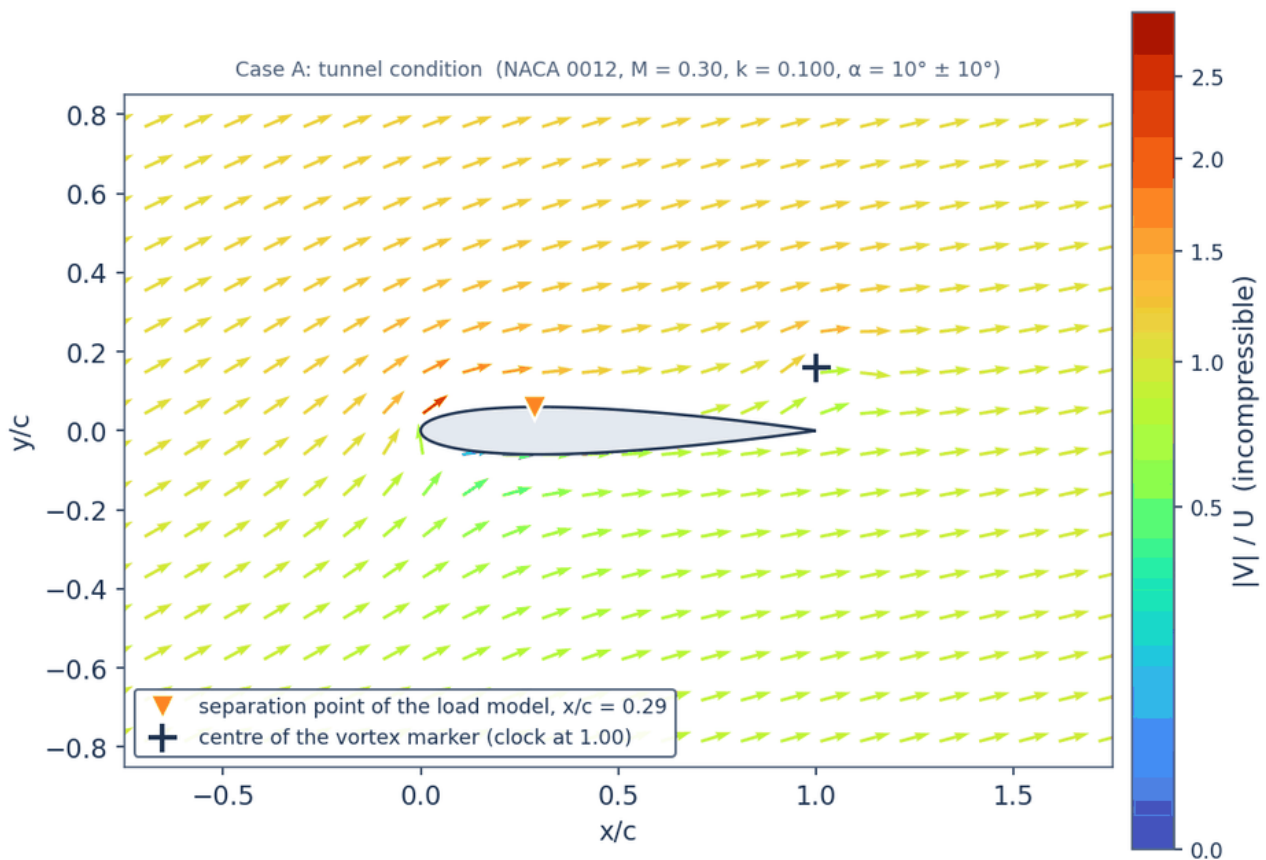
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.91$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_A_validation_rise_a15.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

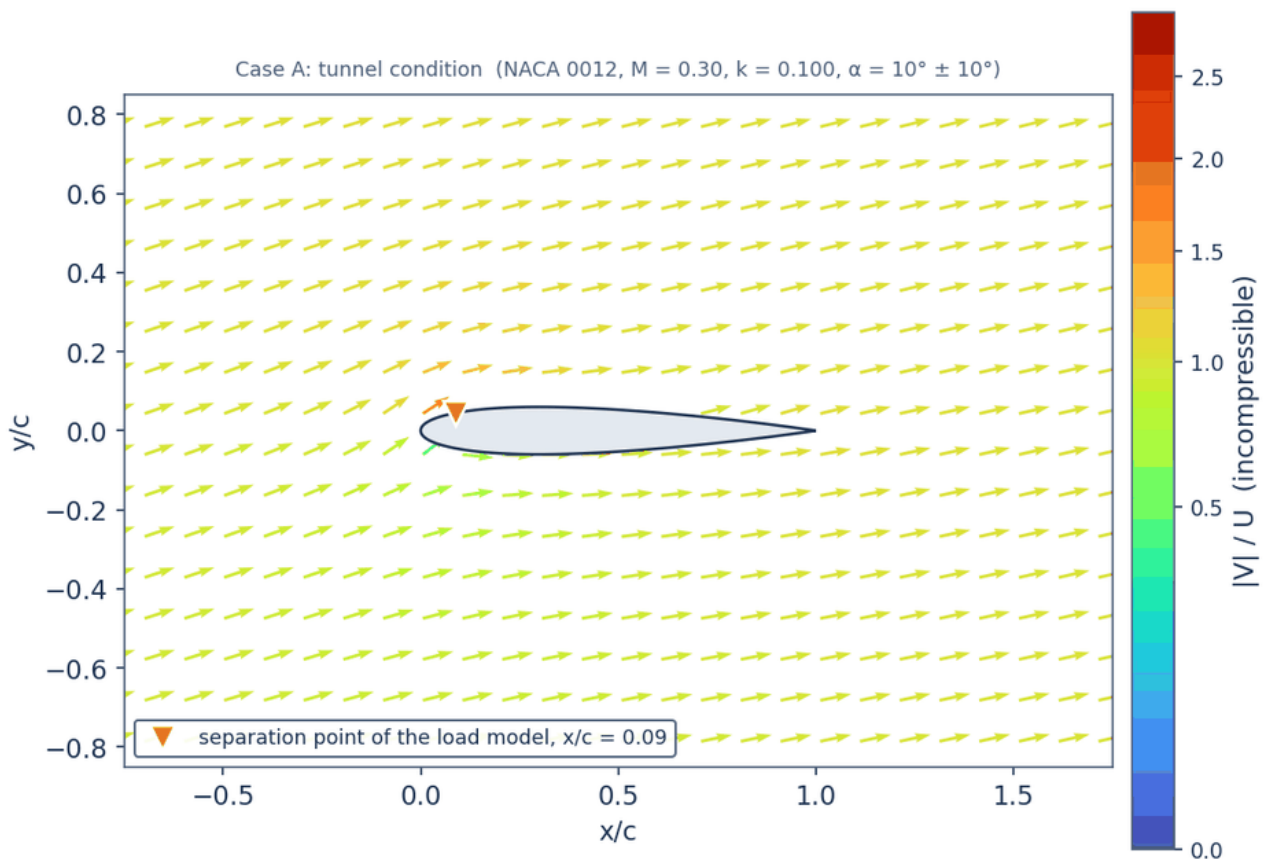
Velocity vectors at greatest vortex lift, $\alpha = 19.6^\circ$



06_postprocessing/plots/contour_vectors_A_validation_dsv_a20.png

Velocity vectors of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

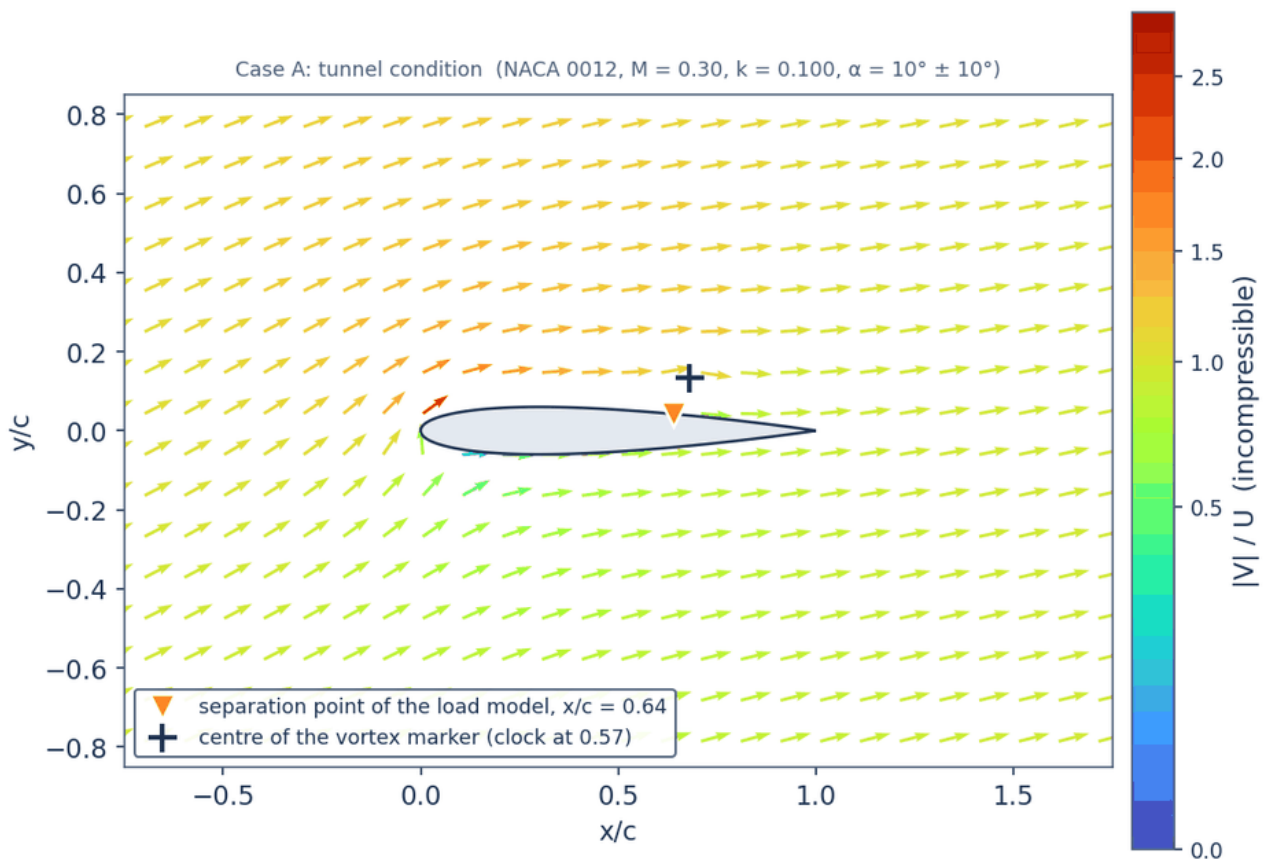
Velocity vectors at fall (downstroke), $\alpha = 15.0^\circ$



06_postprocessing/plots/contour_vectors_A_validation_fall_a15.png

Velocity vectors of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

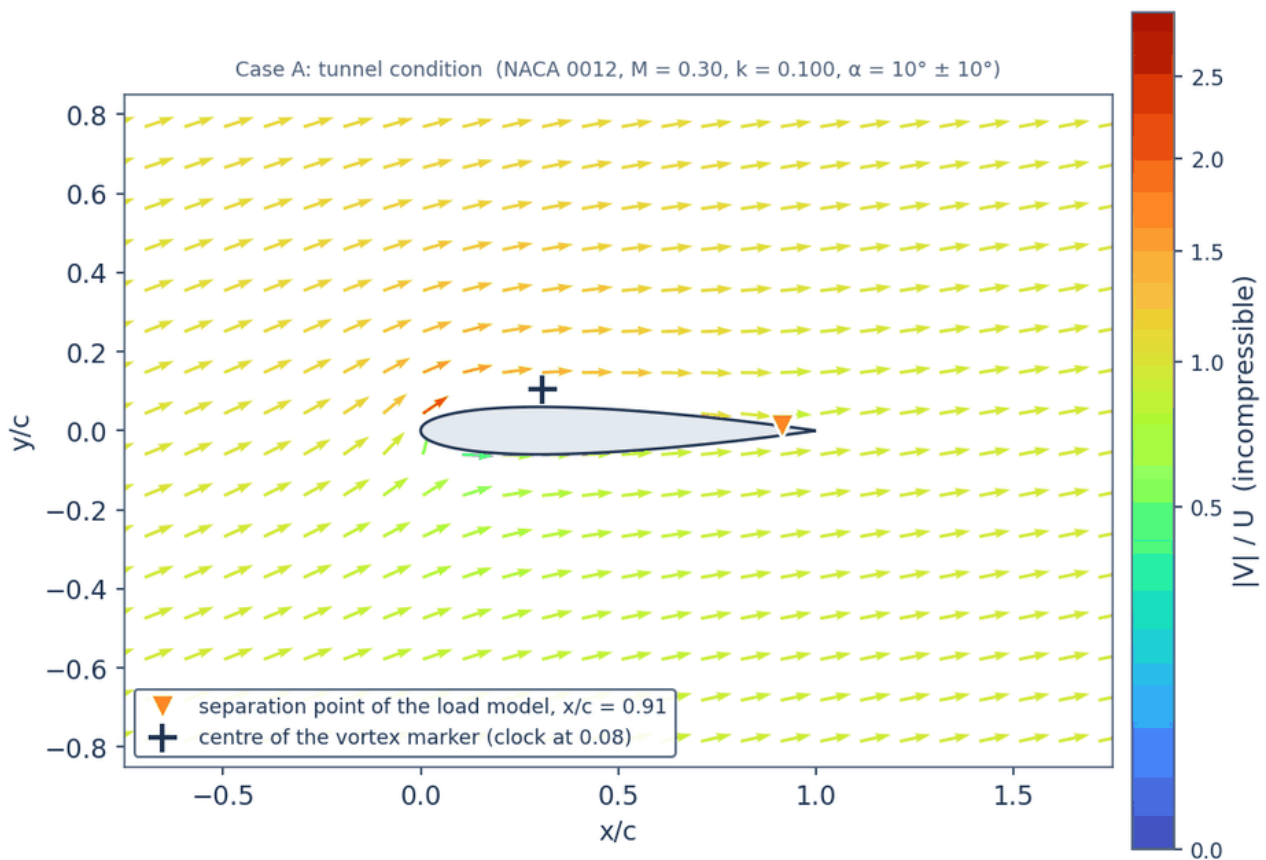
Velocity vectors at greatest lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_vectors_A_validation_peak_a18.png

Velocity vectors of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

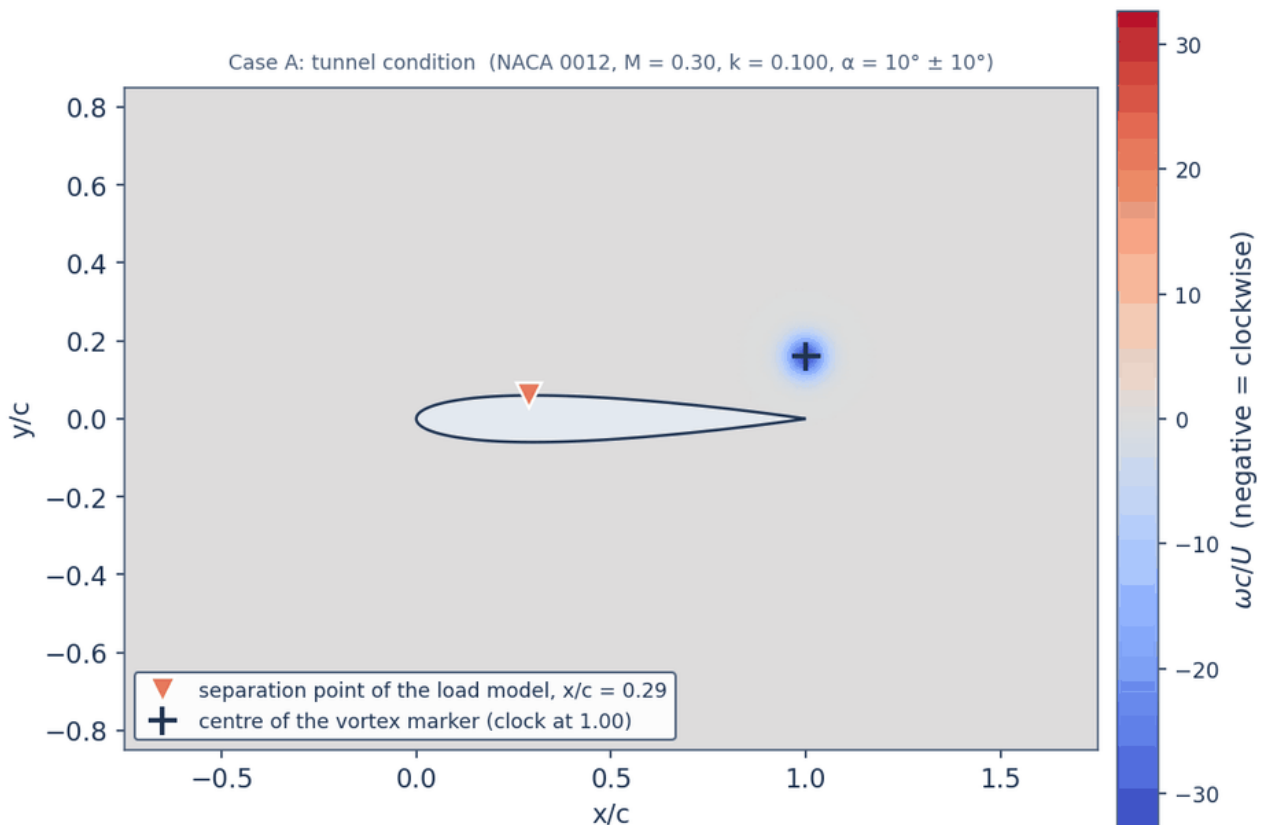
Velocity vectors at rise (upstroke), $\alpha = 15.0^\circ$



06_postprocessing/plots/contour_vectors_A_validation_rise_a15.png

Velocity vectors of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at greatest vortex lift, $\alpha = 19.6^\circ$



Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|\omega c/U| = 32.7$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

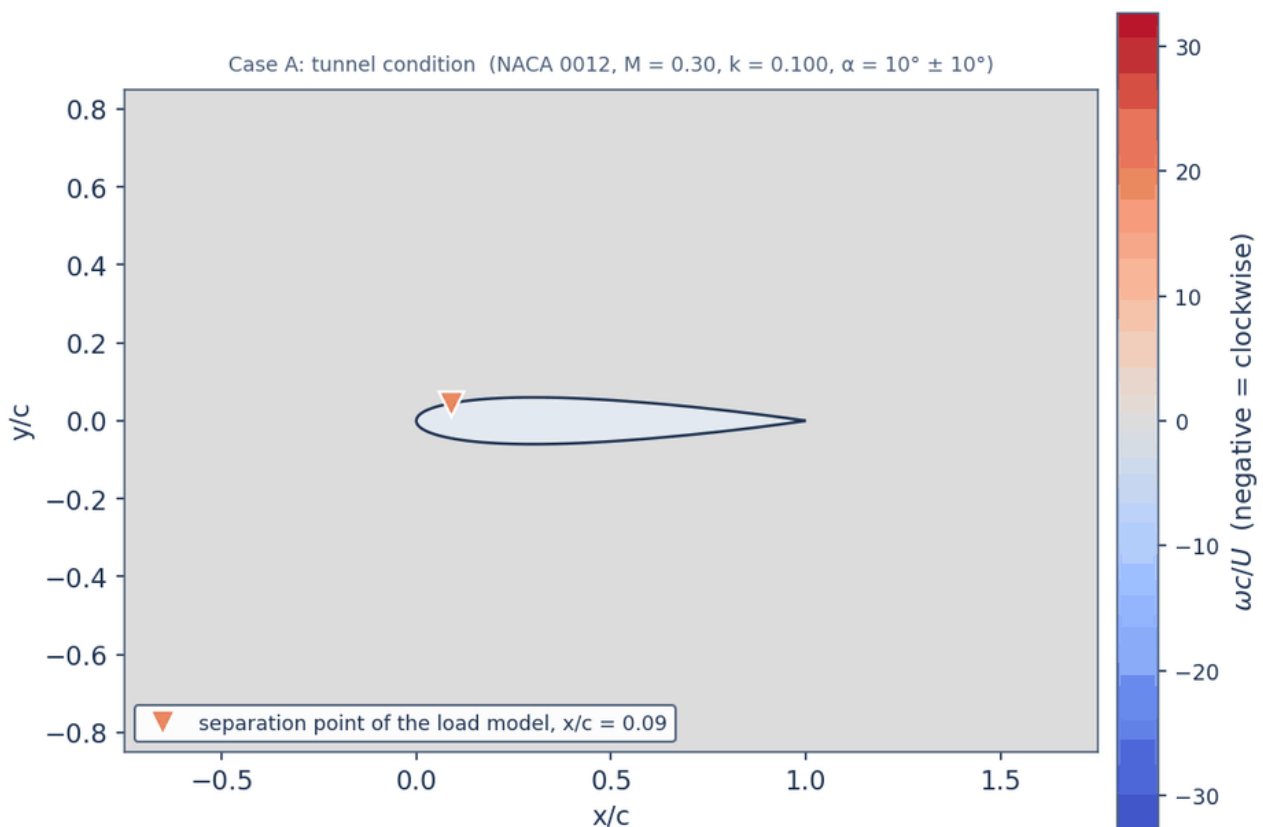
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.29$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_A_validation_dsv_a20.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case A: tunnel condition; instant of greatest vortex normal force (incidence 20° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

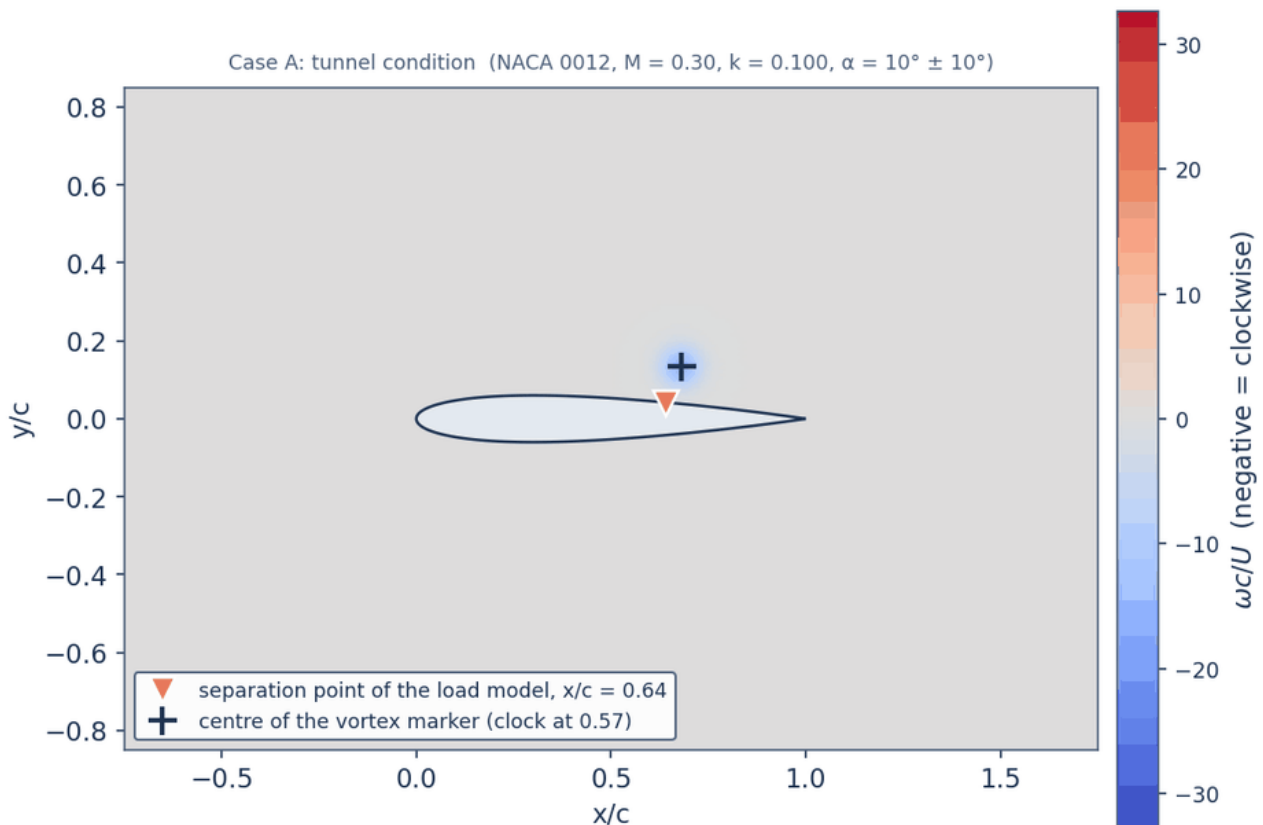
Vorticity at fall (downstroke), $\alpha = 15.0^\circ$



06_postprocessing/plots/contour_vorticity_A_validation_fall_a15.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case A: tunnel condition; downstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at greatest lift, $\alpha = 18.1^\circ$



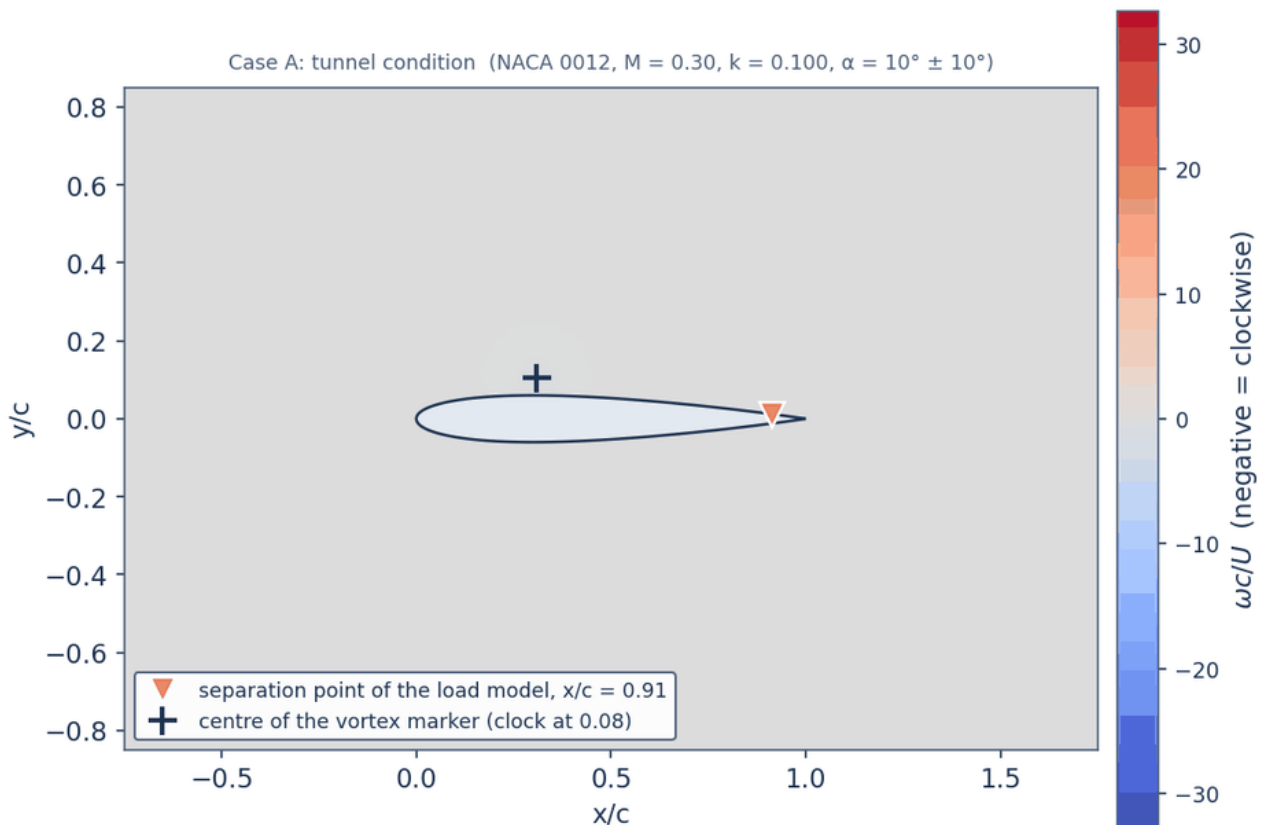
Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|\omega c/U| = 32.7$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.64$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_A_validation_peak_a18.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case A: tunnel condition; instant of greatest lift (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at rise (upstroke), $\alpha = 15.0^\circ$ 

Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|wc/U| = 32.7$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.91$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

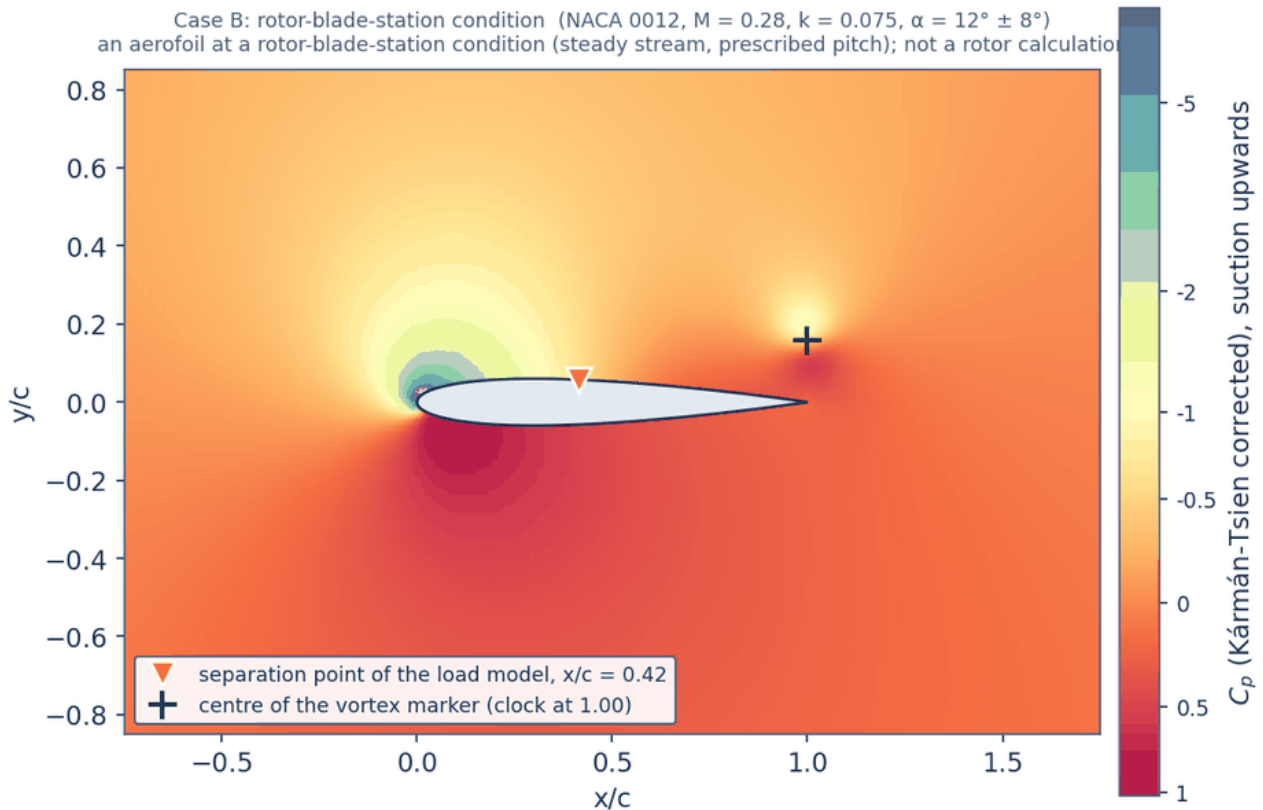
Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_A_validation_rise_a15.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case A: tunnel condition; upstroke, mean incidence plus half the amplitude (incidence 15° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Reconstructed fields, Case B: rotor-blade-station condition (06_postprocessing)

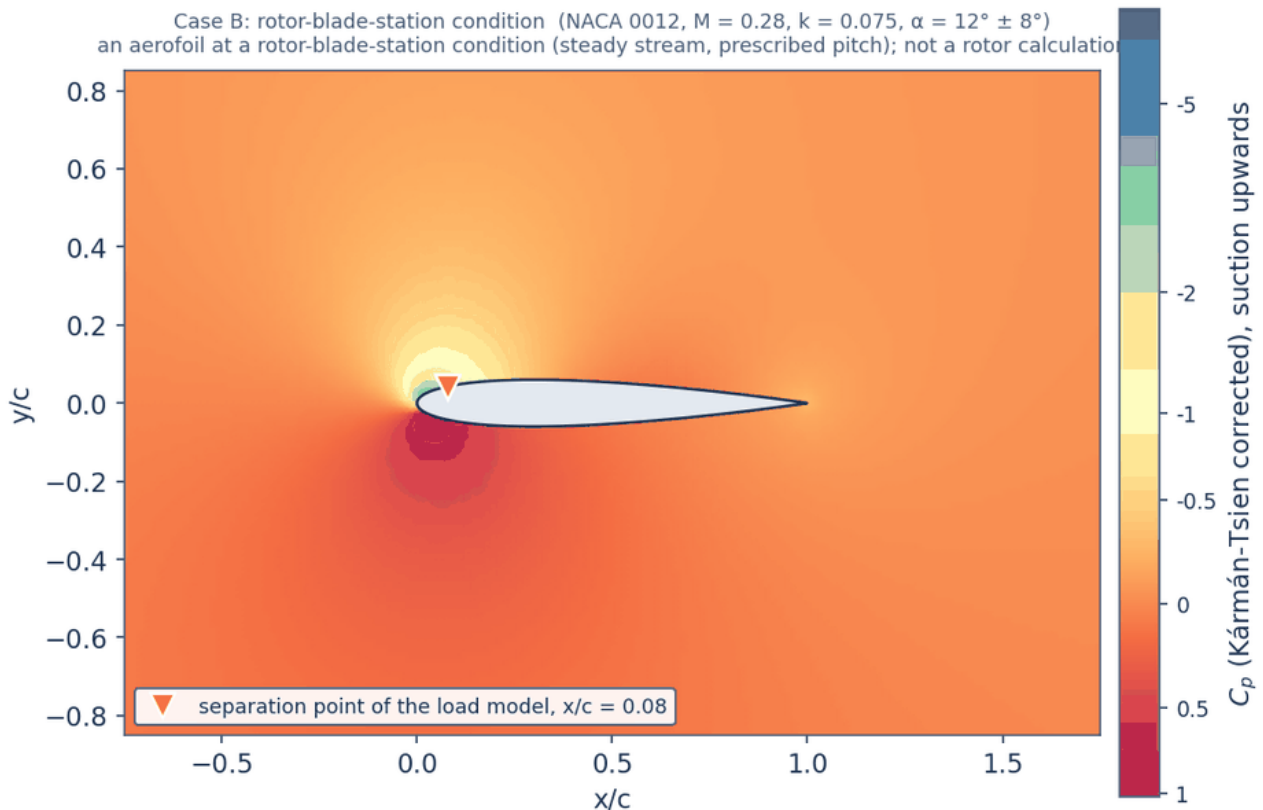
Pressure coefficient at greatest vortex lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_Cp_B_application_dsv_a18.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Pressure coefficient at fall (downstroke), $\alpha = 16.0^\circ$

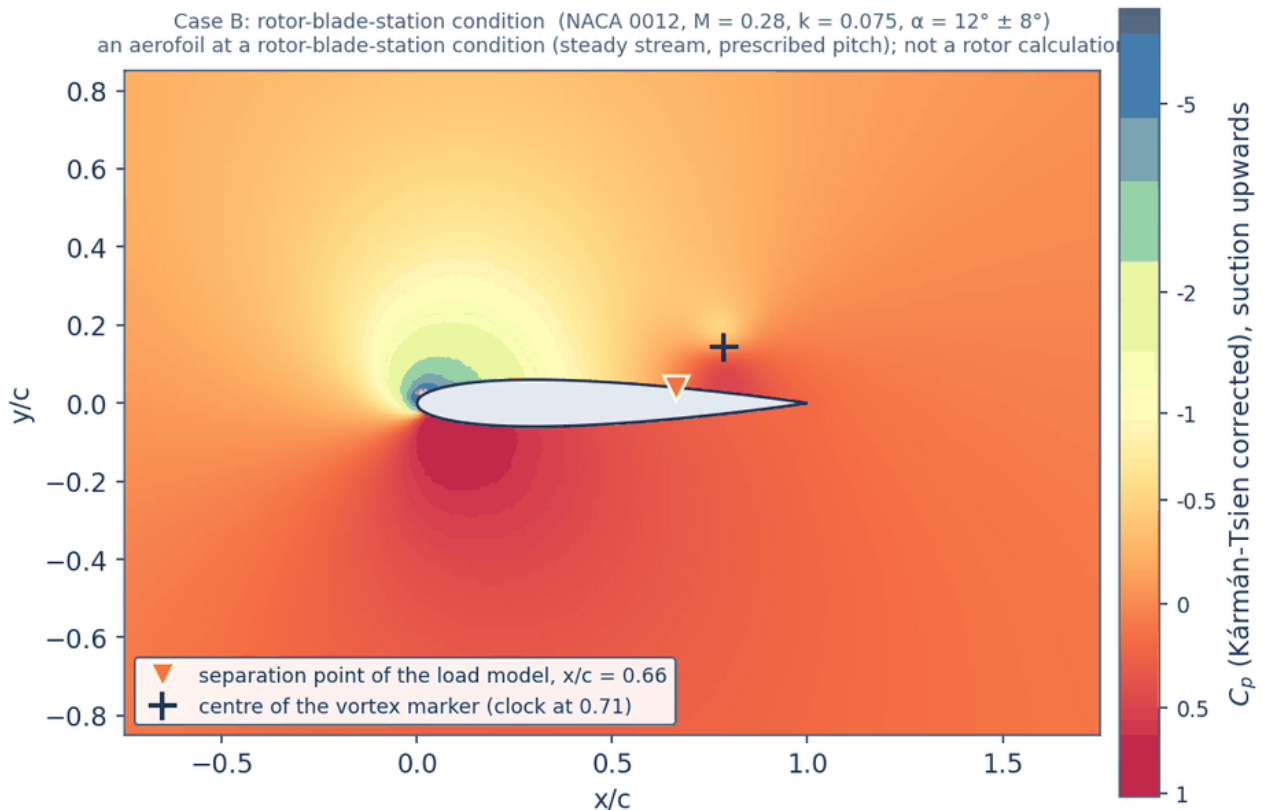


06_postprocessing/plots/contour_Cp_B_application_fall_a16.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name).

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

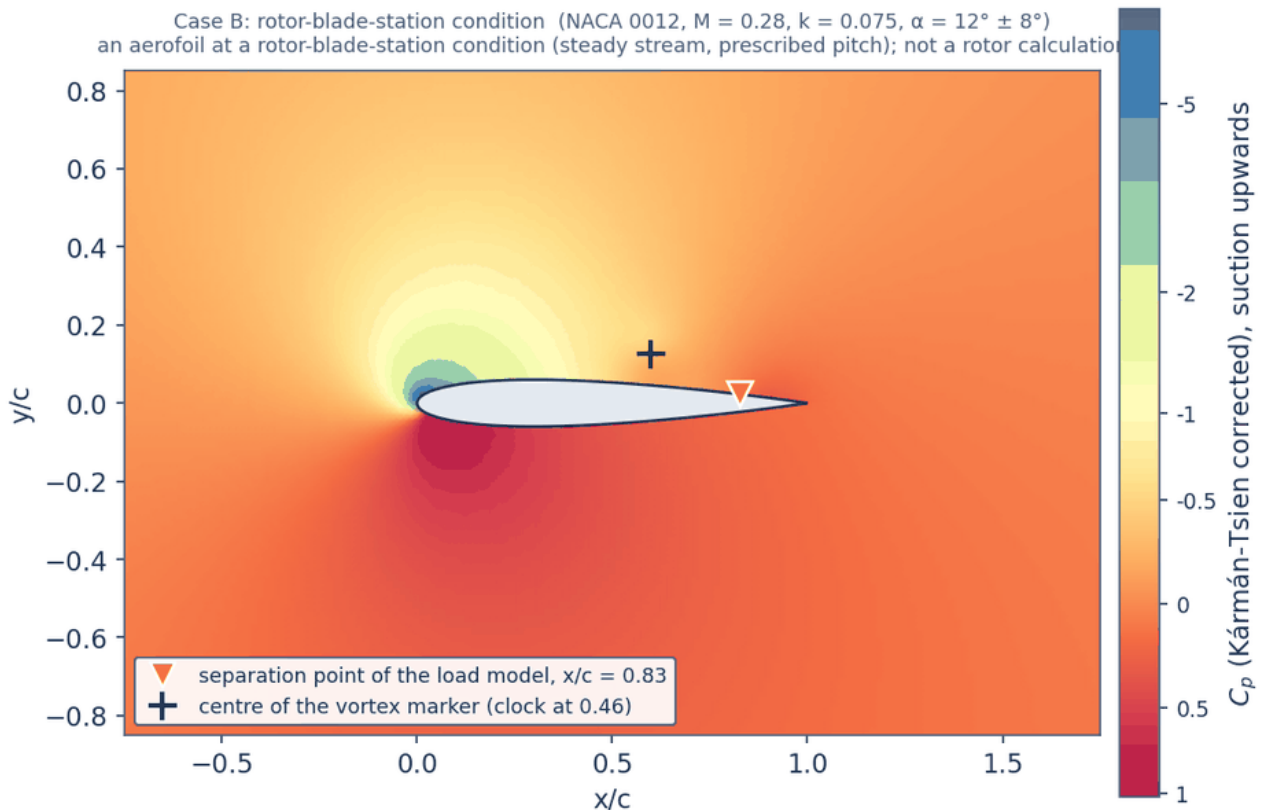
Pressure coefficient at greatest lift, $\alpha = 17.0^\circ$



06_postprocessing/plots/contour_Cp_B_application_peak_a17.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Pressure coefficient at rise (upstroke), $\alpha = 16.0^\circ$

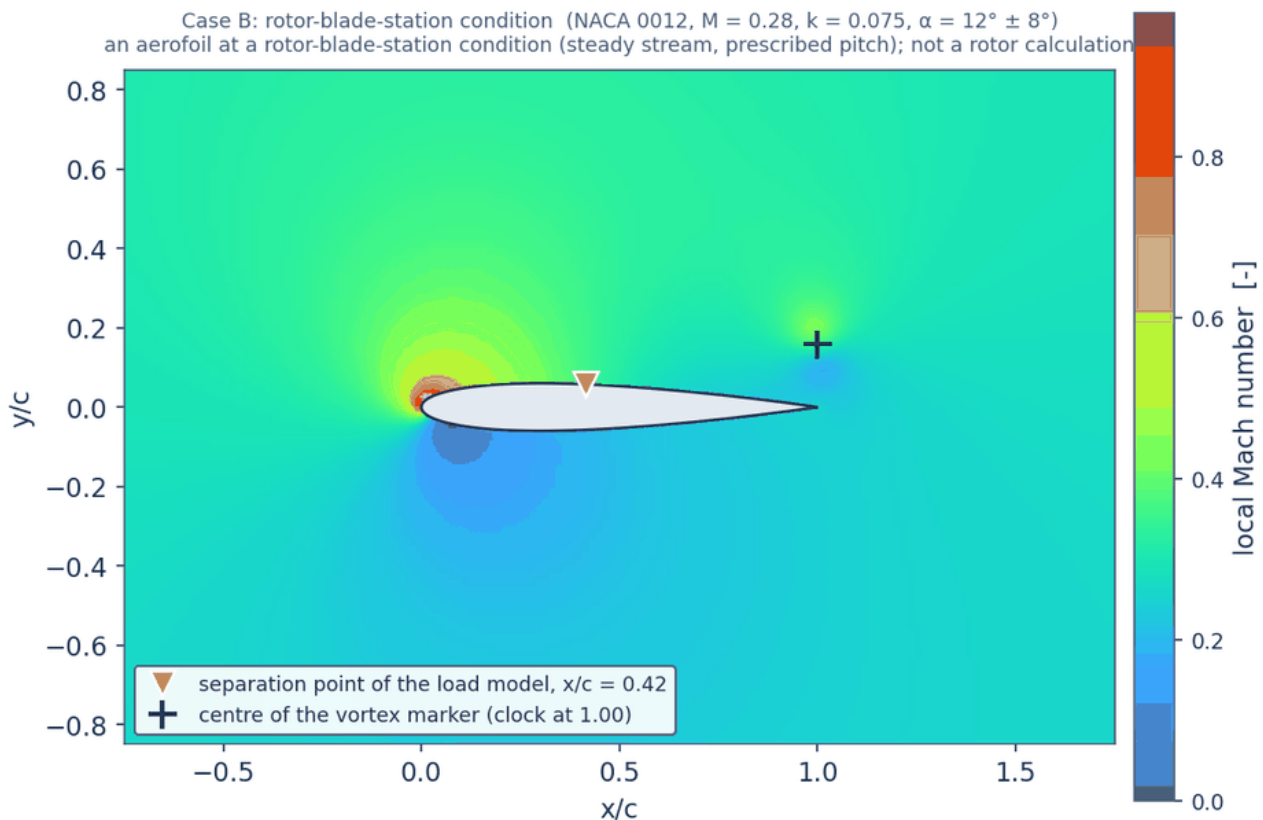


06_postprocessing/plots/contour_Cp_B_application_rise_a16.png

Pressure coefficient (Kármán-Tsien corrected) of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name).

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at greatest vortex lift, $\alpha = 18.1^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 0.98. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 2.3 % of the surface and 0.0006 c^2 of the stored field (0.008 % of it).

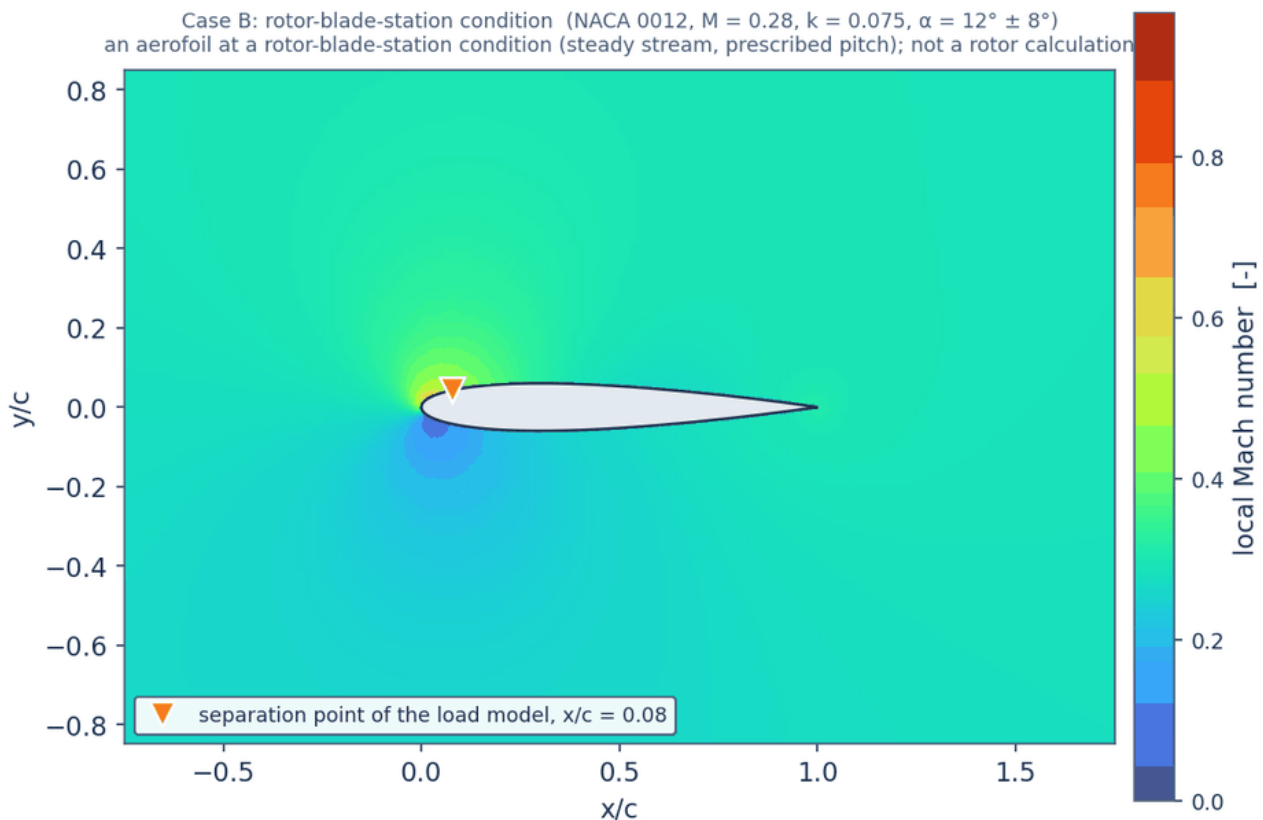
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.42$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_B_application_dsv_a18.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at fall (downstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 0.98. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 0.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it).

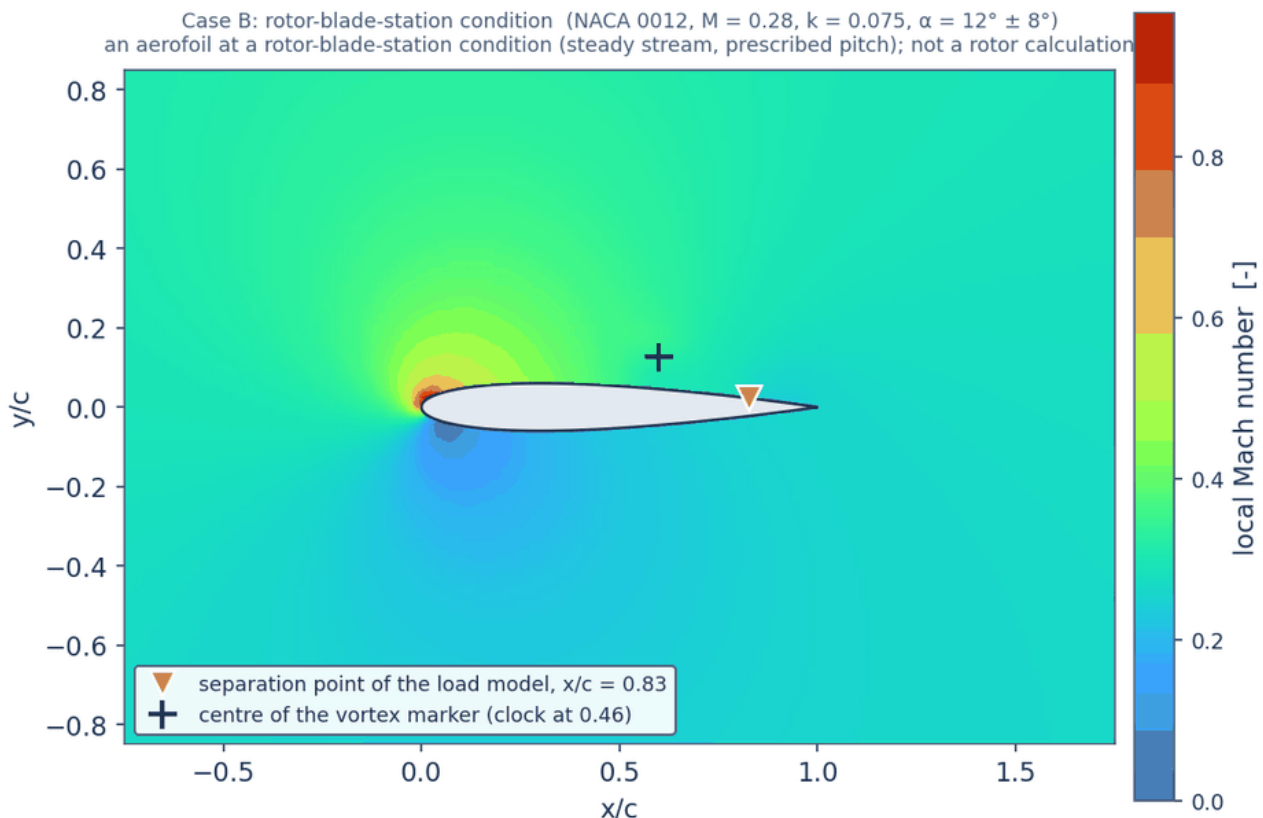
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.08$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Mach_B_application_fall_a16.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Local Mach number at rise (upstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 0.00 to 0.98. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 2.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it).

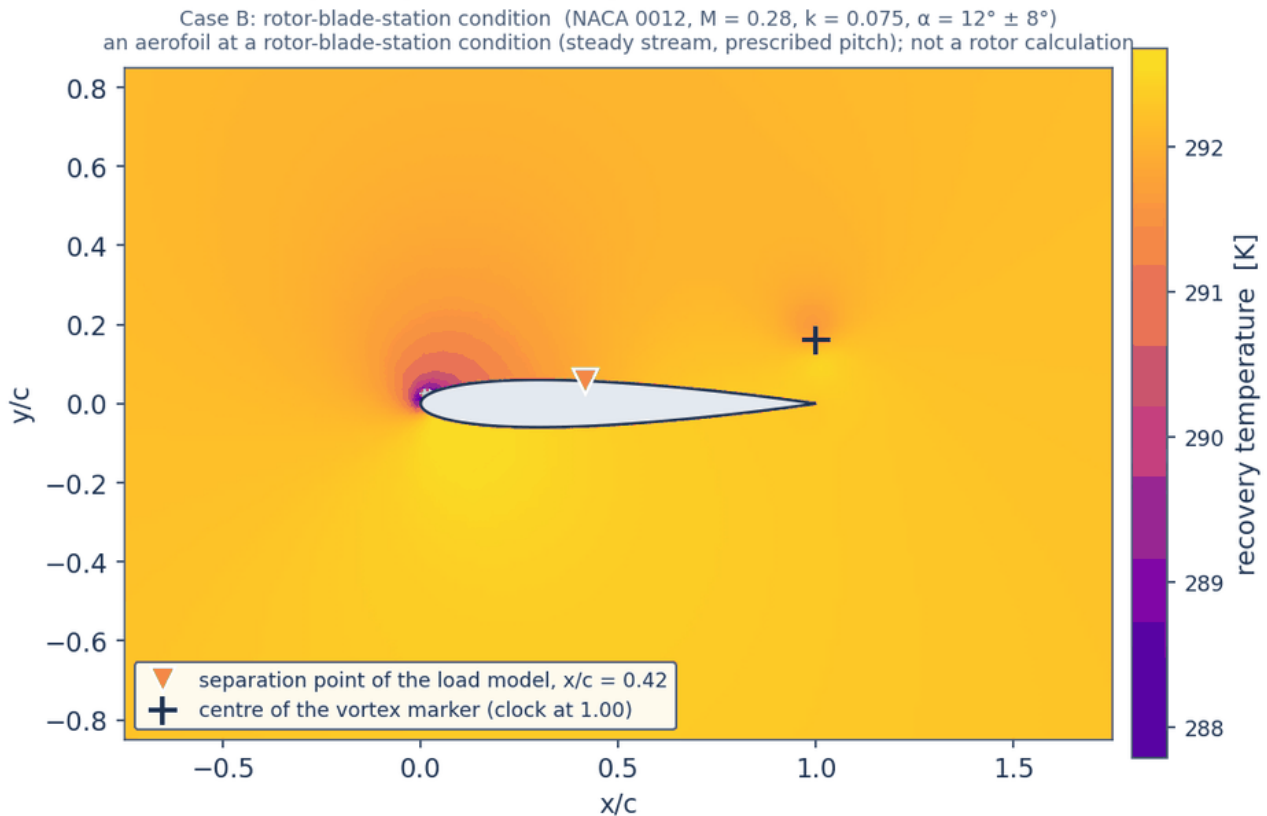
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.83$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06 postprocessing/plots/contour Mach B application rise a16.png

Local Mach number, from the corrected pressure by the isentropic relations of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

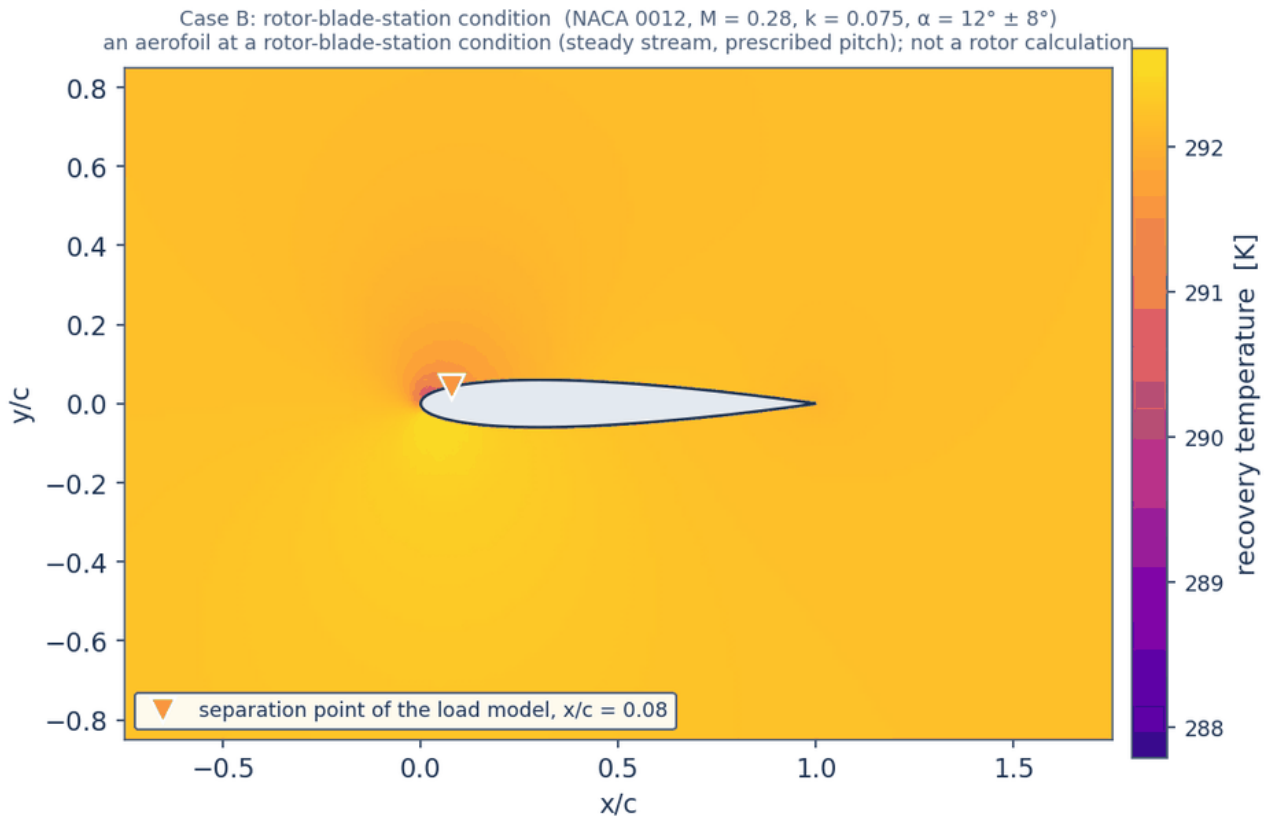
Recovery temperature at greatest vortex lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_Trecovery_B_application_dsv_a18.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

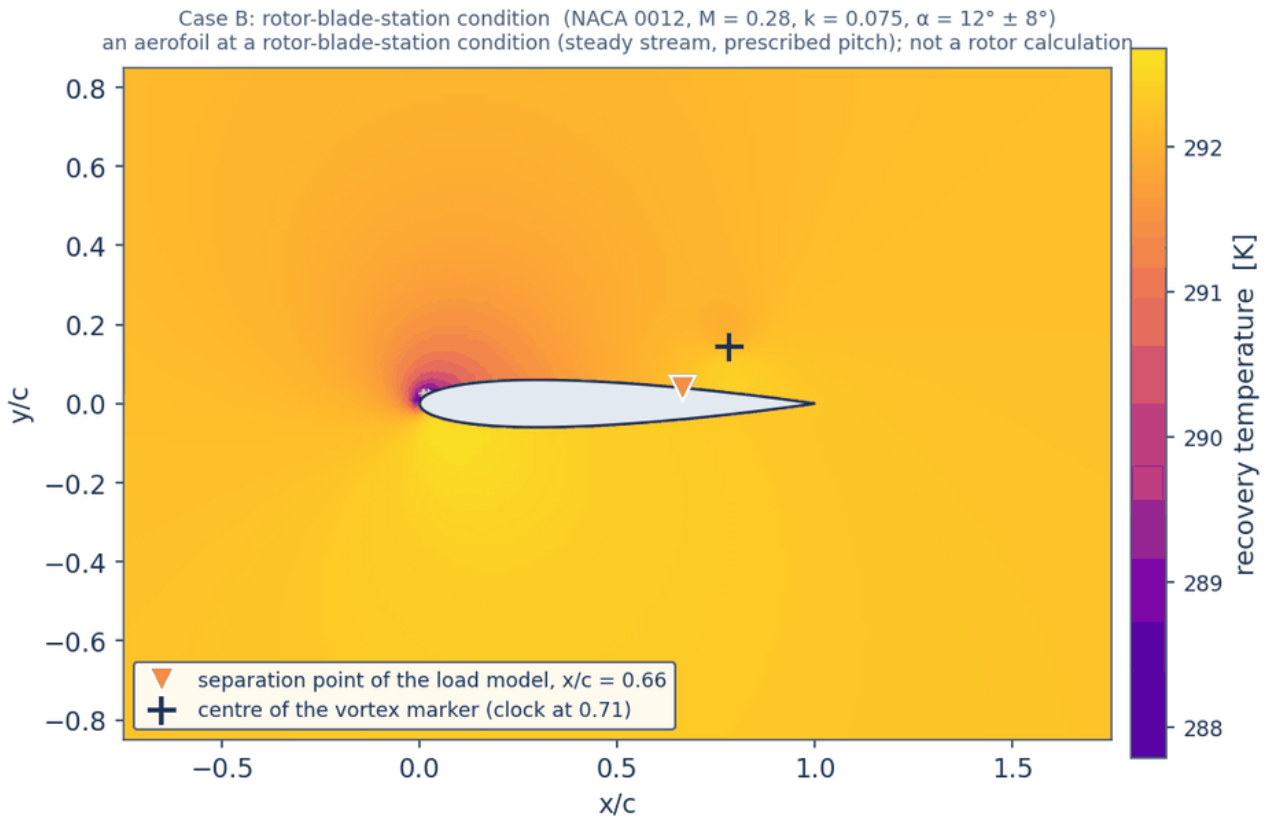
Recovery temperature at fall (downstroke), $\alpha = 16.0^\circ$



06_postprocessing/plots/contour_Trecovery_B_application_fall_a16.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Recovery temperature at greatest lift, $\alpha = 17.0^\circ$

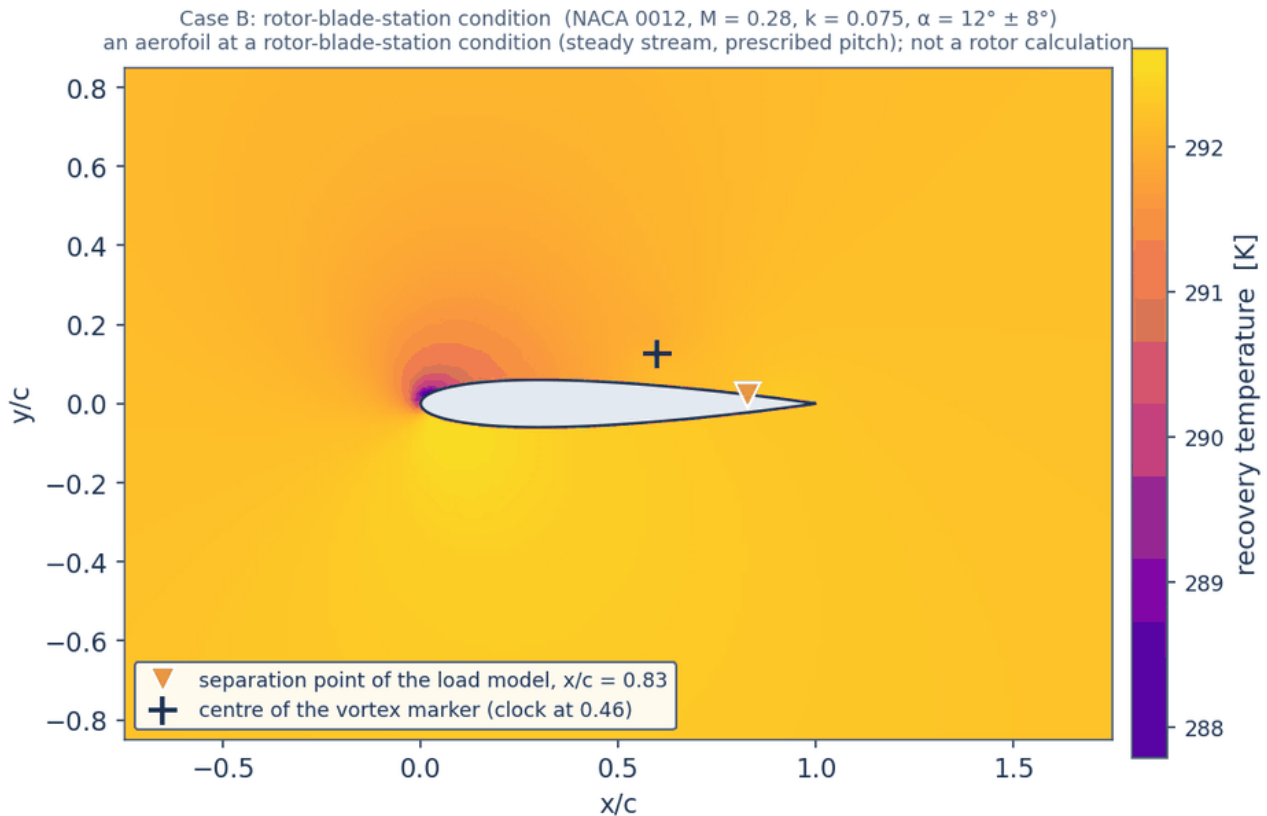


06_postprocessing/plots/contour_Trecovery_B_application_peak_a17.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name).

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

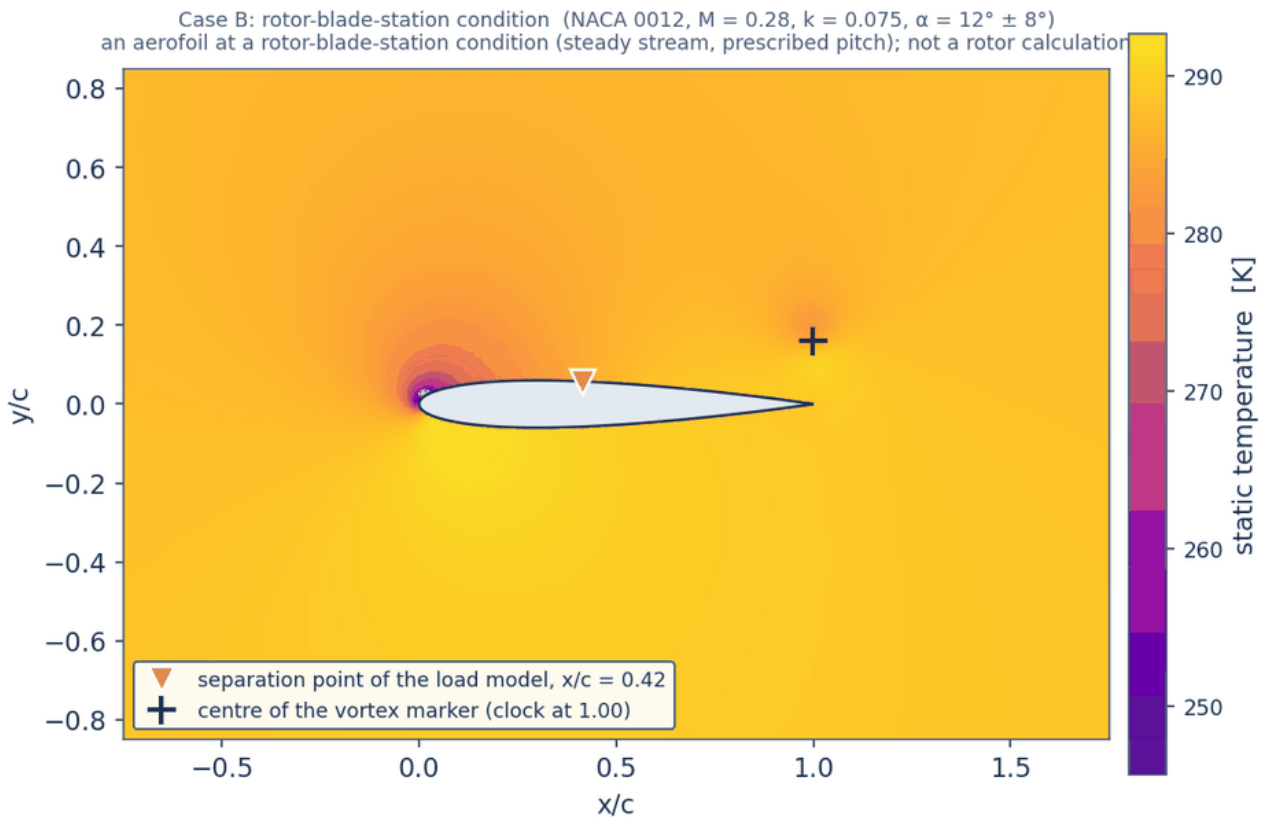
Recovery temperature at rise (upstroke), $\alpha = 16.0^\circ$



06_postprocessing/plots/contour_Trecovery_B_application_rise_a16.png

Recovery temperature (that of an adiabatic wall under a turbulent boundary layer, which the reconstruction does not have; away from the wall, what such a wall would take at the local state) of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

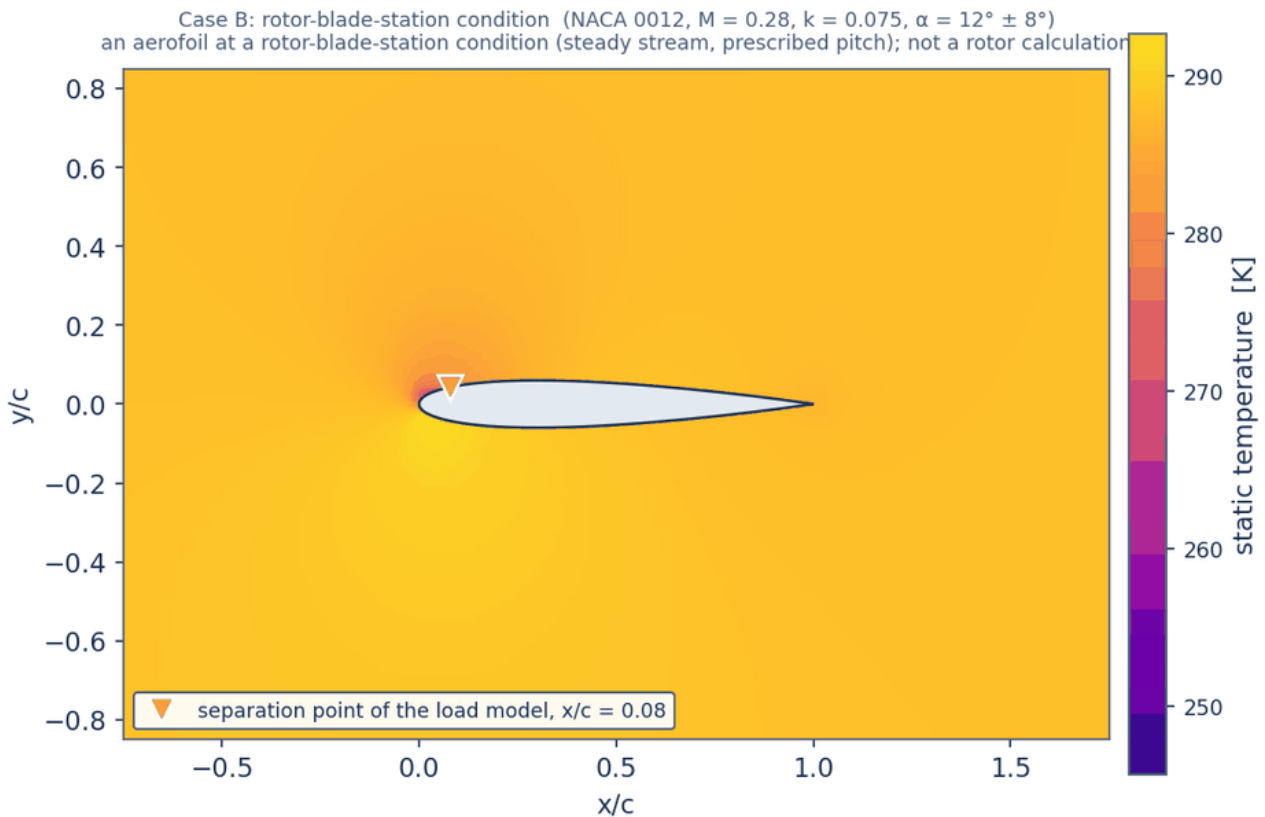
Static temperature at greatest vortex lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_Tstatic_B_application_dsv_a18.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at fall (downstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 245.65 to 292.67. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 0.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

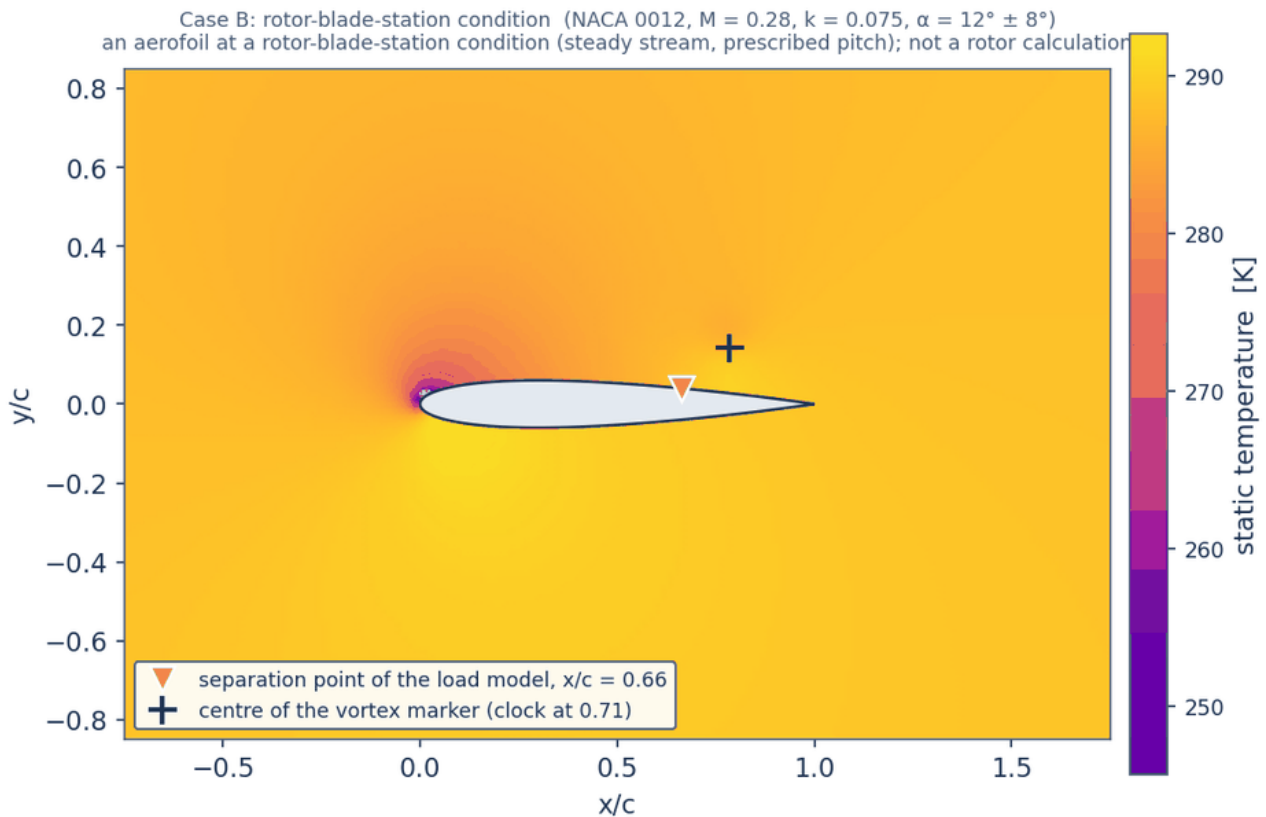
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.08$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Tstatic_B_application_fall_a16.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 0.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at greatest lift, $\alpha = 17.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 245.65 to 292.67. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 2.3 % of the surface and 0.0006 c^2 of the stored field (0.008 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.66$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

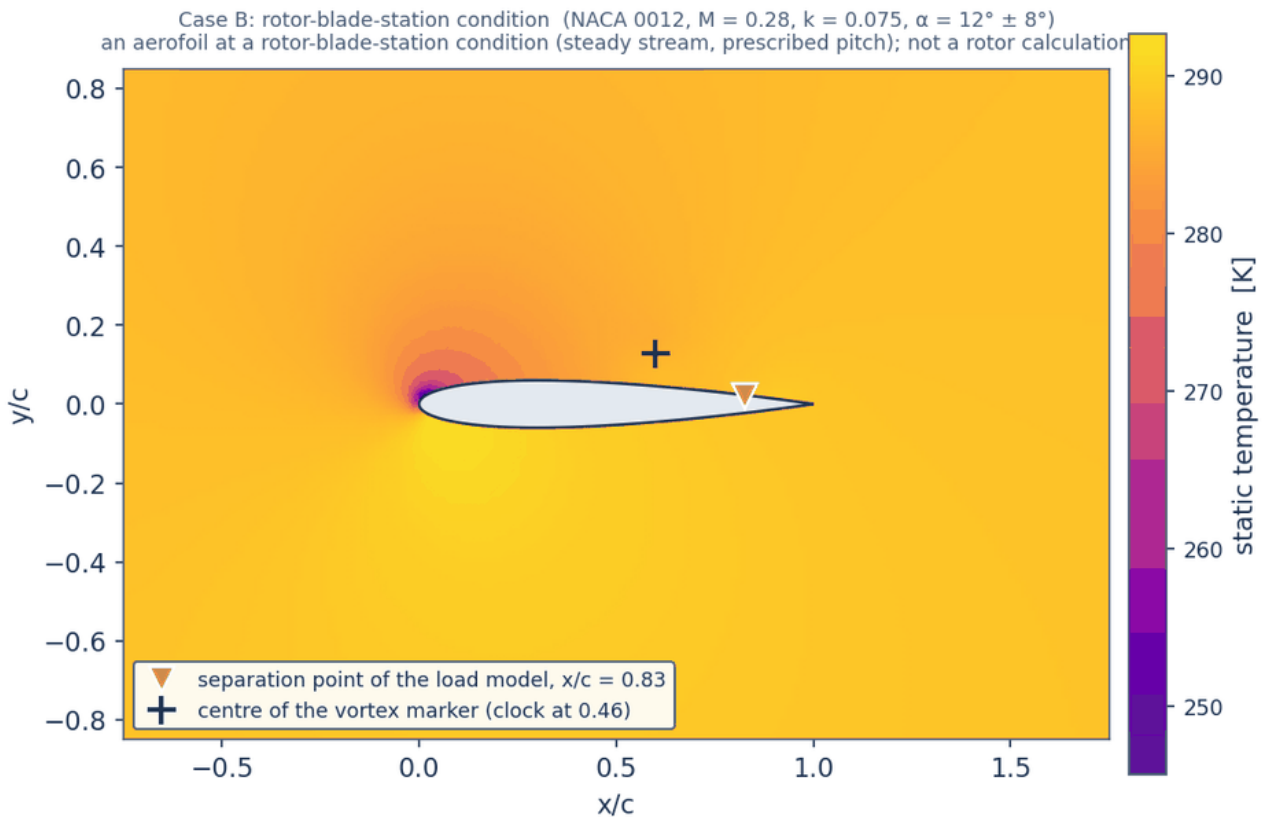
06_postprocessing/plots/contour_Tstatic_B_application_peak_a17.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name).

Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.3 % of the surface and 0.008 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow

reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Static temperature at rise (upstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section and outside the hatched region over the 4 instants of this case: 245.65 to 292.67. Within one grid spacing of the outline the colour is carried over from the nearest node outside. Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. Critical C_p -8.06. At this instant the region is 2.0 % of the surface and 0.0000 c^2 of the stored field (0.000 % of it). Temperatures from the corrected pressure by the isentropic relations; recovery factor 0.896 ($Pr^{1/3}$), the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air).

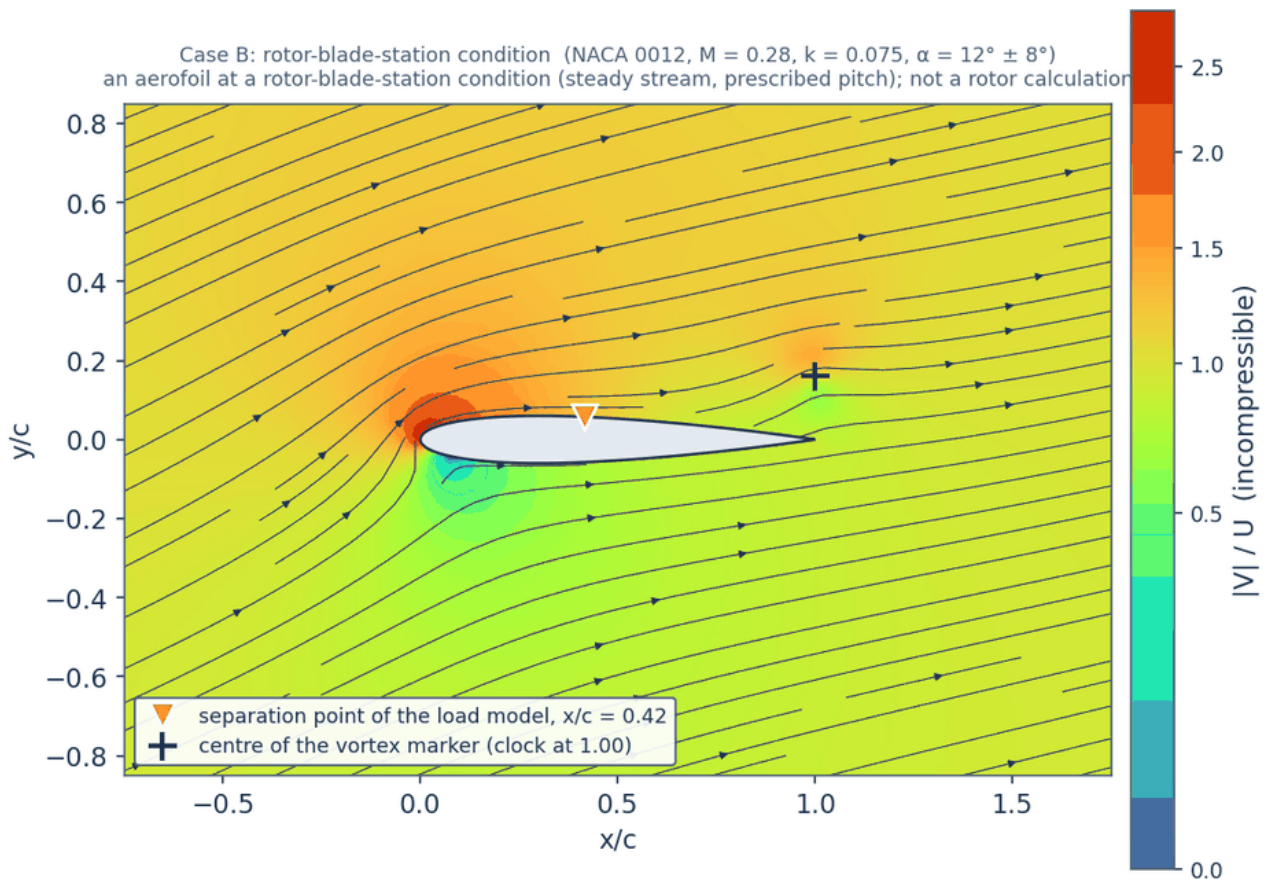
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.83$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_Tstatic_B_application_rise_a16.png

Static temperature, an isentropic estimate from the corrected pressure of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). Hatched: the corrected pressure coefficient is below the critical value (local Mach number above one); the correction does not hold there and the values are not physical. At this instant that region is 2.0 % of the surface and 0.000 % of the stored field. The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

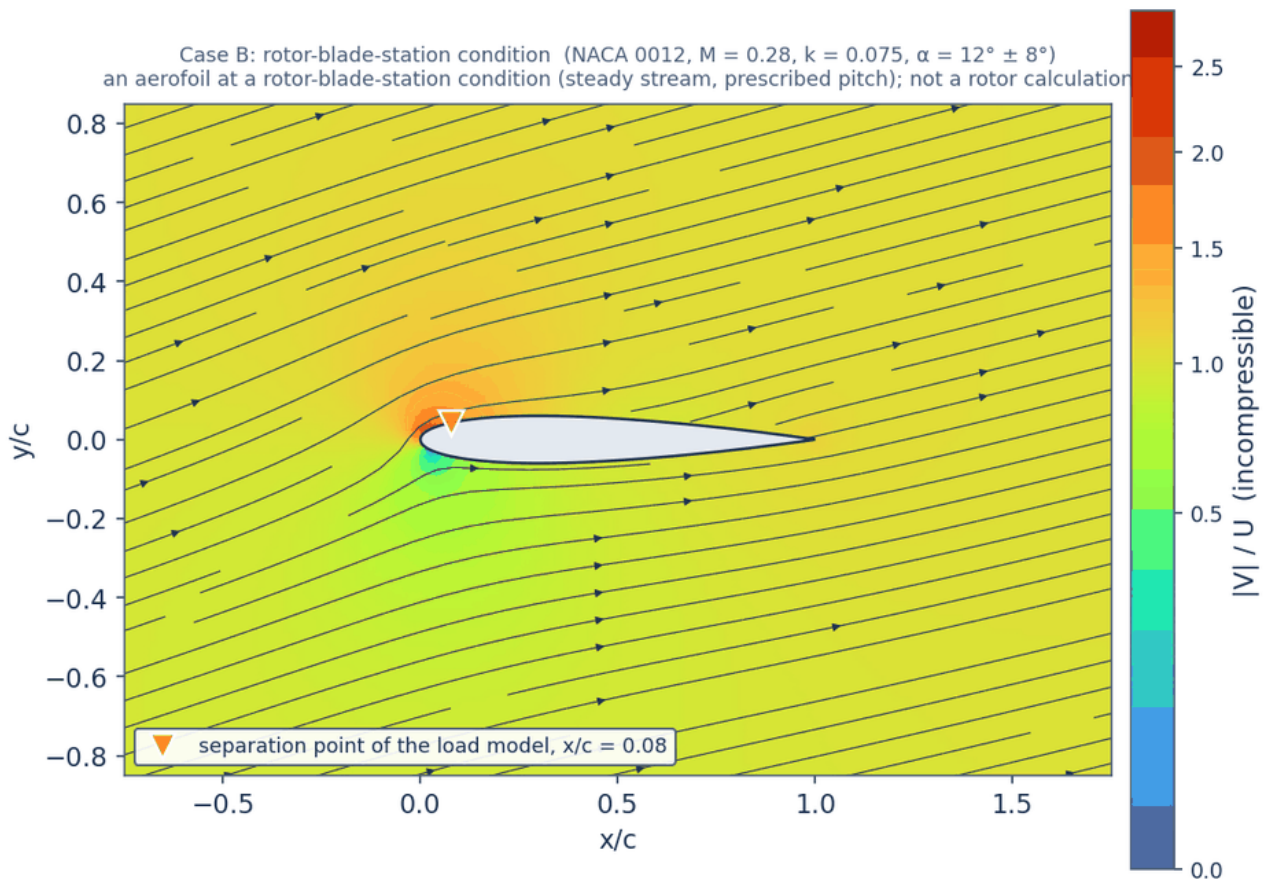
Speed and streamlines at greatest vortex lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_speed_stream_B_application_dsv_a18.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at fall (downstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.86.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

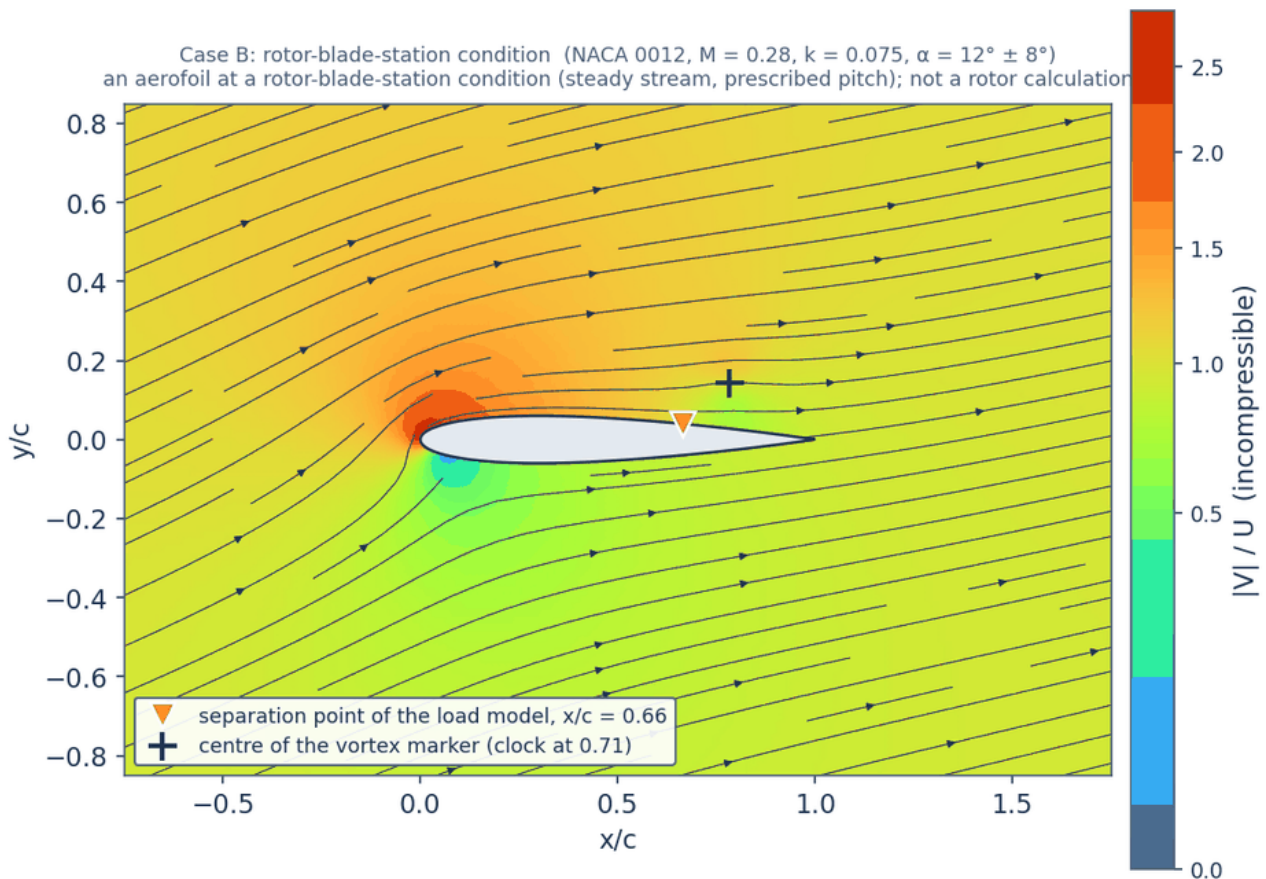
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.08$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_B_application_fall_a16.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at greatest lift, $\alpha = 17.0^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.86.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

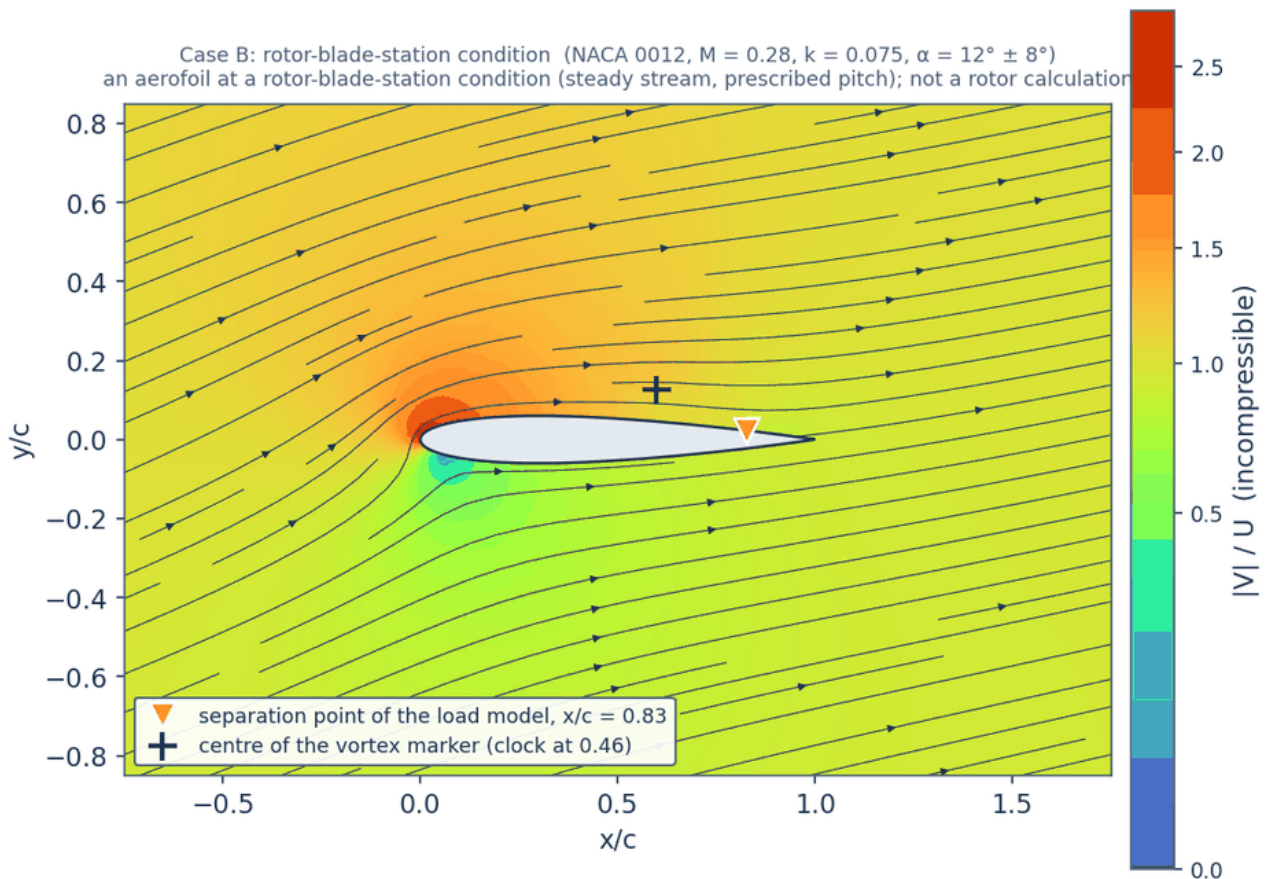
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.66$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_B_application_peak_a17.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Speed and streamlines at rise (upstroke), $\alpha = 16.0^\circ$



Colour scale spans the values at the nodes drawn outside the section over the 4 instants of this case: 0.01 to 2.86.

Within one grid spacing of the outline the colour is carried over from the nearest node outside.

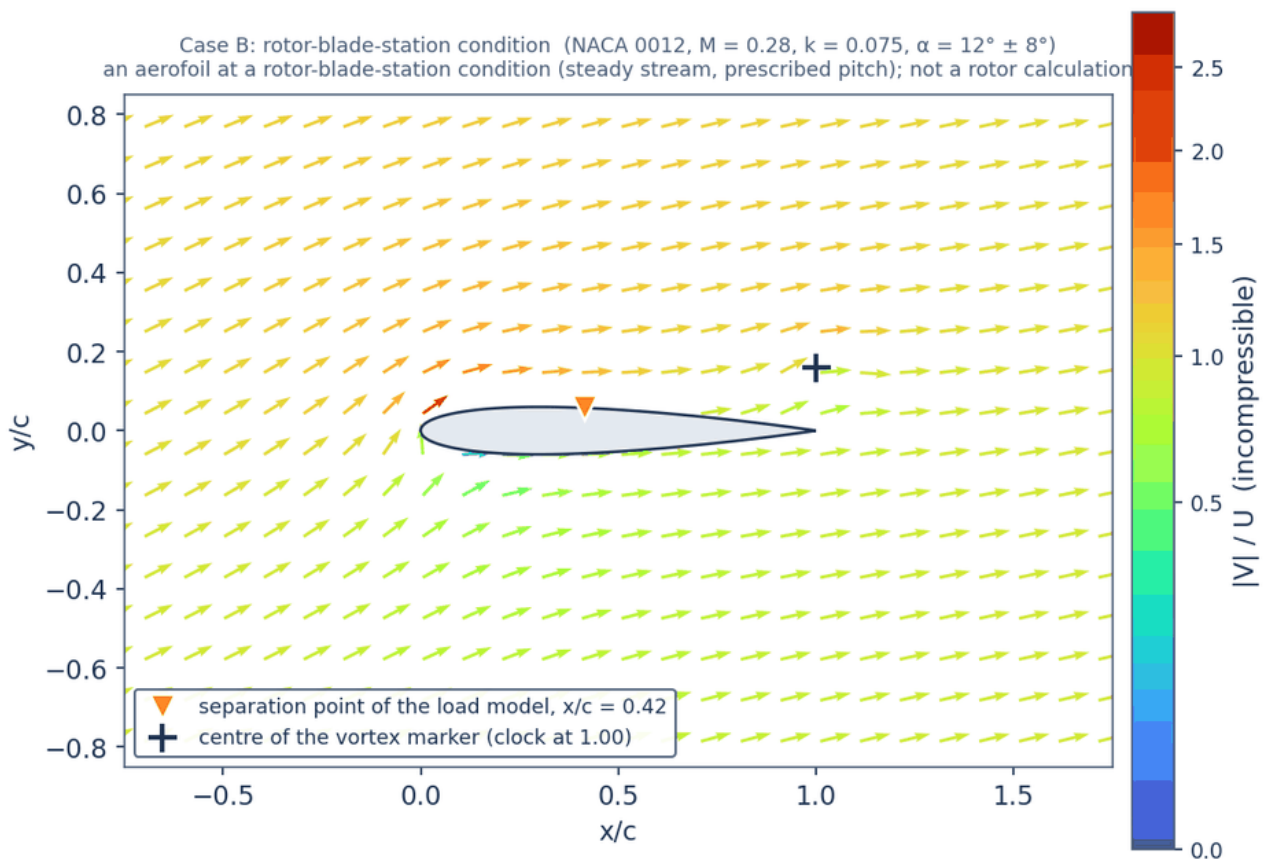
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.83$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_speed_stream_B_application_rise_a16.png

Speed of the incompressible solution and streamlines of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

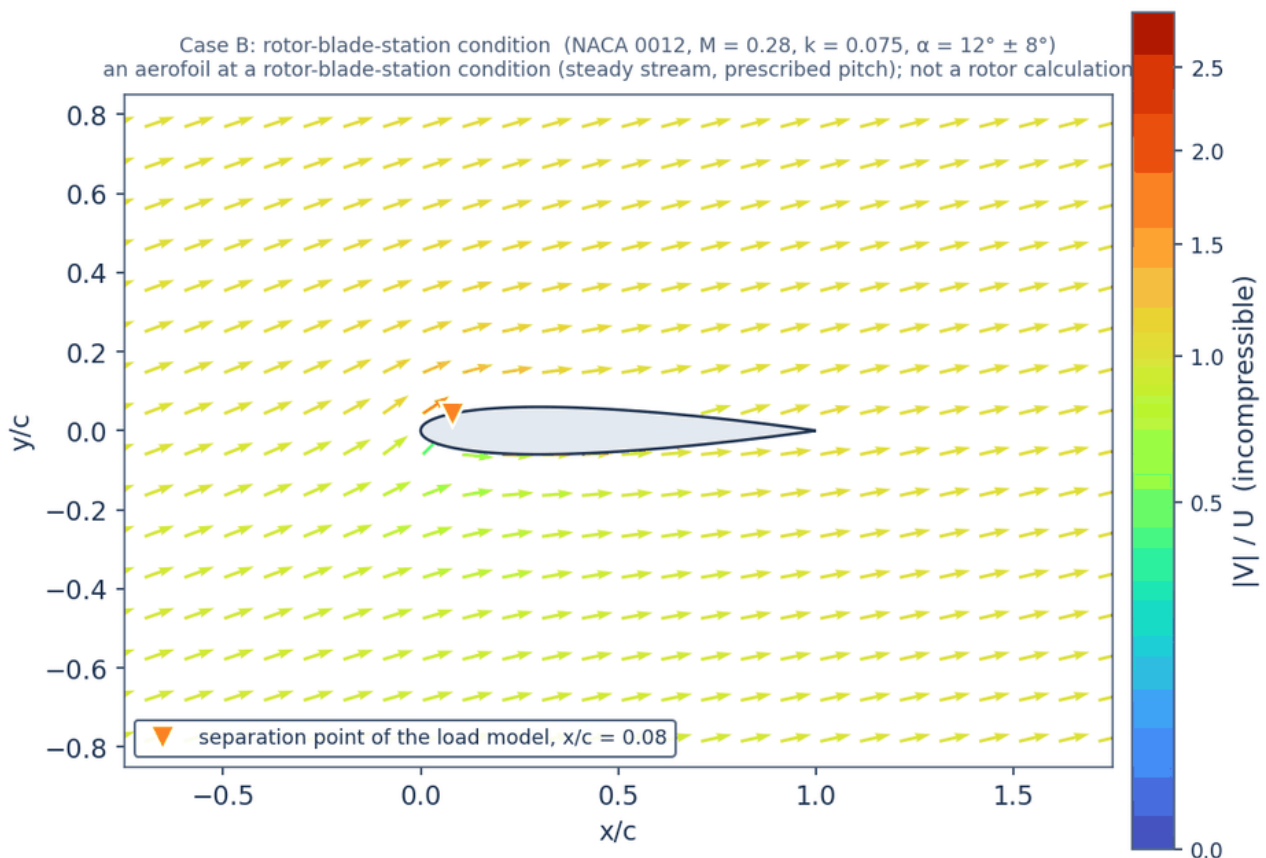
Velocity vectors at greatest vortex lift, $\alpha = 18.1^\circ$



06_postprocessing/plots/contour_vectors_B_application_dsv_a18.png

Velocity vectors of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

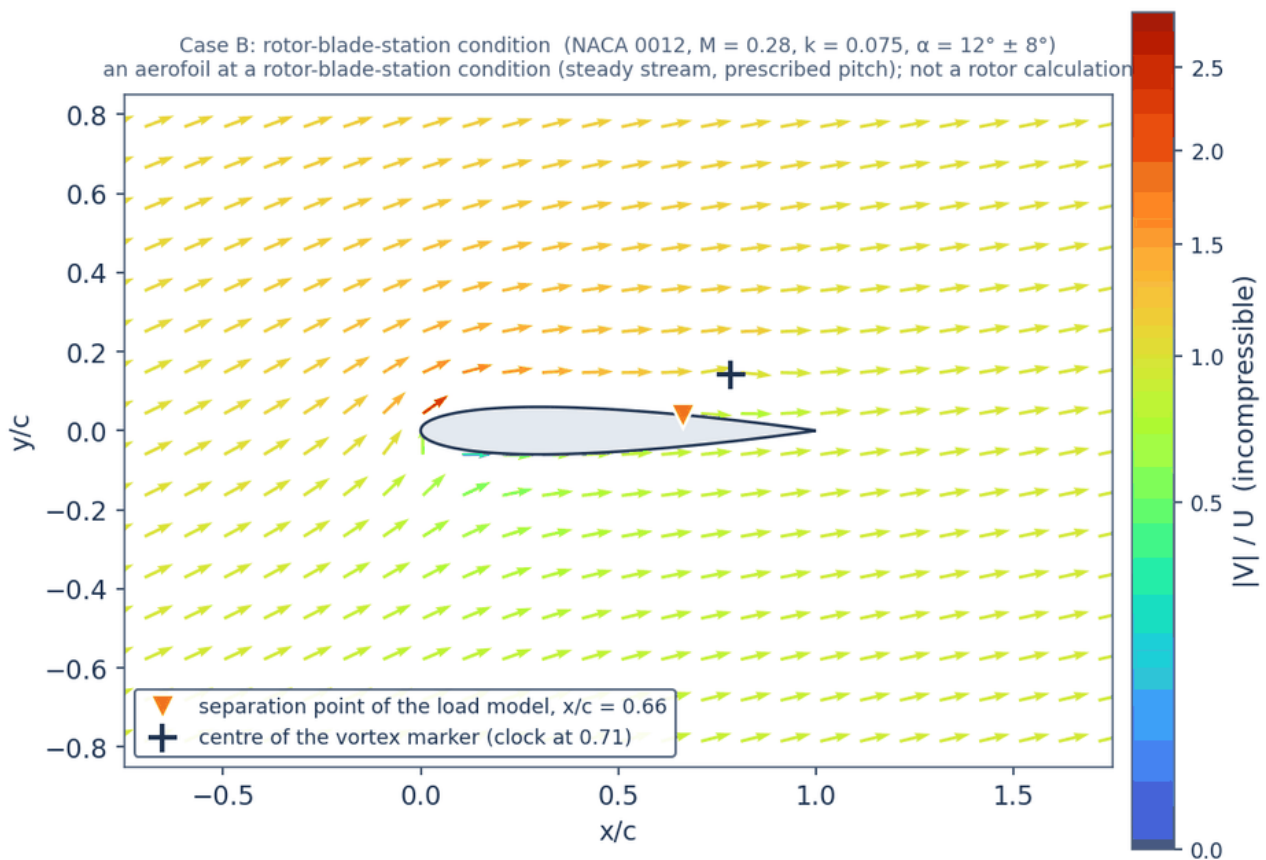
Velocity vectors at fall (downstroke), $\alpha = 16.0^\circ$



06_postprocessing/plots/contour_vectors_B_application_fall_a16.png

Velocity vectors of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

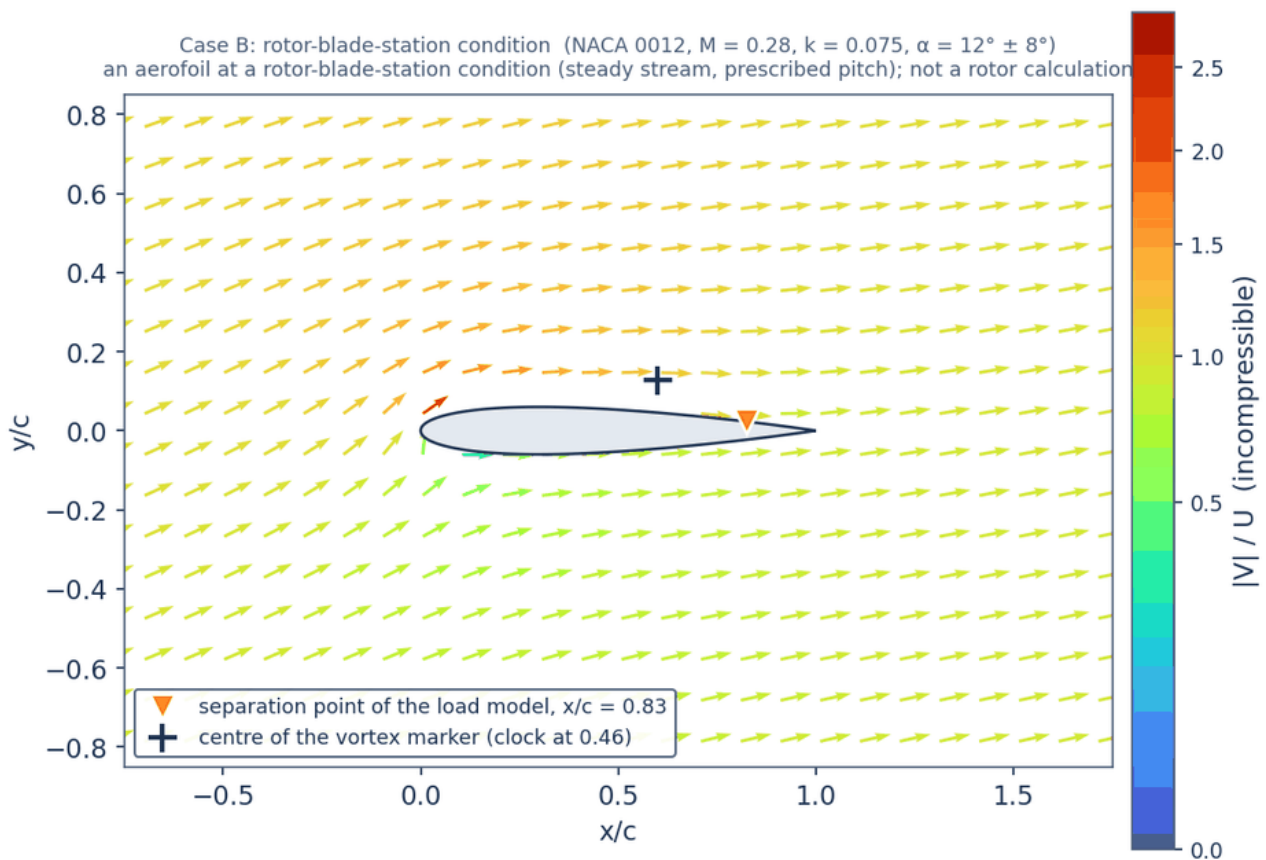
Velocity vectors at greatest lift, $\alpha = 17.0^\circ$



06_postprocessing/plots/contour_vectors_B_application_peak_a17.png

Velocity vectors of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

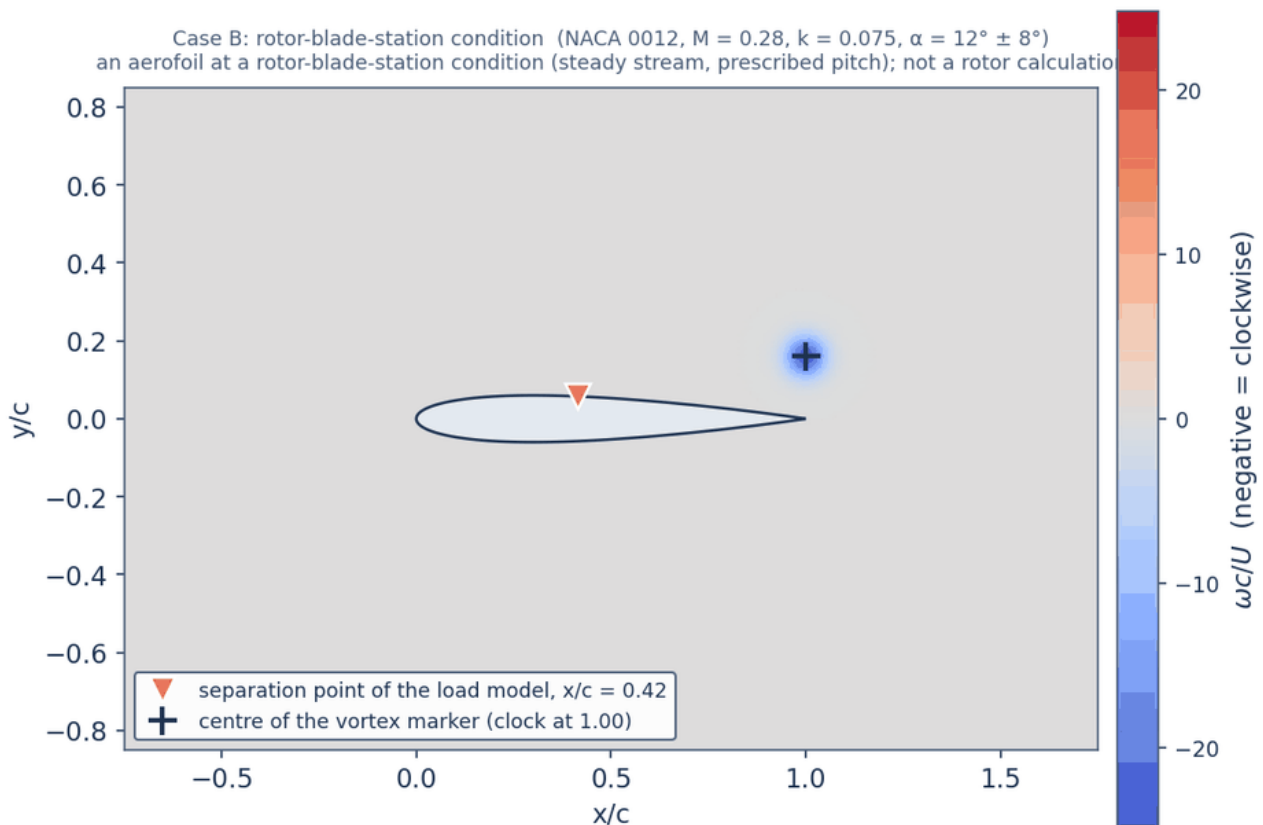
Velocity vectors at rise (upstroke), $\alpha = 16.0^\circ$



06_postprocessing/plots/contour_vectors_B_application_rise_a16.png

Velocity vectors of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at greatest vortex lift, $\alpha = 18.1^\circ$



Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|\omega c/U| = 24.8$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

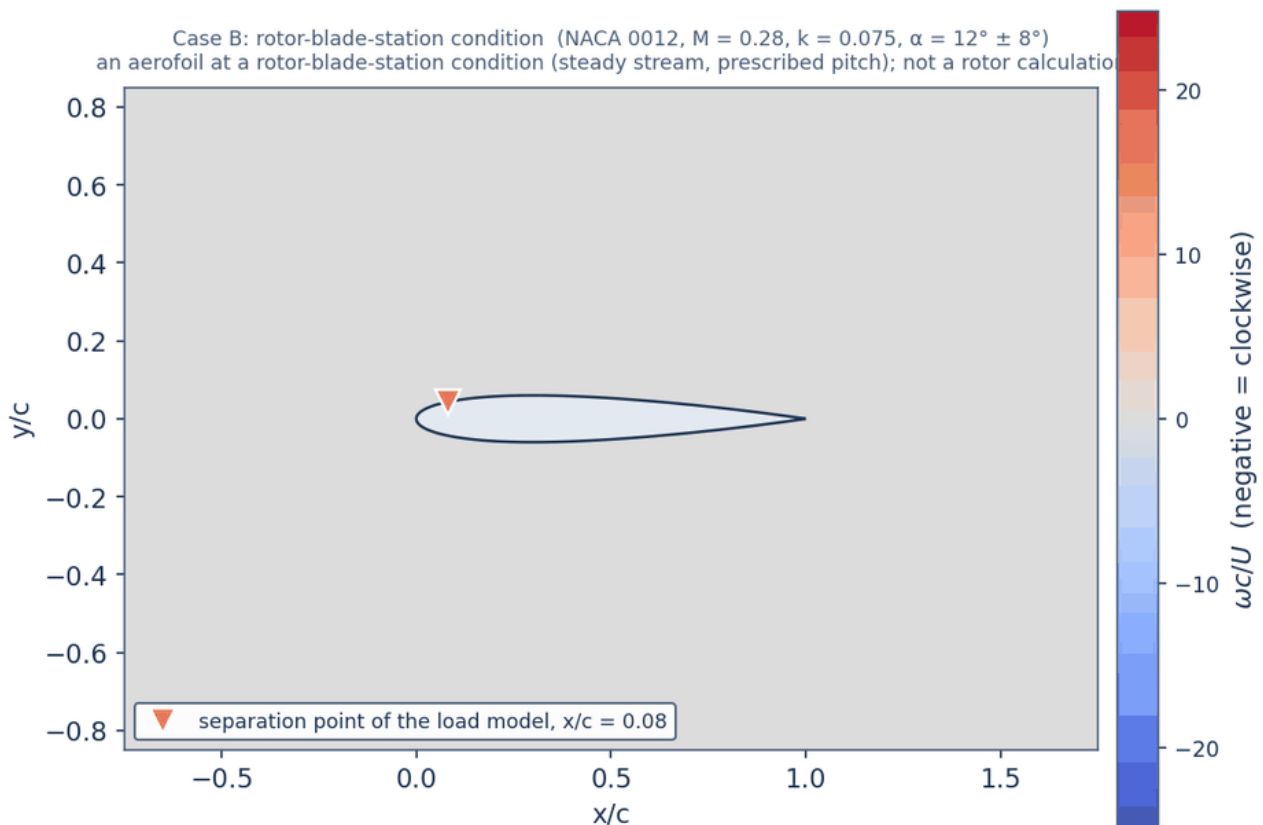
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.42$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_B_application_dsv_a18.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest vortex normal force (incidence 18° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at fall (downstroke), $\alpha = 16.0^\circ$



Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|\omega c/U| = 24.8$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

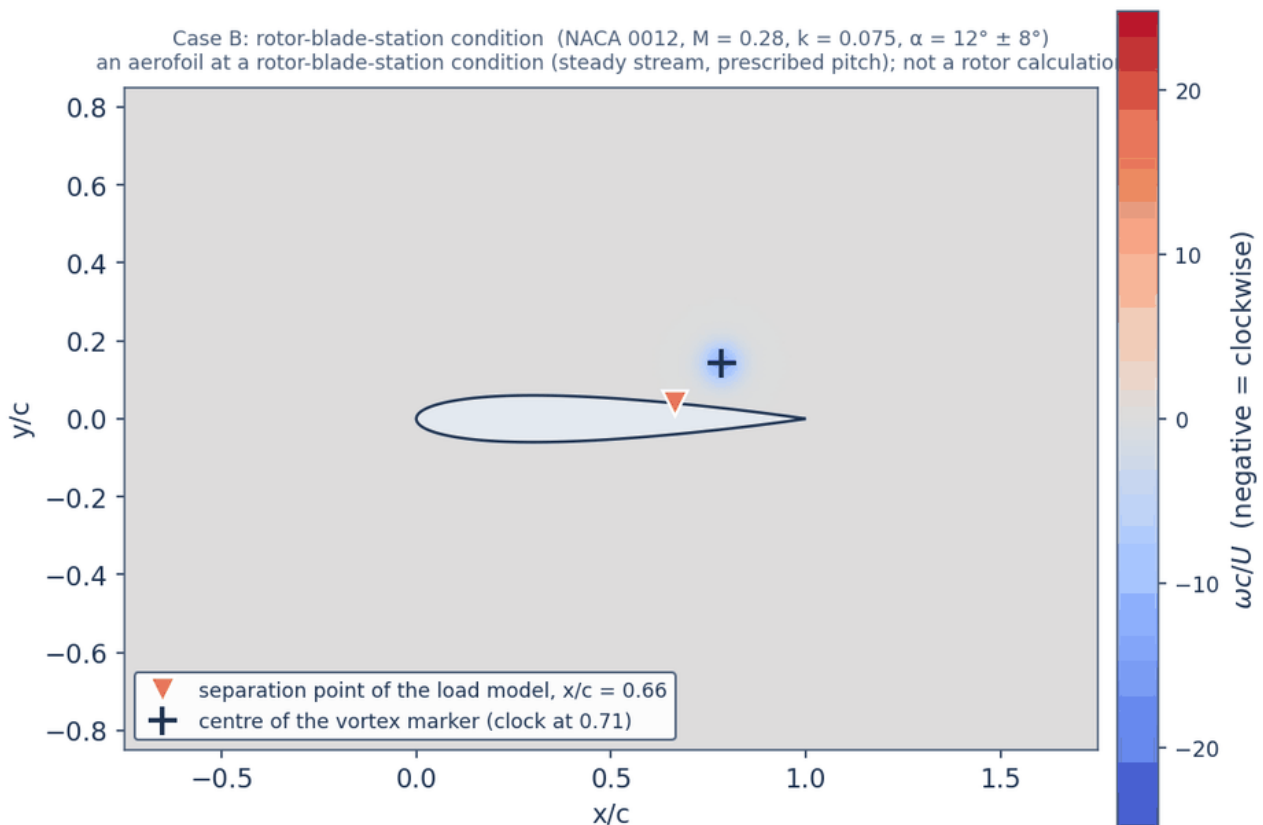
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.08$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_B_application_fall_a16.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case B: rotor-blade-station condition; downstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at greatest lift, $\alpha = 17.0^\circ$



Colour scale: linear, symmetric, set by the largest core vorticity of the vortex marker over the 4 instants of this case ($|\omega c/U| = 24.8$, from its circulation and core radius). The vorticity is zero outside that core: the bound sheet lies on the surface itself and the transpiration sheds none; a map of the singularities, with no boundary-layer or wake vorticity in it.

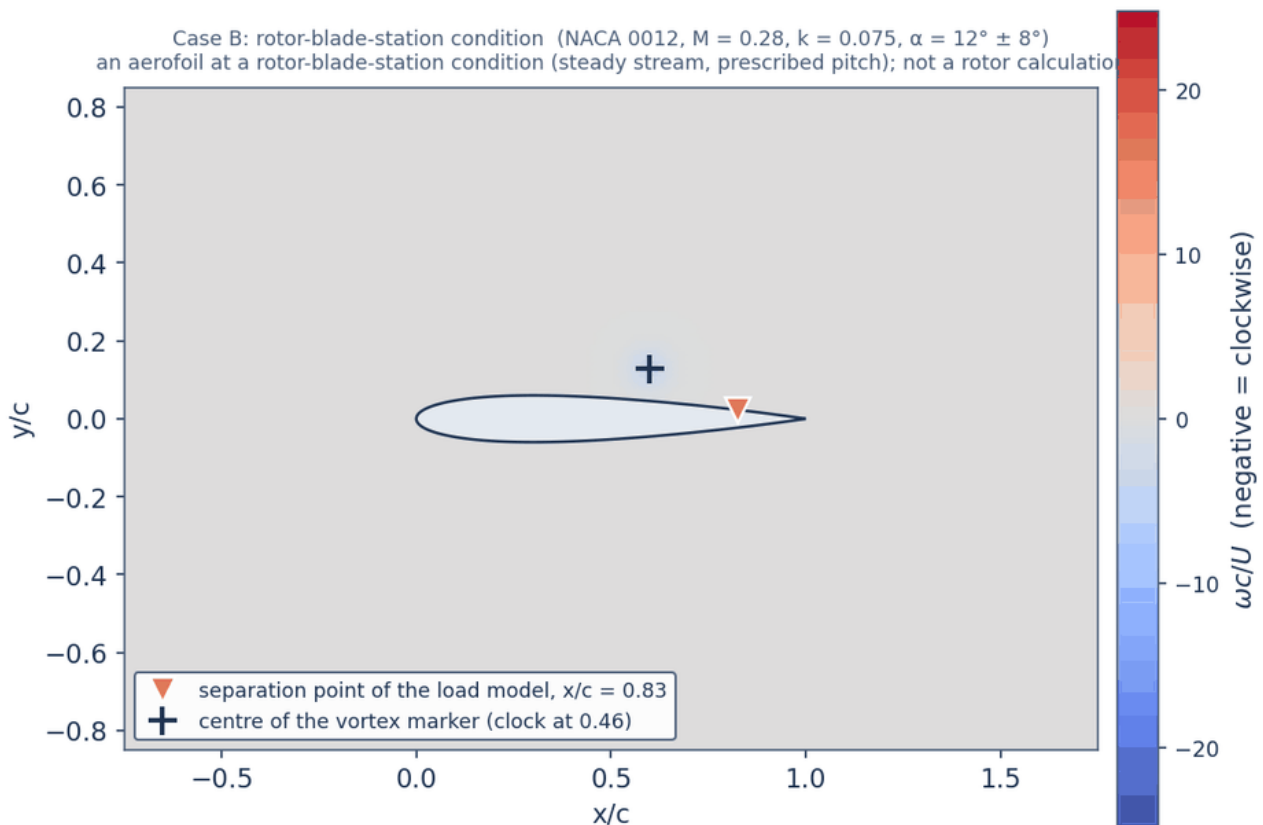
The separated region is not drawn: the triangle marks where the load model puts the separation point ($f = 0.66$), and the flow shown aft of it is not separated flow. Axes fixed to the section; the stream comes from below at the incidence shown.

Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

06_postprocessing/plots/contour_vorticity_B_application_peak_a17.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case B: rotor-blade-station condition; instant of greatest lift (incidence 17° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Vorticity at rise (upstroke), $\alpha = 16.0^\circ$

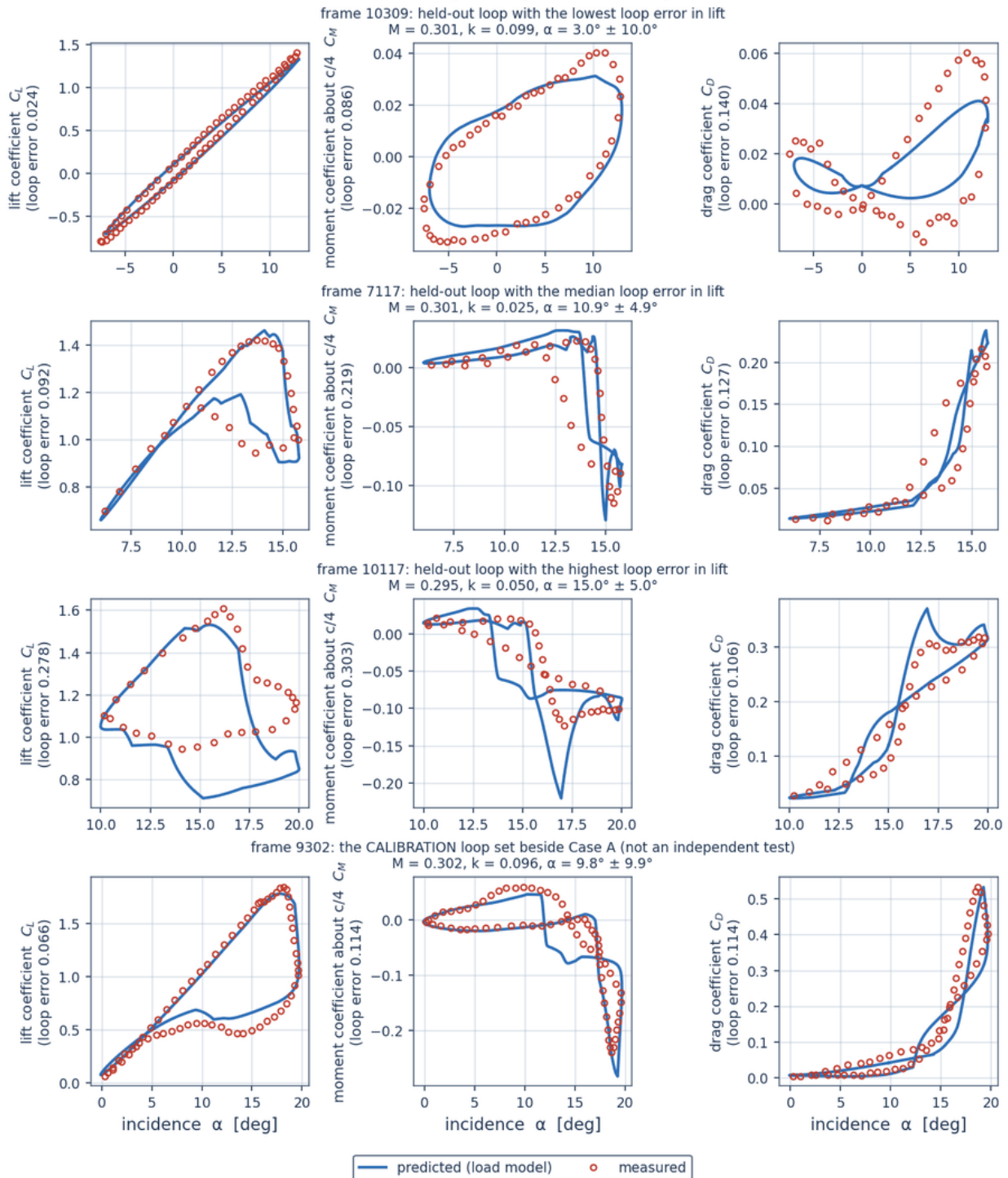


06_postprocessing/plots/contour_vorticity_B_application_rise_a16.png

Vorticity (zero outside the core of the vortex marker) of the reconstructed field. Case B: rotor-blade-station condition; upstroke, mean incidence plus half the amplitude (incidence 16° to the nearest degree, from the file name). The separated region is not drawn; the model's separation point is marked. Axes fixed to the section; the stream comes from below at the incidence shown. Standing note: Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.

Comparison with measurement (06_postprocessing/validation)

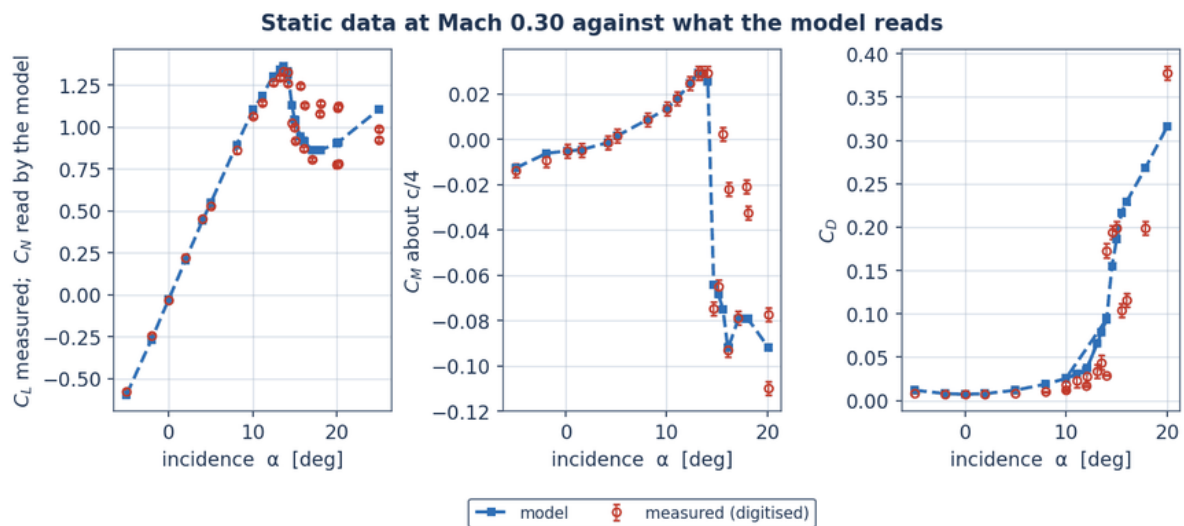
Measured against predicted loops



Loops chosen by rule from results/validation_table.csv: lowest, median and highest loop error in lift of the held-out loops inside the static Mach range, and the calibration loop of Case A. Each is marched at its own measured conditions with the calibrated constants. The loop error is the RMS difference over the measured range, as in that file.

06_postprocessing/validation/fig_validation_loops.png

Measured against predicted loops of lift, moment and drag for loops chosen by rule from results/validation_table.csv: lowest, median and highest loop error in lift of the held-out loops, and the calibration loop of Case A. The same numbers as results/; not a second assessment.

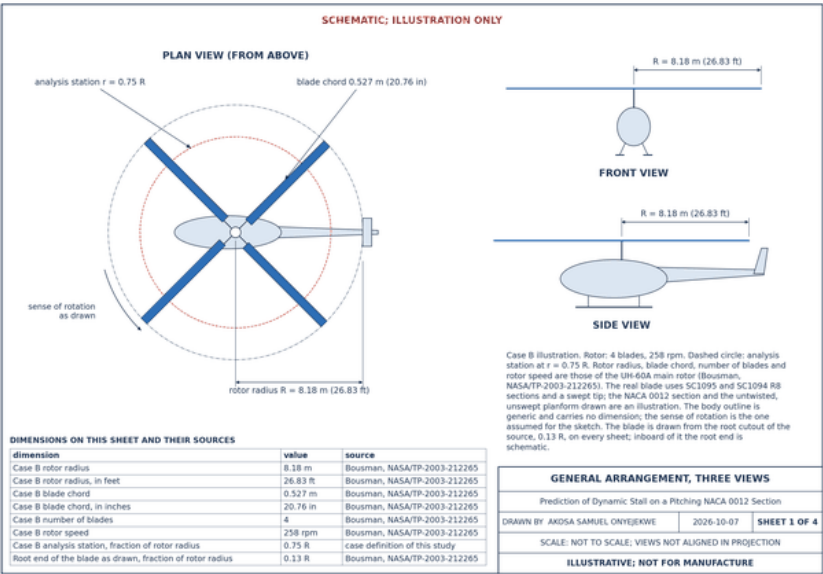


Left: the model reads a static normal force from quasi-steady sweeps of the experiment, which are not corrected for the tunnel walls; the measured lift is from the report's corrected static figure. Middle: static moment. Right: the model has no static drag input; the line is its drag in a slow sweep, which is pressure drag where the measured points are total drag from a wake survey. Sources of the points: data/static_naca0012_M030_*.csv.

06_postprocessing/validation/fig_validation_static.png

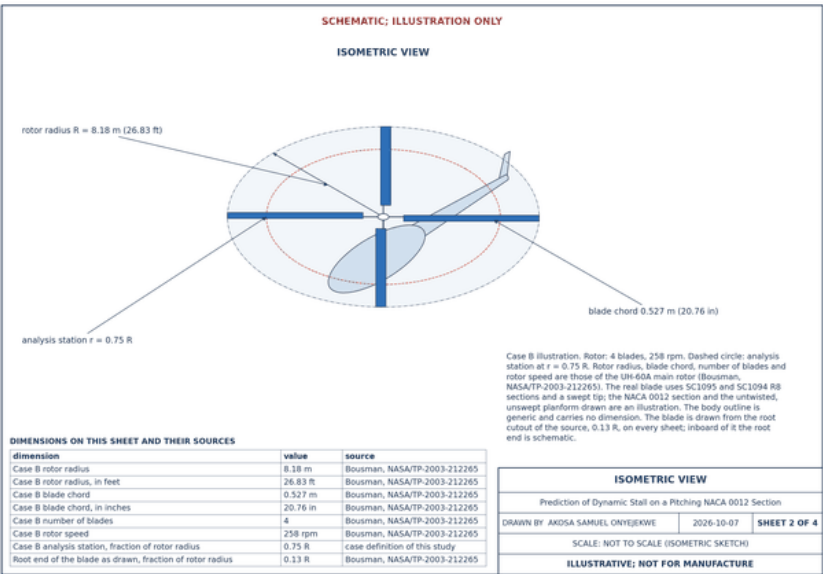
Static data at Mach 0.30 beside what the model reads: its static normal force (set beside the measured lift), its static moment, and its drag in a slow sweep. The same data as results/ and data/; not a second assessment.

Drawings (08_engineering_drawings)



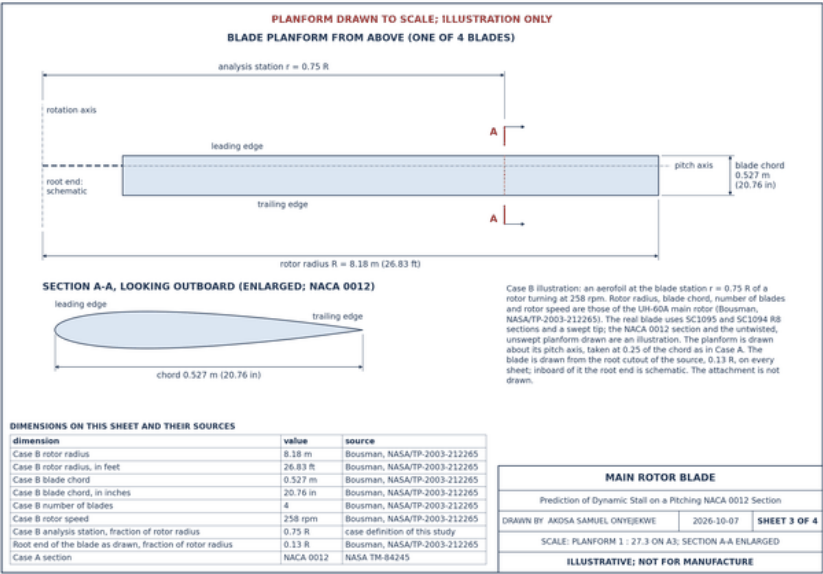
08_engineering_drawings/sheet1_general_arrangement_3view.png

Drawing sheet 1: general arrangement three views. Illustrative; every dimension from the table with its sources; not for manufacture.



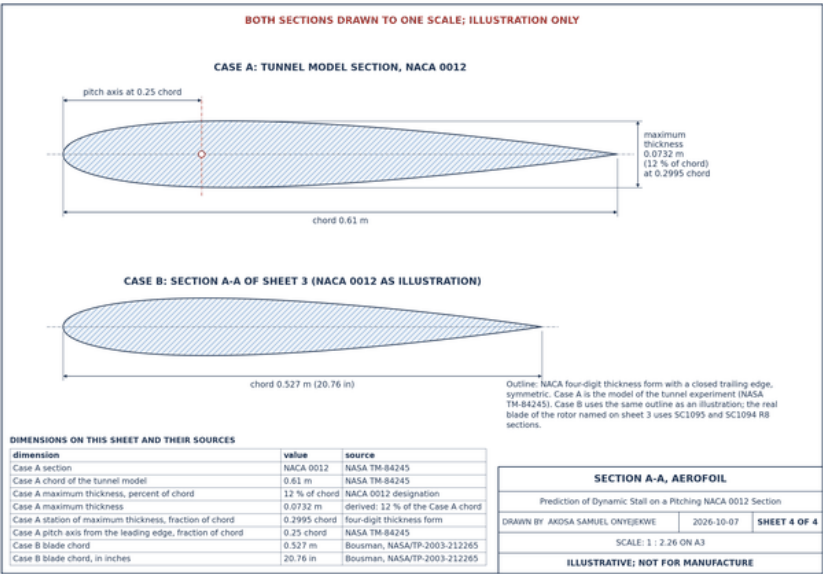
08_engineering_drawings/sheet2_isometric.png

Drawing sheet 2: isometric. Illustrative; every dimension from the table with its sources; not for manufacture.



08_engineering_drawings/sheet3_main_rotor_blade.png

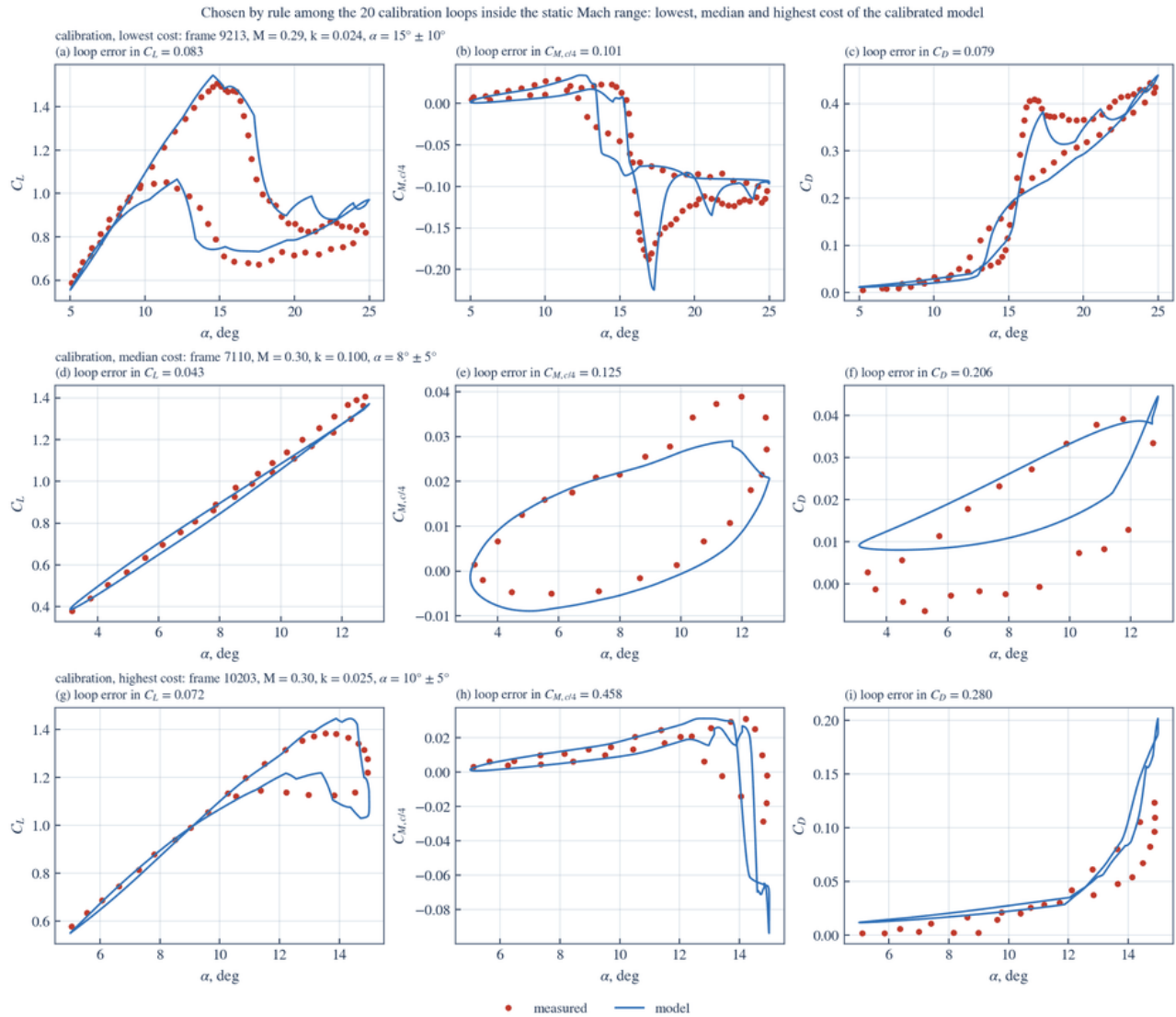
Drawing sheet 3: main rotor blade. Illustrative; every dimension from the table with its sources; not for manufacture.



08_engineering_drawings/sheet4_section_AA_airfoil.png

Drawing sheet 4: section A-A airfoil. Illustrative; every dimension from the table with its sources; not for manufacture.

Calibration and accuracy of the load model (results/figures)



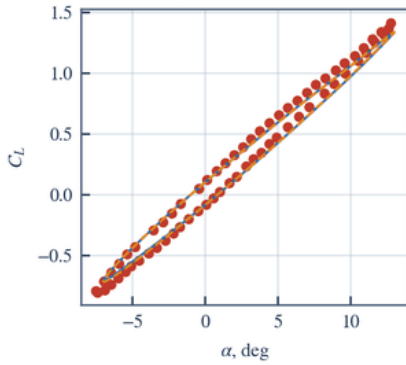
results/figures/fig04_calibration_loops.png

Three calibration loops with their RMS errors. The secondary bumps and slope breaks in the predicted loops come from repeated shedding and the switching of time constants; the measured loops have no counterpart to them.

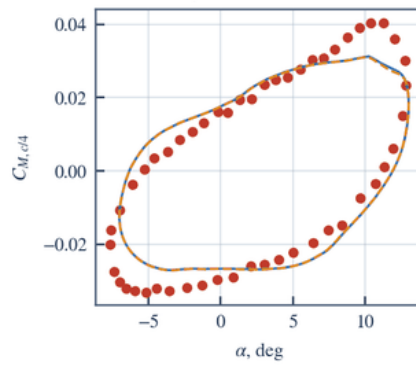
Chosen by rule among the 45 primary held-out loops: lowest, median and highest lift error of the model with tabulated separation

held out, lowest lift error: frame 10309, $M = 0.30$, $k = 0.099$, $\alpha = 3^\circ \pm 10^\circ$

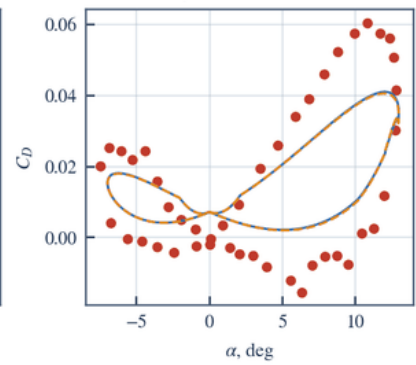
(a) loop error in $C_L = 0.024$
curves differ by at most 0.012



(b) loop error in $C_{M,c/4} = 0.086$
curves differ by at most 0.001

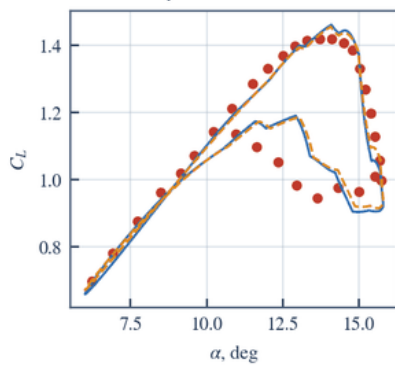


(c) loop error in $C_D = 0.140$
curves differ by at most 0.001

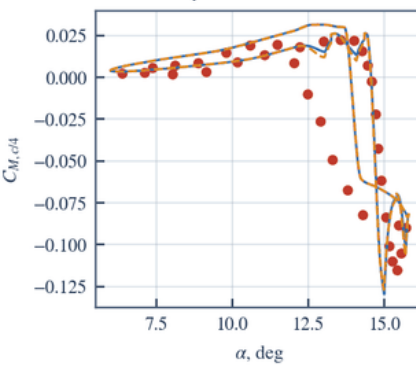


held out, median lift error: frame 7117, $M = 0.30$, $k = 0.025$, $\alpha = 11^\circ \pm 5^\circ$

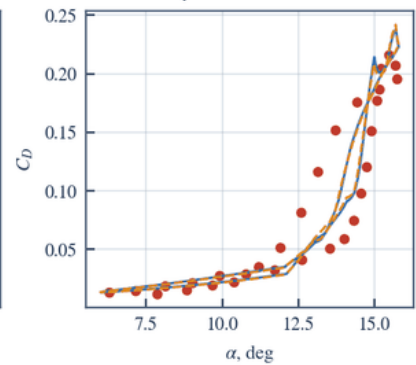
(d) loop error in $C_L = 0.092$
curves differ by at most 0.041



(e) loop error in $C_{M,c/4} = 0.219$
curves differ by at most 0.007

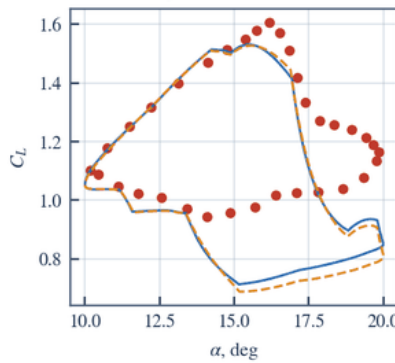


(f) loop error in $C_D = 0.127$
curves differ by at most 0.007

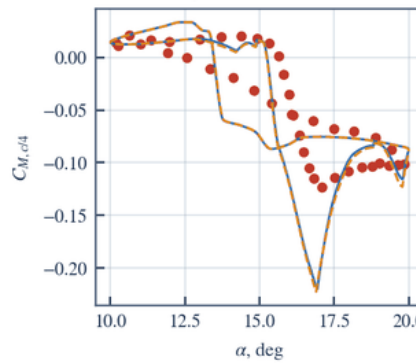


held out, highest lift error: frame 10117, $M = 0.29$, $k = 0.050$, $\alpha = 15^\circ \pm 5^\circ$

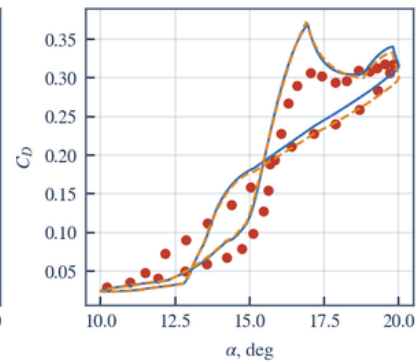
(g) loop error in $C_L = 0.278$
curves differ by at most 0.044



(h) loop error in $C_{M,c/4} = 0.303$
curves differ by at most 0.007



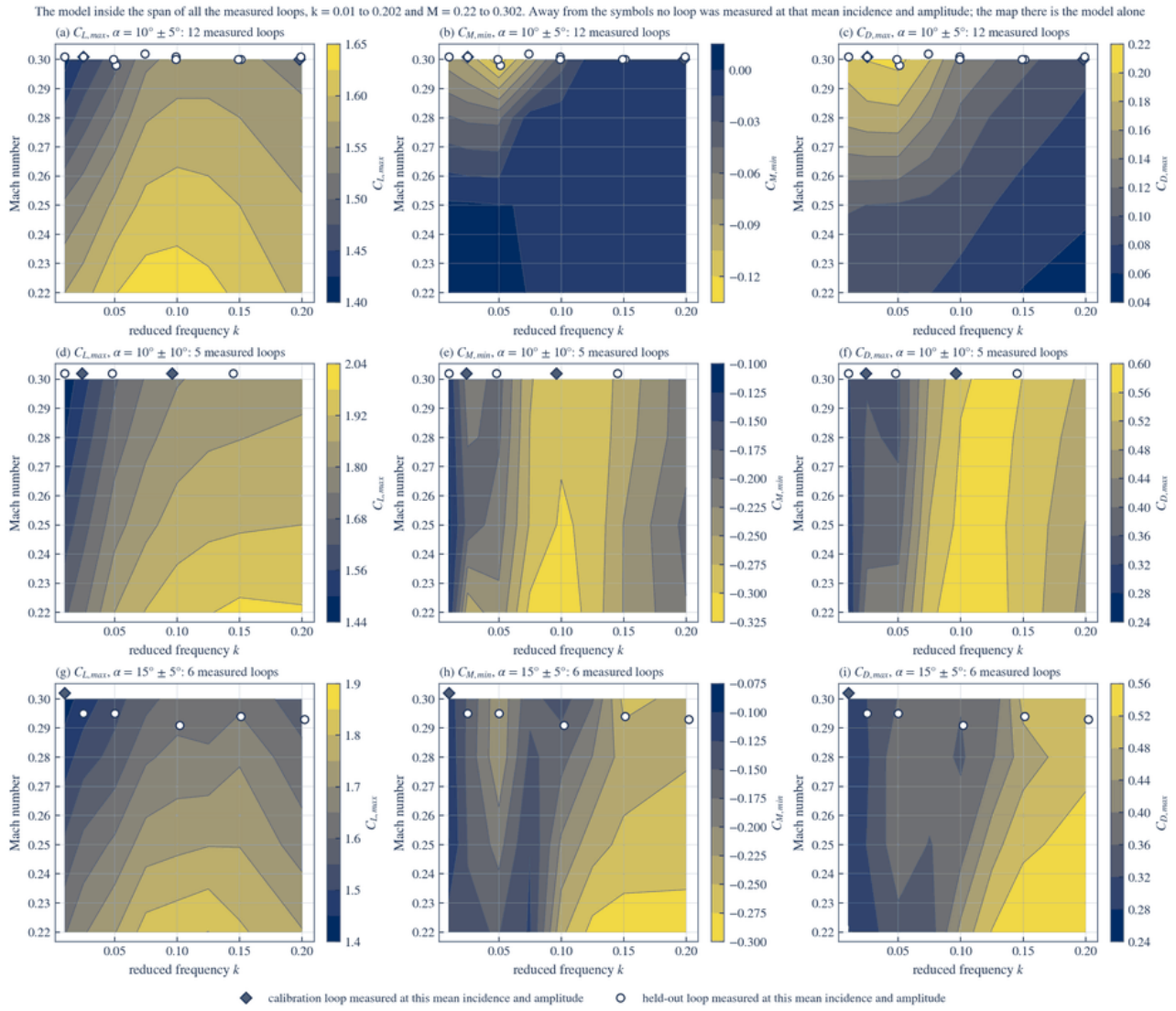
(i) loop error in $C_D = 0.106$
curves differ by at most 0.015



● measured — model, tabulated separation - - - model, fitted separation law

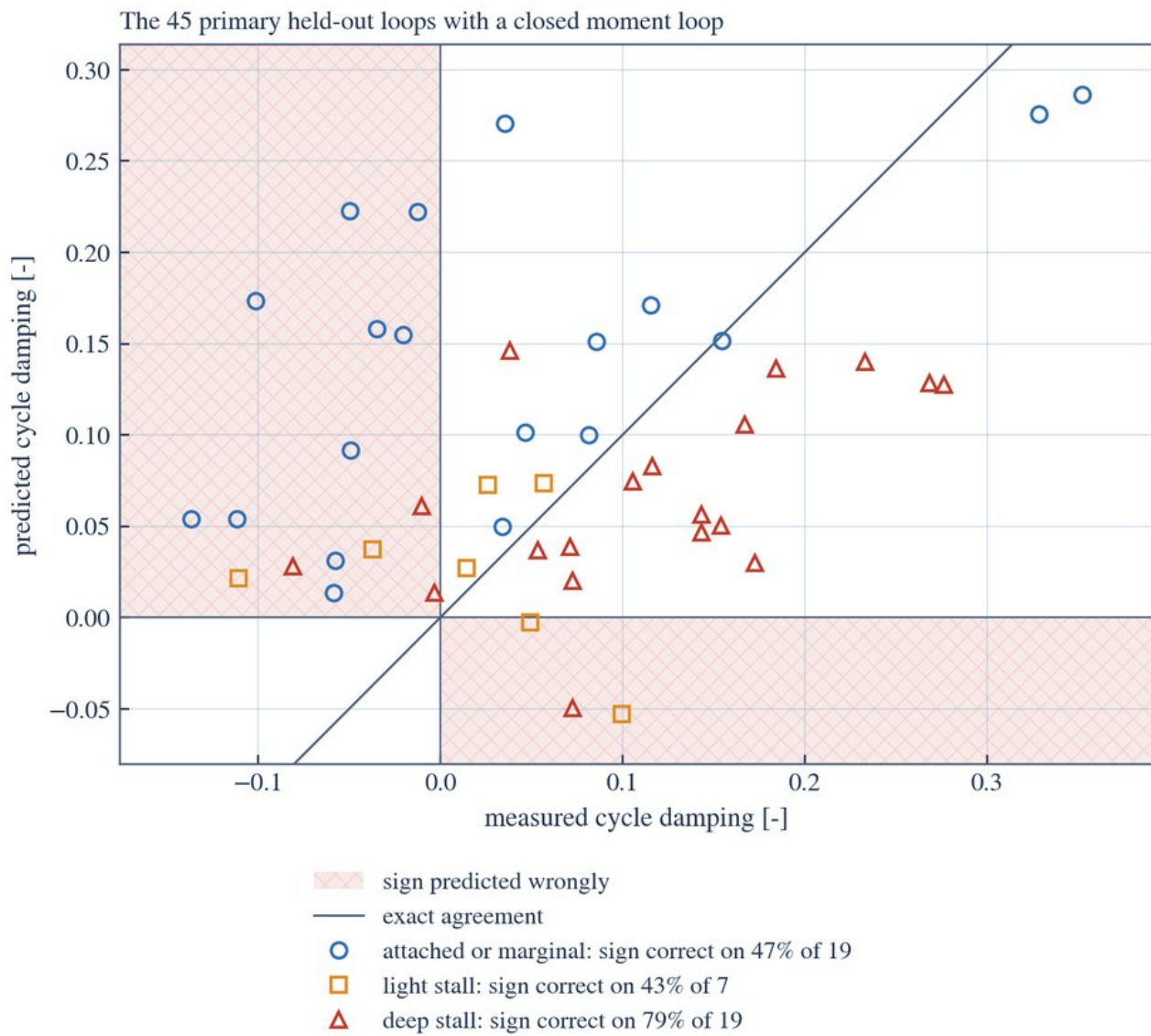
results/figures/fig05_held_out_loops.png

Three held-out loops chosen by rule, both separation laws.



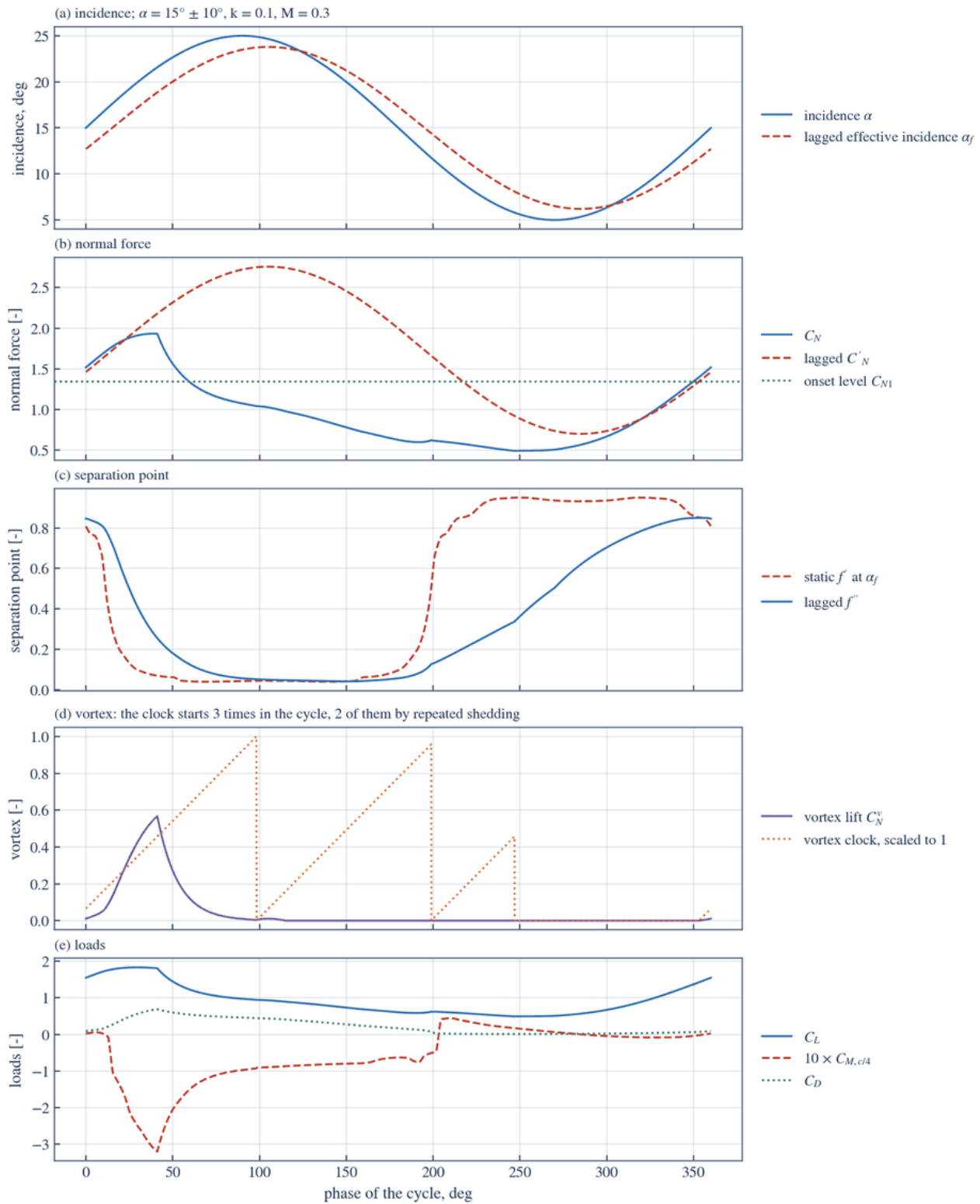
results/figures/fig10_mach_trends.png

Contours of the predicted peaks over reduced frequency and Mach number.



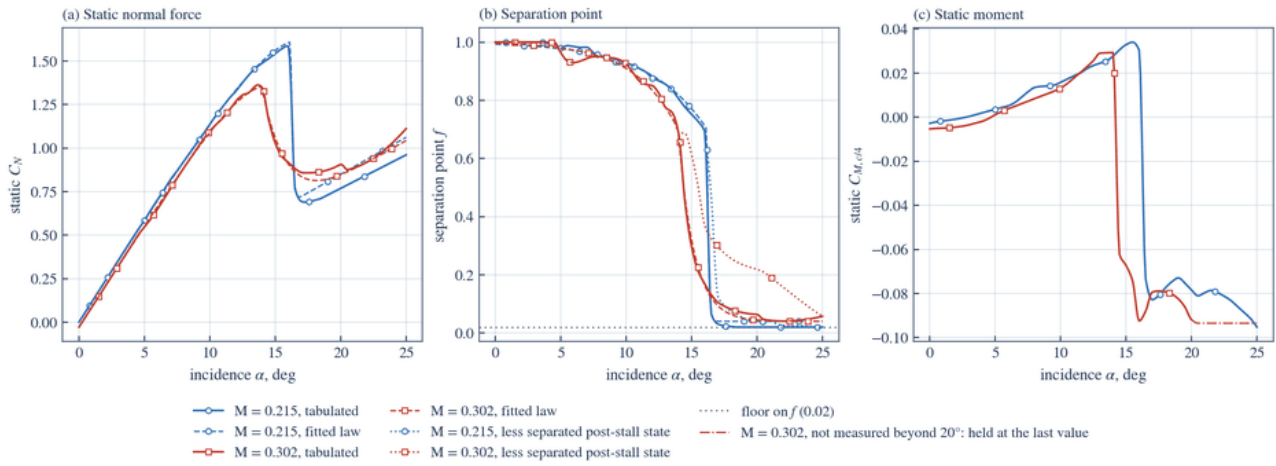
results/figures/fig13_cycle_damping.png

Predicted against measured cycle damping.



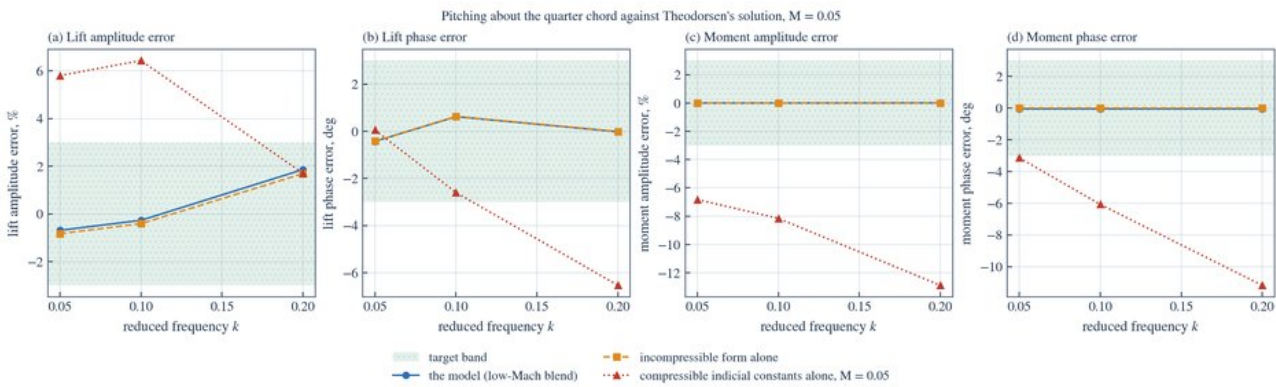
results/figures/fig15_model_states.png

The solver's internal states over one cycle, with the number of times the vortex clock starts in the cycle and how many of those are repeated sheddings.



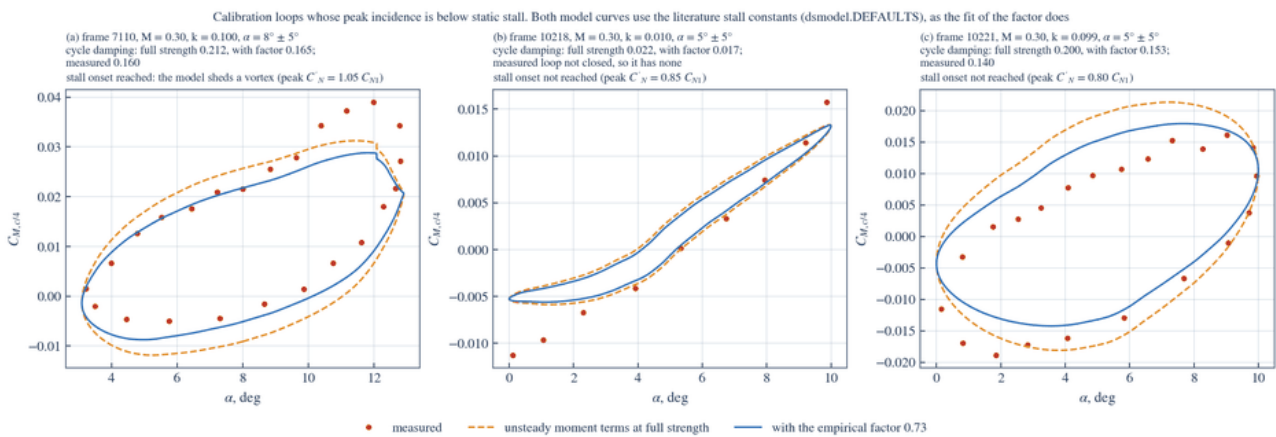
results/figures/fig01_static_inputs.png

Static normal force, separation point and moment at the two Mach stations.



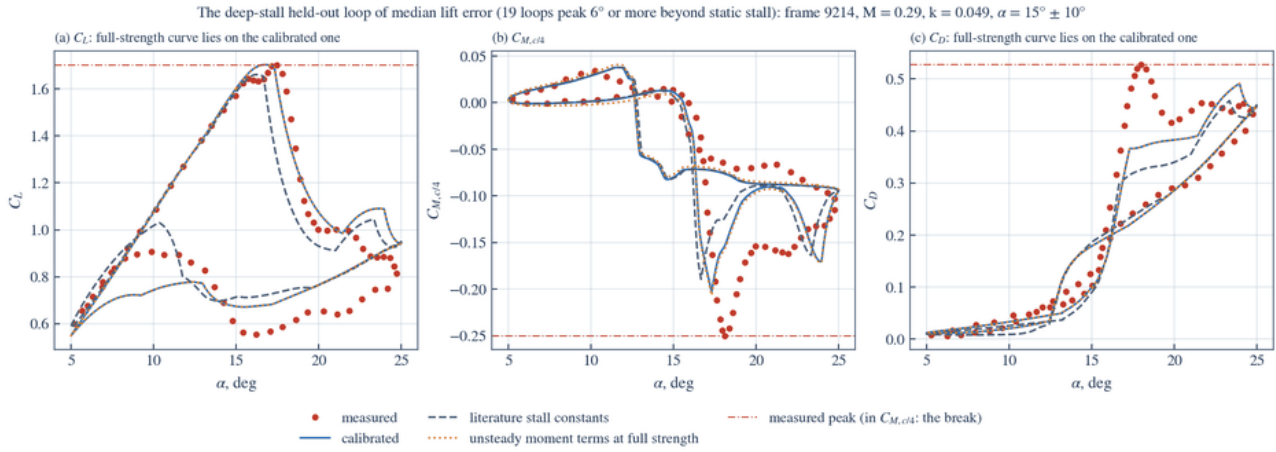
results/figures/fig02_attached_flow.png

Attached-flow loads against Theodorsen's solution.



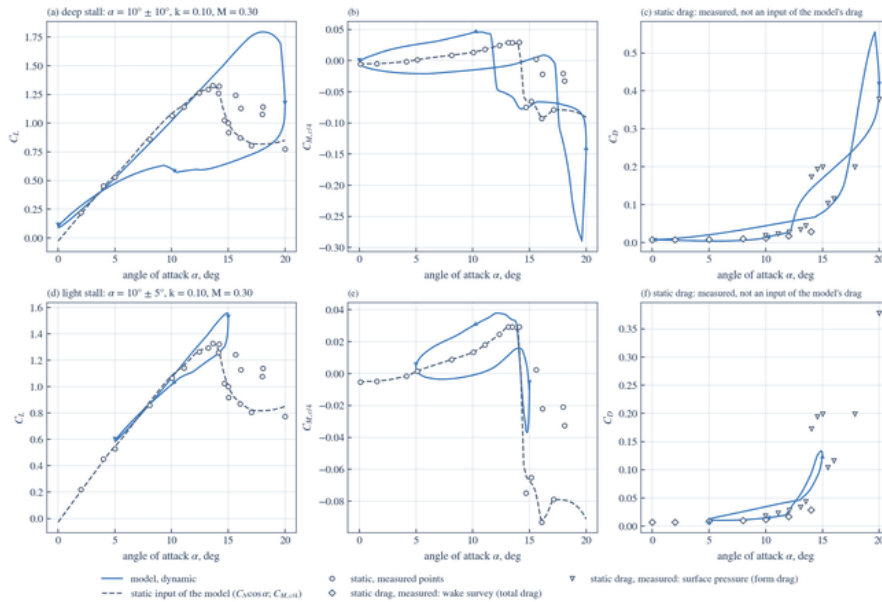
results/figures/fig03_attached_moment.png

Moment loops that peak below static stall: unsteady terms at full strength and with the empirical factor (literature stall constants; frame 7110 reaches the model's onset level, which is the kink near its peak).



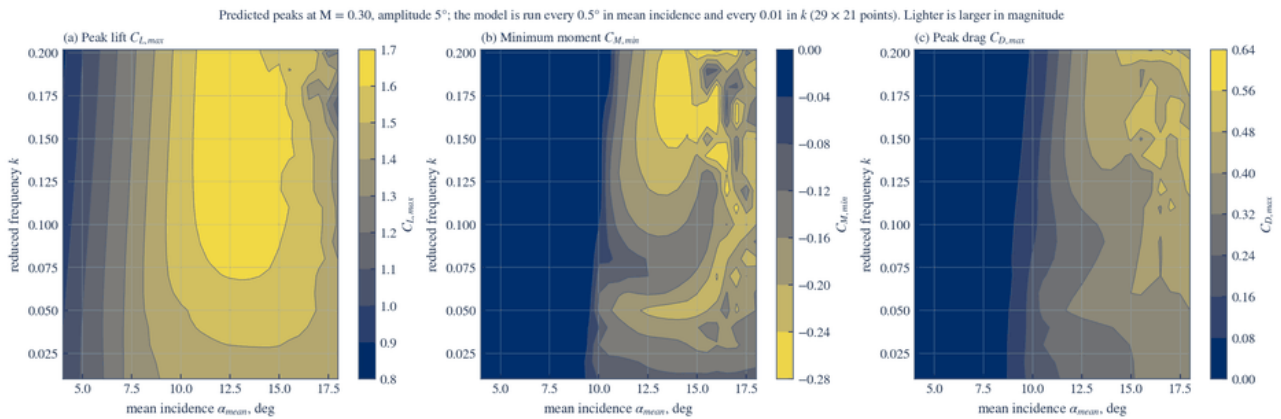
results/figures/fig06_effect_of_constants.png

Effect of the constants on one held-out deep-stall loop.



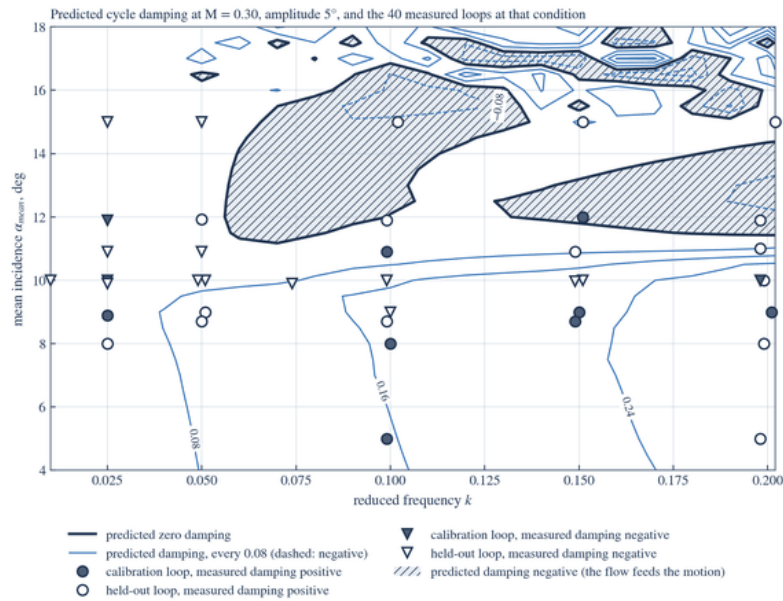
results/figures/fig07_dynamic_and_static.png

Dynamic loops against the static curves.



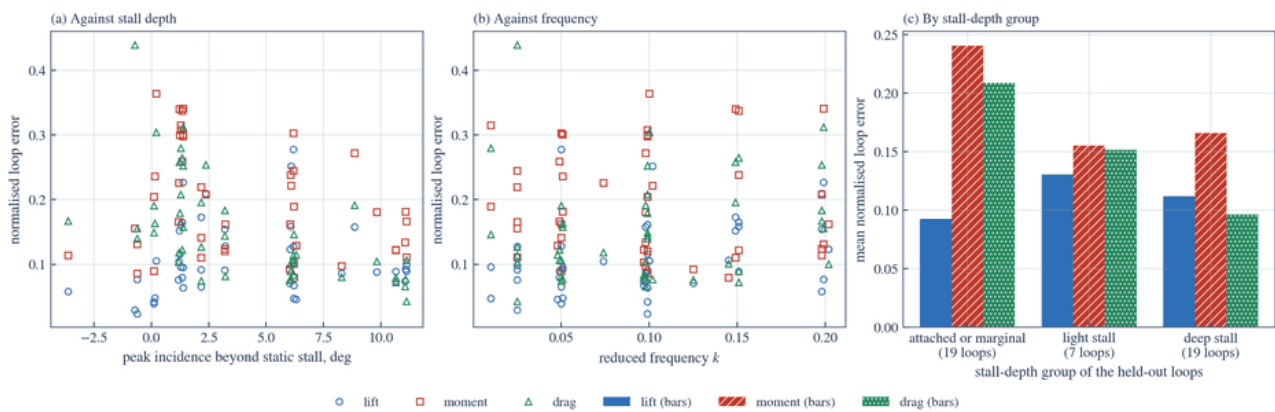
results/figures/fig08_design_space.png

Contours of the predicted peaks over mean incidence and reduced frequency.



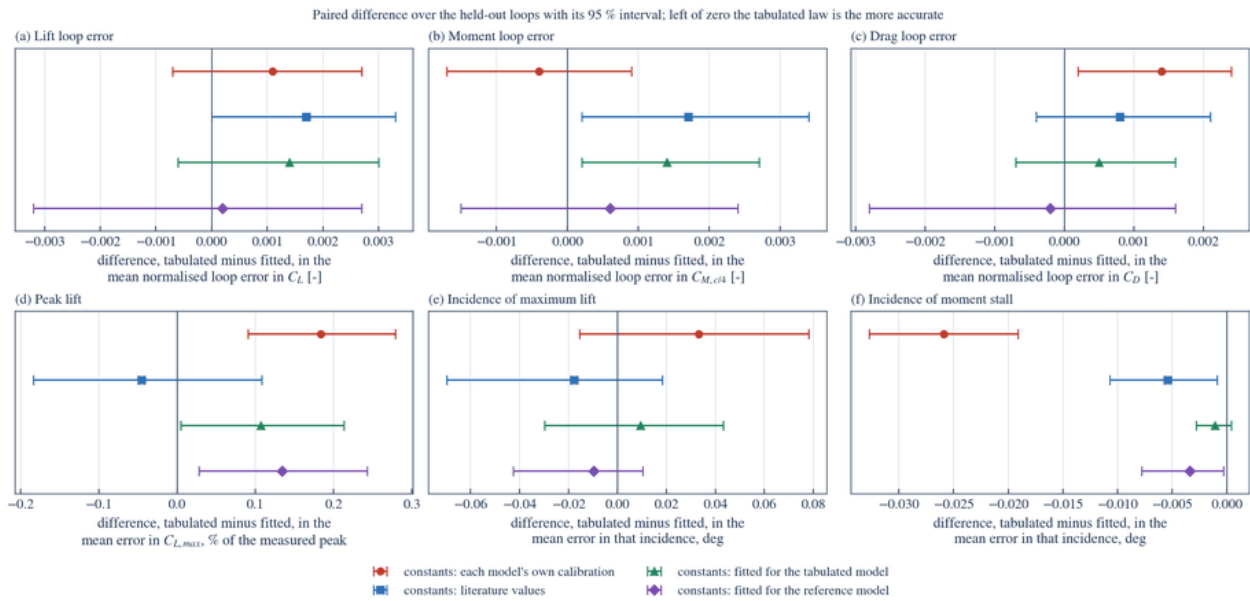
results/figures/fig09_damping_map.png

Cycle damping over the same plane, with the measured loops.



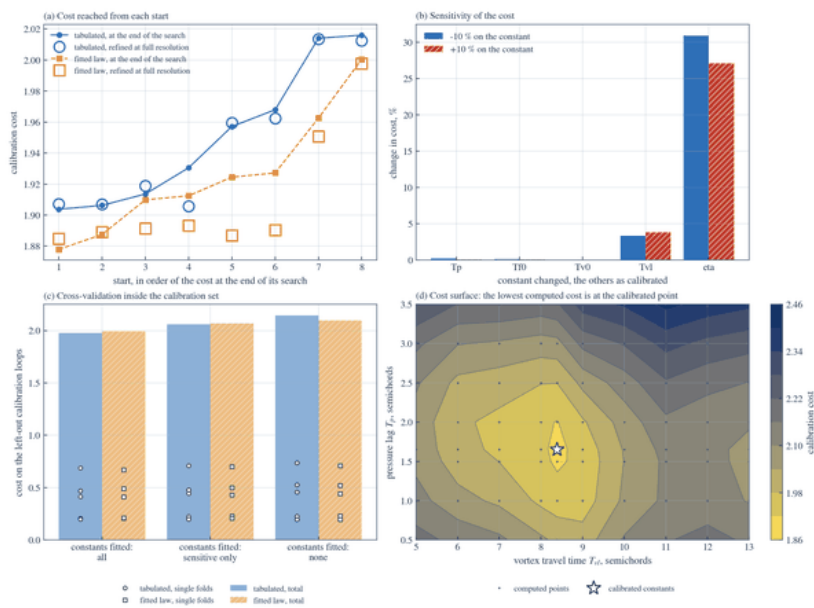
results/figures/fig11_error_by_condition.png

Loop error against stall depth and frequency.



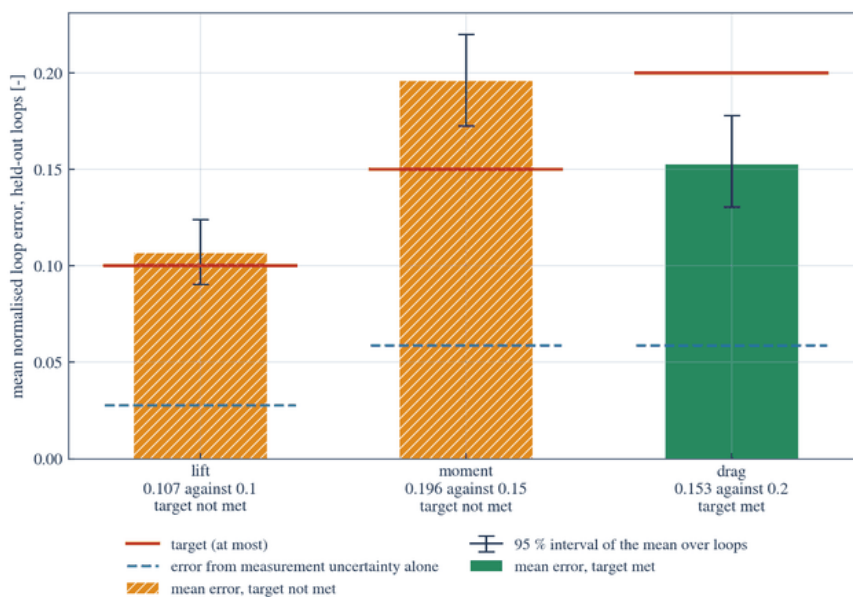
results/figures/fig12_separation_laws.png

Tabulated against fitted separation law, paired differences.



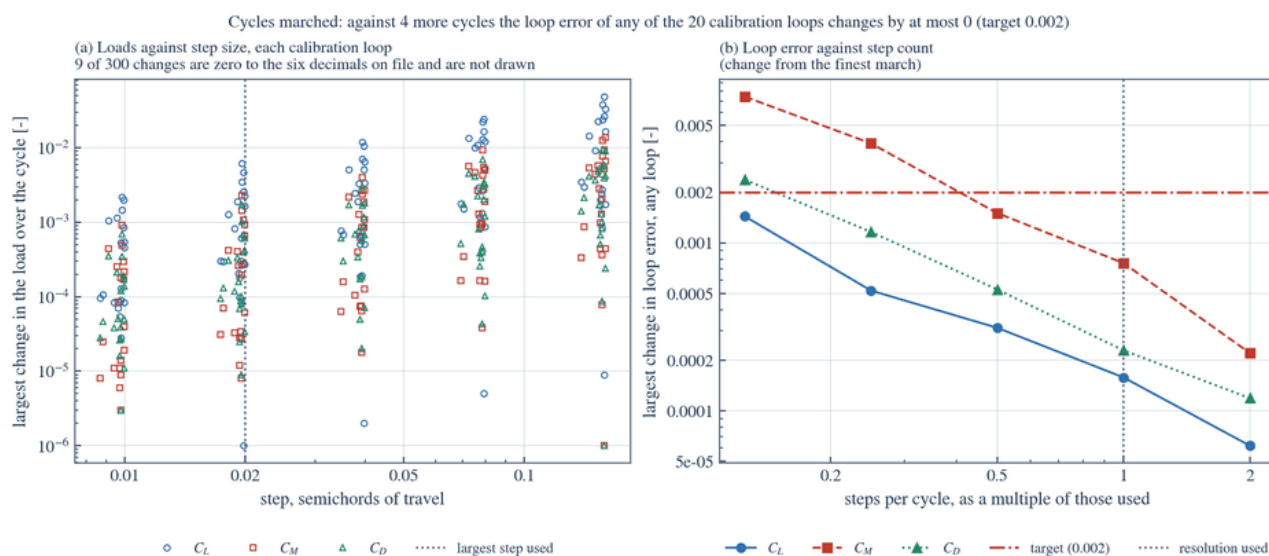
results/figures/fig14_calibration.png

Cost of every start, sensitivity, cross-validation, cost surface.



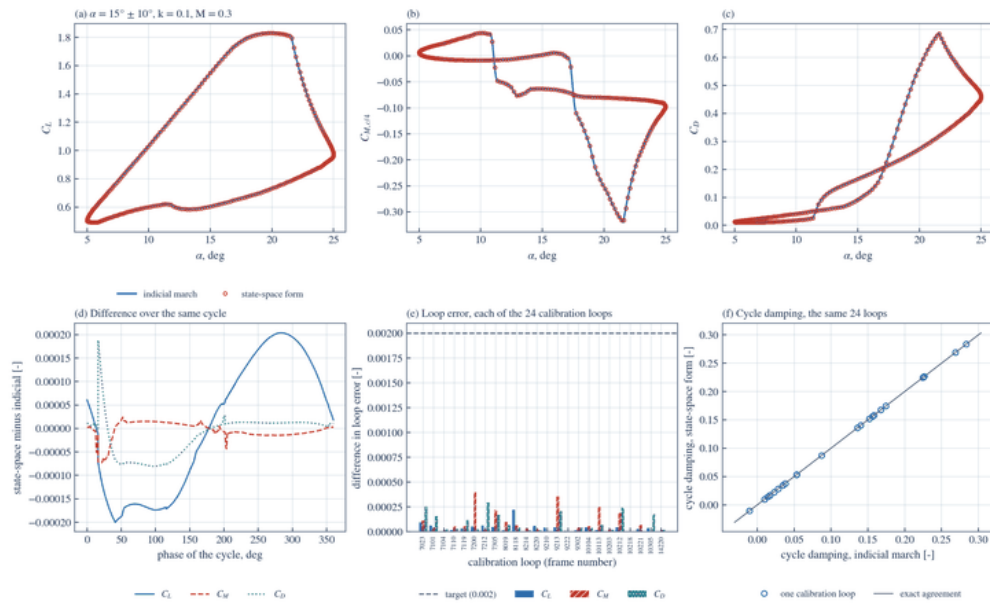
results/figures/fig16_accuracy_summary.png

Held-out errors, 95 % intervals, measurement floor and targets.



results/figures/fig17_convergence.png

Loads and loop errors against step size and cycles marched.



results/figures/fig18_state_space.png

The state-space form against the indicial march.